

ATTAINABLE INFORMATION GEOMETRY IN POSITIVE EXPERIMENTS

QIAN QI

ABSTRACT. We determine finite-memory prediction error in positive experiments through the geometry jointly acquired by the past and observed by the future. For sparse monomial experiments at a fixed horizon, maximal Vandermonde products give the sharp resolution profile uniformly across all additive exponent collisions. The profile is truncated by the attainable past dimension and is realized by one causal finite-state filter. The proof combines unconditional positive-history mass, a whole-image anisotropic cover, and raw-moment updates without reciprocal collision gaps. For command-polynomial likelihoods on an arbitrary compact latent space, an augmented moment rank gives the checkpoint and causal memory exponents for every prior. For one affine acquisition step, the full profile is determined by the physical acquisition–query covariance, uniformly through rank loss. We characterize exact kernels of continuous moment pairings by a positivity condition and the rank of a quotient multiplication pencil, with full-support witnesses obtained by arbitrarily small bounded density perturbations. This separates exact realization from the weaker feasibility of annihilation. A fixed positive experiment realizes arbitrary small covariance matrices; classical analytic orders then determine their joint memory–degeneration law. Circular contrast, operational recovery of scale products, and prior ambiguity give further consequences, with separate numerical resource bounds for the monomial realization.

1. INTRODUCTION

A finite memory in an experiment must preserve the prediction directions that the past has actually acquired and the future can still distinguish. It must also permit every subsequent update without recovering a discarded command. These requirements are not determined by the ambient dimension of a posterior family. When an experiment degenerates, different prediction directions can disappear at different rates, and their attainable dimension can be smaller than the future test space.

The principal result of this paper identifies this geometry uniformly through arbitrary additive exponent collisions in positive monomial experiments. A family of exterior volumes gives the optimal checkpoint error for every integer memory budget. The same family, maximized over checkpoints, gives the error of one causal filter. The constants remain uniform at the collision itself. The proof connects three distinct assertions: unconditional mass acquired by positive histories, an anisotropic cover of the entire reachable image, and stable updates of reachable representatives in coordinates with no reciprocal collision gaps.

We place this result within a structural theory for command-polynomial experiments on arbitrary compact latent spaces. That theory separates rank from resolution, and the dependence on a specified prior from attainment under every full-support prior. Positive affine acquisition provides an exact covariance model for the resolution geometry. A quotient of the product space distinguishes prescribed annihilation from exact kernel realization. The circular detector, the

Date: September 7, 2026.

2020 Mathematics Subject Classification. 62B05, 62C10, 93E11, 68Q19, 13D40.

Key words and phrases. attainable information state, positive experiment, finite-state filtering, exponent collisions, causal memory, exact moment kernels, prior strata.

inverse of the checkpoint budget curve, and prior ambiguity then describe other aspects of the same acquired–observed distinction. Each conclusion below is stated with its own uniformity and resource assumptions.

1.1. The collision-uniform memory law. For the monomial experiment write

$$A = \{0 = a_0 < a_1 < \cdots < a_{r-1} = D\}, \quad W_A = \text{span}\{t^a : a \in A\}, \quad h_A(m) = |mA|,$$

where mA consists of sums of exactly m members of A , including zero entries. A positive finite detector spans W_A . A command in $[\eta, 1 - \eta]^J$, with $0 < \eta < 1/2$, retains or rejects each cell; rejection is a report, and every trial is counted. All report likelihoods are at least $\kappa > 0$. The exact information statements hold on $I = [l, u]$, $0 \leq l < u < \infty$, with a full-support prior. For the uniform collision statements take $I = [0, 1]$, fix $N \geq 2$, and let the one-step exponents vary over a compact subset K of the strictly ordered chamber. One fixed full-column-rank coefficient matrix specifies the positive detector on K , and μ is fixed with full support; it need not have a density.

Choose $H \geq \max\{1, N \max_K a_{r-1}\}$. Keep every nonconstant formal future label $\alpha \in \mathbb{N}_0^r$, $|\alpha| = m$, even when exponents coincide. Put

$$q_m = \binom{m+r-1}{r-1} - 1, \quad x_\alpha = H^{-1}\alpha \cdot a, \quad \mathcal{V}_{m,\ell}(a) = \max_{|F|=\ell} \prod_{\{\alpha,\beta\} \subset F} |x_\alpha - x_\beta|.$$

Thus $\mathcal{V}_{m,1} = 1$; a zero product remains zero. Set

$$\begin{aligned} p_{n,m} &= \min\{n(r-1), q_m\}, \\ \Psi_{n,m}(M, a) &= \max_{1 \leq \ell \leq p_{n,m}} (\mathcal{V}_{m,\ell}(a)/M)^{2/\ell}, \\ \Xi_N(M, a) &= \max_{1 \leq n < N} \Psi_{n, N-n}(M, a). \end{aligned}$$

Endpoint profiles are zero. The checkpoint query is a uniformly selected member of a fixed finite spanning menu of physical m -trial products, selected independently after the memory index is formed. Only the selected query is executed. Commands during acquisition are independent uniform commands for the unconditional criterion. A checkpoint code may read the complete acquired prefix before retaining one of M labels. A causal code must update that label after each report without reading discarded commands. Regret is the excess squared loss for the selected query; its unconditional and worst-history versions are compared below. Persistent labels, the read-only clock, and the fixed program are specified in Definition 2.4.

Theorem 1.1 (Intrinsic attainable resolution and causal memory). *Under these hypotheses there are constants $0 < c \leq C < \infty$, depending only on the fixed experiment, K , N and μ , with the following properties for every known calibration $a \in K$ and every integer $M \geq 1$.*

At each checkpoint $n + m = N$, the optimal M -message squared-prediction regret is between $c\Psi_{n,m}(M, a)$ and $C\Psi_{n,m}(M, a)$, both unconditionally under independent uniform acquisition commands and in the minimax sense over admitted histories. The query is sampled independently after the index is formed, from the ordered products of one fixed attainable failure-factor basis; all its m trials are counted.

There exists one deterministic stage-compatible transducer, allowed to depend on a and M , storing at most M persistent labels after every report, whose regret at every checkpoint and on every history is at most $C\Xi_N(M, a)$. Conversely every such transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under the same common exploration law. Independent coding randomization does not remove the lower bound. The clock and fixed real-valued program are read-only as in Definition 2.4.

The constants are uniform across all additive collision strata of K . No calibration-blind codebook or uniformity over all full-support priors is part of the assertion.

The exact continuous-state dimension is separately $d_A(n, m) = \min\{n(r-1), |mA| - 1\}$, as stated in Theorem 2.5. The finite-resolution formula specifies how these directions disappear. Its proof joins three assertions not contained in an ambient spectral estimate. Actual interior failure factors have a separated binomial product tangent. Pairing it with complete Newton–Hermite prefixes loses exactly the evidence direction under normalization and supplies a positive-history minorization. The entire posterior image has bounded semialgebraic format, so its thin-rectangle covering bound truncates at the same attainable dimension. Finally, raw-moment updates have a Lipschitz recurrence without inverse collision gaps, and quantization of reachable representatives realizes one causal filter. These are established in Lemmas 5.2 and 4.4 and Theorems 5.3 and 5.4.

1.2. From attained directions to one causal filter. The scales in the preceding theorem belong to the interaction of past and future. A future Vandermonde determinant alone supplies neither history mass nor a streaming algorithm. The proof first pairs the actual binomial product tangent with complete Newton–Hermite prefixes. After removal of evidence, every attainable prefix has uniformly positive unconditional mass. The past truncation $n(r-1)$ therefore enters the lower bound before quantization.

For the upper bound, every report-word posterior image is a rational image of its command cube. Bounded semialgebraic format and its attainable dimension control a cover of the whole image in a thin rectangle. This step does not require a single global regular chart. For the causal part, the representative is a reachable raw-moment vector. Multiplication by the next likelihood closes the remaining test space, and its positive evidence denominator bounds the update on posterior-mixture segments. The finite-horizon error recurrence includes every preceding quantization error. Thus the theorem joins local acquisition, global geometry, and temporal compatibility, rather than using any one of them as a substitute for the others.

The structural and affine results below explain which parts of this argument are general and which depend on monomial acquisition. In particular, a dimension law has no collision-uniform constants unless additional geometry is proved, while an affine covariance law does not by itself supply the multi-step Newton flags.

1.3. Structural rank and prior strata. Let X be a compact metric space, let μ be a probability on X , and fix a finite horizon N . At time i the command belongs to a finite-dimensional cube, the report alphabet is finite, and

$$L_{i,y}(u, x) = \sum_{\alpha \in F_{i,y}} u^\alpha c_{i,y,\alpha}(x), \quad c_{i,y,\alpha} \in C(X), \quad L_{i,y} \geq \kappa > 0, \quad \sum_y L_{i,y} = 1.$$

The polynomial condition concerns u , not x . In particular, finite detectors with affine retention gates belong to this class for arbitrary continuous detector cells. Future word likelihoods span a finite-dimensional test space. Choose a finite spanning list of actual queries, each with a fixed positive selection probability, and write $p_n(h)$ for the weighted vector of their conditional success probabilities after a history of length n .

For each past report word a , let P_a be its likelihood product as a function of the commands, set $Z_a = \mu P_a$, and put

$$Y_a = (Z_a, Z_a p_n)^t, \quad d_n = \max_a \text{rank}_{\mathbb{R}(u)}[Y_a, DY_a] - 1.$$

All entries are command polynomials with finitely many actual prior moments as coefficients. The evidence column is adjoined before normalization; it is not counted as an acquired prediction

direction. Theorem 6.2 proves, whenever $d_n > 0$,

$$R_{M,n}^{\text{av}} \asymp R_{M,n}^{\text{max}} \asymp M^{-2/d_n}, \quad \mathfrak{R}_{M,N}^{\text{av}} \asymp \mathfrak{R}_{M,N}^{\text{max}} \asymp M^{-2/d_*}, \quad d_* = \max_{1 \leq n < N} d_n > 0.$$

Here R is checkpoint regret and $\mathfrak{R}$ is the largest checkpoint regret of one causal filter. The average criterion uses independent uniform commands and includes report evidence; the maximum criterion ranges over all admitted histories. A zero rank gives a finite reachable set, not necessarily a singleton. If all checkpoint ranks vanish, a finite exact causal filter exists.

This formula classifies each specified experiment without assuming an accessible flag or an ordered-scale representation of its risk. Theorem 6.4 further identifies every locus $d_n \leq k$ by the vanishing of finitely many polynomials in the prior moments: they are the command coefficients of the $(k+2)$ -minors of the augmented matrices. Full-support priors fill the relative interior of the finite moment body. Simultaneous maximal ranks hold off a proper algebraic set there, and can be reached by arbitrarily small bounded-density perturbations of any full-support prior. Exceptional priors are not excluded from the classification; their actual ranks enter the formula. Uniform constants hold on compact constant-rank moment sets. A rank formula by itself does not assert uniformity across changing ranks.

1.4. Affine acquisition and realizable degeneration. The first full-class resolution theorem applies to positive affine acquisition. Let $f \in C(X)^b$ with $\sup_x \sum_i |f_i(x)| \leq a < 1$, choose a uniform command $u \in [-1, 1]^b$, and observe a report $y \in \{-1, 1\}$ with likelihood

$$L_y(u, x) = \frac{1}{2}(1 + yu^t f(x)).$$

For any finite menu of future query probabilities, let h be its vector weighted by the square roots of the query probabilities and define

$$C_\mu = \text{Cov}_\mu(h, f).$$

The physical command cube fixes the input units. No singular whitening is performed. If $s_1 \geq \dots \geq s_d \geq 0$ are the singular values of C_μ , Theorem 7.1 proves

$$R_M^{\text{av}} \asymp R_M^{\text{max}} \asymp \max_{1 \leq \ell \leq d} \left(\frac{s_1 \cdots s_\ell}{M} \right)^{2/\ell}.$$

The constants depend only on a, b and the number of queries. They are uniform over priors, query weights, multiplicities, and all rank losses. The assertion includes the one-label exact case $C_\mu = 0$.

The proof derives this geometry before addressing quantization. With $v = yu$ and $m = \mu f$, the exact prediction vector is

$$p = \mu h + C_\mu \theta, \quad \theta = \frac{v}{1 + m^t v}.$$

The law of v , including both reports and their evidence, has density $(1 + m^t v)/2^b$. Its projective image contains a fixed ball, is contained in another fixed ball, and has a uniformly positive density on the inner ball. Thus the same covariance image controls both global approximation and unconditional acquisition mass. This is not an inference from an ambient observation matrix. As a concrete interior rank change, acquisition through x and prediction through x^2 under $d\mu_t = (1 + tx) dx/2$ on $[-1, 1]$ have covariance proportional to t . Proposition 7.5 obtains the full uniform $t^2 M^{-2}$ law, including zero at $t = 0$, while every prior still has full support.

A second conclusion concerns rank degeneration inside a dominated prior family. For each pair b, q , one fixed positive acquisition and one fixed physical query menu admit priors μ_A with

$$\frac{1}{2}\mu_0 \leq \mu_A \leq \frac{3}{2}\mu_0, \quad C_{\mu_A} = \frac{A}{8b\sqrt{q}}, \quad \sum_{j,i} |A_{ji}| \leq \frac{1}{2}.$$

Thus arbitrary small matrix germs, and not only a scalar cancellation, are realized by genuine priors in one experiment (Theorem 12.1). Along an analytic arc let A_ℓ be the least vanishing order among its ℓ -minors. Theorem 12.3 gives, uniformly in small $t > 0$ and all integer $M \geq 1$,

$$R_{M,1}(t) \asymp \max_{1 \leq \ell \leq r} t^{2A_\ell/\ell} M^{-2/\ell}.$$

The differences $\alpha_\ell = A_\ell - A_{\ell-1}$ are ordered nonnegative integers. Every such finite list occurs under the same physical bounds, and the full checkpoint budget curve determines the A_ℓ operationally. In particular the phase exponent along $M = \lceil t^{-\gamma} \rceil$ is $\min_\ell 2(A_\ell + \gamma)/\ell$. This is a complete arcwise classification for affine acquisition, not a simultaneous normal form for all polynomial history maps in several parameters.

1.5. Universal attainment and exact kernel constraints. Theorem 8.1 addresses a different quantifier. At an interior positive history let $\tilde{T} = \text{span}\{P\} + \text{im } DP$ and $\Phi = \text{span}\{1, \phi_1, \dots, \phi_p\}$. When both spaces have dimension $p+1$, normalized acquisition has rank p for *every* full-support prior if and only if

$$\det[v_i(x_k)]_{i,k=0}^p \det[\phi_j(x_k)]_{j,k=0}^p$$

has one weak sign on X^{p+1} and is nonzero somewhere. Here v is any basis of $\tilde{T}$. Necessity follows by approximating atomic probabilities by full-support probabilities; sufficiency follows from determinant integration. Separate Chebyshev properties are sufficient instances, not necessary hypotheses. In particular, $\tilde{T} = P\Phi$ gives prior-uniform full rank on arbitrary compact latent spaces. The square hypothesis is explicit; no reduction of every rectangular pairing to this criterion is asserted.

For finite-dimensional $E, F \subset C(X; \mathbb{R})$ with $\dim F \leq \dim E$, Theorem 9.2 characterizes full pairing rank under every full-support prior by

$$0 \neq f \in F \implies \text{there exists } g \in E \text{ with } fg \geq 0, \quad fg \not\equiv 0.$$

Theorem 9.3 characterizes subspaces that can be contained in a pairing kernel, and constructs full-support priors attaining the minimum rank. Exact realization of a specified subspace requires a further condition. Adjoining the positive history product before normalization converts pairing rank into actual prediction rank. This yields the unconditional mass conclusion without a square reduction. The five-point experiment in Proposition 9.6 satisfies the rectangular criterion but has no fixed universally nonsingular two-dimensional tangent subspace containing evidence.

The exact-kernel question admits a finite multiplication criterion. For a specified $U \subset F$ set

$$W_U = \text{span}(EU), \quad V = \text{span}(\{1\} \cup EF).$$

Annihilation is feasible under full support precisely when $W_U \cap C(X)_+ = \{0\}$. Choose a basis of V/W_U beginning with $[1]$, and express the pairing on $E \times (F/U)$ in this basis. This gives an affine matrix pencil $\mathcal{H}_U(z)$ depending only on pointwise multiplication modulo W_U . Theorem 10.2 proves

$$\exists \mu \text{ of full support with } \ker H_\mu = U \iff W_U \cap C(X)_+ = \{0\}, \quad \text{rank}_{\mathbb{R}(z)} \mathcal{H}_U = \dim(F/U).$$

The determinantal condition is a polynomial identity test on the full affine quotient, not an existential restatement on an unknown prior slice. Positive density perturbations give local

coordinates on that slice and attain its maximal rank arbitrarily close to every feasible prior. The common forced kernel is $\{f \in F : Ef \subset W_U\}$, which need not equal U . Proposition 10.4 computes both the forced kernel and the exact criterion in a family of positive block experiments by the common zeros of a polynomial subspace. This includes examples where annihilation is feasible but exact realization fails, and explicit full-support witnesses for the subspaces that are exactly realizable. Corollary 10.5 returns this distinction to the actual normalized prediction rank and its unconditional history mass.

1.6. Contrast, inversion, and unresolved prior information. A circular experiment with Haar prior has the positive cells $\frac{1}{4}(1 \pm \tau \cos \theta)$ and $\frac{1}{4}(1 \pm \tau \sin \theta)$. Its four lookup probabilities vary continuously in $[\eta, 1 - \eta]^4$. For every fixed horizon and all sufficiently small known $\tau \geq 0$, the checkpoint law is

$$H_{n,m}(M, \tau) = \max_{1 \leq j \leq \min(n,m)} \tau^{2(j+1)} M^{-1/j}.$$

The j th harmonic is attenuated by τ^j when acquired and again by τ^j when queried. The real physical scales are therefore $\tau^2, \tau^2, \tau^4, \tau^4, \dots$, not just a count of Fourier coefficients. Theorem 13.1 proves a uniform causal realization through weighted Fourier updates without inverse contrast. The contrast interval and constants belong to the fixed horizon, not to an unbounded-horizon assertion.

Within these established profile families the complete checkpoint curve has an inverse. For $R(M)$ define

$$\mathcal{I}_\ell(R) = \inf_{M \in \mathbb{N}, M \geq 1} MR(M)^{\ell/2}.$$

Theorem 14.2 and Corollary 7.4 identify this transform with the initial scale products up to uniform constants. Integer budgets require a factor-two sandwich; the underlying real-budget identity is exact. Repeated scales and zero products are included. This inverse uses the entire unbounded budget curve, not finitely observed errors, and does not recover a latent model or exact optimal constants. Nor is it applied to a maximum over checkpoints or to a curve with a positive uncertainty floor.

Prior uncertainty has its own geometry. For monomials, the local ambiguity body after a positive history has successive width orders $\varepsilon d_1, \dots, \varepsilon d_{q_m}$, where d_i are the future Leja scales. There is no past-dimension truncation of this ambiguity body. After an exact prefix of k raw future moments, the remaining squared radius is $\asymp \varepsilon^2 d_{k+1}^2$. The normalization of prior tilts is explicit in Proposition 15.3. For an imperfect common moment name of error δ , the separate dominated-family and interior-slack hypotheses of Theorem 16.1 give the sharp joint law $\Xi_N(M, a) + \delta^2$. The lower bound uses fixed genuine prior alternatives and overlapping history laws, not alternatives chosen after seeing a history.

1.7. Classical inputs and the scope of the conclusions. The analytic inputs include semialgebraic dimension and metric entropy, determinant integration, total positivity, and divided differences [31, 32, 7, 12, 16, 18, 17]. The finite square-pairing criterion is the strong-compatibility result of Banaji–Pantea [33]; the finite rectangular sign-vector alternative is present in Müller et al. [34]. The exact specializations are recorded in Section L.1. The product-character expansion used to perturb the covariance is classical Fourier analysis on the finite cube [36, Section 2.2]. Analytic invariant orders and their singular-value interpretation are classical as well [35]. We do not claim any of these inputs as an independent advance in their respective theories.

What must additionally be proved in the experiment is the passage from these inputs to attainable prediction error with the stated quantifiers. For monomials this is the collision-uniform conjunction of positive history mass, the dimension-truncated whole-image cover,

Result	Acquisition and latent space	Uniformity	Memory conclusion
Collision law	Monomial, multiple steps; interval	Fixed full-support prior; fixed horizon and compact ordered chamber; all additive collisions	All integer budgets; checkpoint and one causal filter
Structural rank	Command-polynomial; arbitrary compact space	Every specified prior; uniform on compact constant-rank moment sets	Rank exponent; checkpoint and one causal filter
Affine profile	One affine step; arbitrary compact space	Across all covariance rank losses, for fixed physical bounds and dimensions	Full checkpoint profile; not a general multi-step profile
Exact kernels	Continuous finite product spaces	Full-support annihilation slice; arbitrarily small bounded positive tilts	Exact local rank and acquired mass; global and causal claims require more
Circular law	Multiple steps; circle and Haar prior	Fixed horizon and known sufficiently small contrast, including zero	Paired scales; checkpoint and one causal filter
Prior ambiguity	Monomial moment uncertainty	Stated domination and interior-slack hypotheses	Widths; joint memory and common-moment error

TABLE 1. Hypotheses and conclusions. In every memory statement the query is chosen after the label is formed. Persistent labels are charged; known calibration, clock, and the fixed program follow the stated read-only convention. No discarded command tape is free.

and one stage-compatible causal filter. For affine acquisition, Bayes' formula gives the global covariance image and its actual probability law before quantization. The analytic-arc law is a consequence of that physical classification and classical matrix orders, not a separate normal-form theory. The kernel theorem supplies the missing exact-realization condition together with positive prior witnesses; the underlying separation and determinantal arguments remain elementary.

The manuscript first develops the monomial experiment, its attainable transversality, and the collision-uniform law. It then gives the general rank and affine classifications, the square, rectangular, and exact kernel criteria, and the associated positive-history and analytic consequences. The circular classification, operational inverse, and prior-uncertainty geometry follow. The observation algebra, confluent specializations, sequential decision results, and numerical constructions appear with their proofs in the appendices.

The separate finite implementation theorem, Theorem H.9, supplies integer transition tables and dyadic outputs under its explicit numerical interface. Advice, program length, transient workspace, and persistent labels are distinct resources. Sharpness is asserted for persistent memory; sufficient bounds are supplied for the other resources. The intrinsic results do not presume computability of unspecified real prior moments. Detailed comparisons with finite control approximations and information-state methods are in Appendix L.

2. CALIBRATED EXPERIMENTS AND THEIR OPERATIONAL QUOTIENT

All spaces carry their Borel sigma-algebras. Deterministic continuation, transition and decision rules are Borel measurable; randomized rules are Borel probability kernels. Every

external random seed is independent of the hidden parameter and the acquired history unless explicitly charged as part of that history.

Let $k_1, \dots, k_K \in W_A$ be strictly positive on I , satisfy $\sum_j k_j = 1$, and span W_A . On each trial a fresh raw symbol has these probabilities conditional on the same fixed parameter t . The raw trials are conditionally independent. A command $g \in \mathcal{G} = [\eta, 1 - \eta]^K$ accepts raw symbol j with probability g_j , using an independent prepared uniform variable. A rejected symbol is reported as $\dagger$. Thus the report alphabet is $\mathcal{X} = \{1, \dots, K, \dagger\}$ and its likelihood factors are

$$(1) \quad f_{g,j} = g_j k_j, \quad f_{g,\dagger} = F_g = \sum_j (1 - g_j) k_j.$$

A single command is applied at every possible parameter value. Every trial is counted, including every rejected report. There is a constant $\kappa > 0$ such that $\kappa \leq f_{g,x}(t) \leq 1$ for all g, x, t . Indeed one may take $\kappa = \eta \min_{j,t} k_j(t)$.

Lemma 2.1 (Attainable interior and realizability). *The failure family is precisely the image of the rejection cube under $u \mapsto \sum_j u_j k_j$. It contains a neighborhood of the constant $1/2$ in W_A . Conversely, every finite exponent set A as above has a positive normalized r -cell realization with this property.*

Proof. The displayed map is a surjective linear map onto W_A . Choose a linear right inverse $L : W_A \rightarrow \mathbb{R}^K$. For every v satisfying $\|Lv\|_\infty < 1/2 - \eta$, rejection vector $u = \mathbf{1}/2 + Lv$ realizes $1/2 + v$ inside the cube. This proves both the interior statement and that the same physical command implements it. For existence, choose $\delta > 0$ with $\delta \sum_{a \in A \setminus \{0\}} u^a < 1/r$ and set

$$k_a(t) = 1/r + \delta t^a \quad (a \neq 0), \quad k_0(t) = 1/r - \delta \sum_{a \neq 0} t^a.$$

They are positive and sum to one. Their span contains one and then each $t^a = (k_a - 1/r)/\delta$, so has dimension r . These are calibrated categorical experiments. No mechanical origin is asserted for this general realization; the separate Lorentz realization is derived in Section G. $\square$

A history $h = (g_1, x_1, \dots, g_n, x_n)$ has posterior

$$(2) \quad P_h(t) \mu(dt), \quad P_h = \frac{\prod_{i=1}^n f_{g_i, x_i}}{\mu(\prod_{i=1}^n f_{g_i, x_i})}.$$

All commands are known inputs when received. They cease to be available once discarded by a memory-limited filter. A continuous history encoding means continuity in $(g_1, \dots, g_n)$ separately for each fixed report word. It need not be injective on histories and need not identify all histories with the same posterior.

Definition 2.2 (Common future-only continuation). At a checkpoint a continuation has the same fixed task data in the two experiments. Its commands are functions only of the reports and commands *since that checkpoint*, and any fresh independent random seed. Its first command is the same in both experiments. It may not read the original prefix, a prefix-dependent seed, or any other uncharged historical tape. Two posteriors are m -equivalent when all such length- m continuations have the same future report law.

A separate payoff automaton referring to past observations is task memory, not something removed by posterior equivalence. Losses that inspect t directly are also not included without adding their test functions. This quantifier matters: two different constant failure commands leave the same prior, but a rule that reads and repeats the original command does not meet Definition 2.2.

Proposition 2.3 (Exact future test space). *The linear span of length- m future likelihood tests is*

$$(3) \quad W_A^m = \text{span}\{t^s : s \in mA\}.$$

Two posteriors are m -equivalent if and only if their moments at the exponents in $mA \setminus \{0\}$ agree. A finite list of actually executable, nonadaptive all-failure probes spans the same space.

Proof. Choose finitely many attainable failures spanning W_A , using Lemma 2.1. All their m -fold products are executable all-failure event probabilities. Multilinearity makes their span the span of all products of m members of W_A , which is exactly (3). Independence is conditional on t ; one does not replace the shared parameter by fresh parameters between trials.

For any fixed future word a deterministic continuation chooses a known sequence of commands along that word. Its likelihood is a product of m members of W_A , even when the chosen sequence differs on other branches. Equality of the stated moments therefore implies equality of all word probabilities. There are finitely many report words. Integrating an independent continuation seed proves the randomized case. Conversely, equivalence includes all the spanning all-failure probes; their predictions determine all the moments in (3). The constant moment is one. Only the selected probe is executed and all its m trials are consumed, regardless of earlier nonfailure reports. $\square$

A fixed positive nonconstant denominator or an arbitrary nonmonomial factor space is not silently substituted in (3). Such models have their own product and test pairings. The theorem below addresses every sparse monomial space W_A , not every finite-dimensional space of continuous functions.

2.1. Persistent state and prediction loss.

Definition 2.4 (Finite-state filter). An M -state filter has transitions

$$i_{n+1} = U_n(i_n, g_{n+1}, x_{n+1}), \quad i_n \in \{1, \dots, M\}.$$

It reads only its previous index, the current command and current report. Past commands, past reports and retained command-generation or update seeds are unavailable outside the index. Fresh private coins may be used and then discarded. At a checkpoint the decoder reads only i_n and a newly revealed independent query. The known clock permits stage-dependent transition and output tables. Real command inputs and read-only real calibration constants are allowed; arithmetic workspace, execution time and finite-precision implementation of these fixed functions are not the memory resource of this definition. In particular, this is a finite-state transducer, not an algorithm that repeatedly rereads an exact prefix.

For each $1 \leq n < N$ fix a finite spanning probe list $H_{n,1}, \dots, H_{n,s_n^q}$ as in Proposition 2.3, using $m = N - n$ trials. To avoid confusion, s_n^q is the number of queries, not the exact state s_n . After the checkpoint state is formed, choose an independent uniform query J . Announce $a \in [0, 1]$, execute that probe using its m fresh trials, and let Y be its all-failure indicator. All N trials are counted. The conditional squared-loss excess over the full-history predictor is

$$(4) \quad \mathcal{R}(a, p) = \frac{1}{s_n^q} \sum_j (a_j - p_j)^2, \quad p_j = \mu(P_h H_{n,j}).$$

This follows by expanding $\mathbb{E}[(a - Y)^2 \mid h, J]$; decoder randomization is averaged on the right. Only one selected checkpoint and one probe are executed in a run. Simultaneous bounds over checkpoints do not count the unexecuted probes as physical trials.

2.2. Exact information and continuous state dimension.

Theorem 2.5 (Sparse finite-horizon information law). *Suppose the positive finite detector spans W_A and every lookup entry ranges over $[\eta, 1 - \eta]$, where $0 < \eta < 1/2$. For $n, m \geq 0$ put*

$$(5) \quad d_A(n, m) = \min\{n(r - 1), h_A(m) - 1\}.$$

Two attainable posteriors after n reports have the same law under every common future-only continuation of length m if and only if their moments at the exponents in mA agree. The prediction map on all-failure command histories has generic rank $d_A(n, m)$, and it has a smooth attainable patch of this dimension with a continuous section to histories. The smallest Euclidean dimension of a sufficient encoding continuous on each known-report history stratum is exactly $d_A(n, m)$.

For a total budget N , these dimensions can be attained simultaneously by a causal exact-real state. Its maximal size is

$$(6) \quad D_A(N) = \max_{0 \leq n \leq N} d_A(n, N - n).$$

Only prior moments with exponents in NA are needed for its updates.

3. ATTAINABLE TRANSVERSALITY AND THE EXACT MEMORY PROFILE

We first prove the positivity statement that prevents a prior-dependent loss of rank. No density assumption on the prior is needed.

Lemma 3.1 (Strict mixed-moment positivity). *Let $\alpha_1 < \dots < \alpha_s$ and $\beta_1 < \dots < \beta_s$ be nonnegative real numbers. For every full-support probability on I ,*

$$(7) \quad \det[\mu(t^{\alpha_i + \beta_j})]_{i,j=1}^s > 0.$$

Consequently a rectangular moment matrix indexed by two increasing finite sets of distinct nonnegative exponents has rank the smaller cardinality.

Proof. For increasing positive points $t_1 < \dots < t_s$, the generalized Vandermonde determinant $\det[t_j^{\alpha_i}]$ is positive. Here is a self-contained sign argument. Put $t_j = e^{x_j}$. A nonzero combination of s exponentials with distinct ordered exponents has at most $s - 1$ distinct zeros: factor out the first exponential, differentiate, and use Rolle's theorem inductively. Thus the determinant cannot vanish on the connected region $x_1 < \dots < x_s$. When the x_j coalesce to x_0 from ordered nearby points, its first nonzero Taylor term has the sign of the Wronskian

$$e^{x_0 \sum_i \alpha_i} \prod_{i < j} (\alpha_j - \alpha_i)$$

times the ordinary positive Vandermonde in the x_j , divided by positive factorials. Its sign is therefore positive everywhere in that region. The same holds for the β determinant.

Expanding determinants and applying Fubini gives the finite determinant integration identity

$$\det[\mu(t^{\alpha_i + \beta_j})] = \frac{1}{s!} \int_{I^s} \det[t_j^{\alpha_i}] \det[t_j^{\beta_i}] \prod_{j=1}^s \mu(dt_j).$$

The integrand is nonnegative, by permutation symmetry and continuity at zero or repeated points. Choose s disjoint ordered subintervals in $(l, u) \cap (0, \infty)$. Each has positive prior mass by full support; on their product the integrand is strictly positive. The integral is strictly positive. Applying this to any square minor proves the last assertion. This is the classical total-positivity mechanism [7], with its needed form proved explicitly here. $\square$

Lemma 3.2 (A monomial tangent inside the attainable cube). *For every $n \geq 1$, choose distinct, sufficiently small positive numbers $c_1, \dots, c_n$. The failure factors $F_i = (1 + c_i t^D)/2$ are simultaneously attainable at interior commands. At this tuple the differential image of the unnormalized product map $W_A^n \rightarrow \text{span}\{t^s : s \in nA\}$ is*

$$(8) \quad T_{A,n} = \text{span}\{t^b : b \in B_{A,n}\},$$

where

$$(9) \quad B_{A,n} = \{jD : 0 \leq j \leq n\} \cup \bigcup_{a \in A \setminus \{0, D\}} \{a + jD : 0 \leq j < n\}.$$

In particular $\dim T_{A,n} = n(r-1) + 1$, and the product itself belongs to this tangent space.

Proof. Attainability follows by applying the fixed right inverse in Lemma 2.1 to $c_i t^D/2$ and taking all c_i small enough. The factors and their implementing commands lie in an open subset of the allowed cube. Constants 2^{-n} do not change a differential image, so omit them in the remaining calculation. Put $z = t^D$, $P(z) = \prod_i (1 + c_i z)$ and $P_i(z) = P(z)/(1 + c_i z)$. These n polynomials form a basis of the polynomials in z of degree at most $n-1$. Indeed evaluation at $z = -1/c_j$ makes all P_i except P_j zero, and $P_j(-1/c_j) \neq 0$.

The product differential is $\sum_i \dot{F}_i \prod_{j \neq i} F_j$. Variations in the constant direction give all polynomials in z through degree $n-1$; variations in t^D give their multiples by z . Together they give all z^j , $0 \leq j \leq n$. Variations in an interior exponent a give precisely $t^a z^j$, $0 \leq j < n$. The listed exponent sets are disjoint: two numbers $a + jD$ and $a' + j'D$ with $0 < a, a' < D$ agree only if $a = a'$ and $j = j'$, and none is a multiple of D . Their total cardinality is $(n+1) + n(r-2) = n(r-1) + 1$. Every summand of the differential has this form, proving equality rather than only inclusion in (8). The product is obtained by varying one factor along itself, so is in the tangent. $\square$

Theorem 3.3 (Generic normalized pairing rank). *For fixed $n, m \geq 1$, the map from n all-failure commands to the moment vector $(\mu(t^s P_h))_{s \in mA \setminus \{0\}}$ has generic rank $d_A(n, m)$ in (5). This holds for every full-support prior, including priors without a density. At an interior attainable tuple it has a smooth $d_A(n, m)$ -dimensional local image and a continuous local section to histories. A projection onto $d_A(n, m)$ of the moment or spanning-probe coordinates maps such a section onto an open Euclidean ball.*

Proof. Let $P = \prod_i F_i$, $Z = \mu P > 0$, and include the constant test in the linear map

$$L : T_{A,n} \longrightarrow \mathbb{R}^{h_A(m)}, \quad LQ = (\mu(t^s Q))_{s \in mA}.$$

At the tuple from Lemma 3.2, Lemma 3.1 gives

$$\text{rank } L = \min\{n(r-1) + 1, h_A(m)\}.$$

Let e_0 extract the coordinate of exponent zero, and put $p = L(P)/Z$. The derivative of the normalized prediction map on this tangent is

$$(10) \quad Q \longmapsto Z^{-1}(LQ - p e_0(LQ)).$$

Now $e_0 p = 1$ and $p \in L(T_{A,n})$. The linear map $v \mapsto v - p e_0 v$, restricted to $L(T_{A,n})$, has kernel exactly $\text{span}\{p\}$ and image in the zero-constant-coordinate hyperplane. It loses *exactly* one rank. Deleting that identically zero coordinate therefore gives rank $\min\{n(r-1), h_A(m) - 1\}$.

At every tuple the normalized product map is invariant under independent rescaling of its n nonzero factors, so its rank is at most $n(r-1)$. The number of nonconstant tests gives the other upper bound. Thus the attained rank is globally maximal. The command-to-factor linear map is surjective at every interior command; the same maximal rank is attained in command

coordinates. Its Jacobian entries are rational functions of the commands with strictly positive polynomial denominator Z^2 . A maximal minor nonzero at the displayed tuple has a nonzero polynomial numerator. Its zero set is a proper real algebraic set with empty interior and zero Lebesgue measure. This proves genericity throughout the command cube, not only along the binomial family.

Choose $d = d_A(n, m)$ independent output coordinates and d command coordinates at that tuple, and freeze the complementary command coordinates. The inverse function theorem gives a local inverse for the selected output projection; shrink to a ball with compact closure. The full image of this section is a smooth local graph. A spanning list of probes is an injective linear image of the moment coordinates, so the same conclusion holds after choosing independent probe coordinates. All section histories are feasible all-failure histories. Their probability conditional on each parameter is at least η^n ; no failure evidence has been dropped. $\square$

Notice that when $d < h_A(m) - 1$ the attainable image can be curved. The proof does not replace it by its affine hull. The lower-bound ball is in a selected output projection, and its lift is a local graph, not necessarily a flat ball in the ambient prediction space.

Theorem 3.4 (Minimal continuous encoding and causal attainment). *The minimal dimension stated in Theorem 2.5 is $d_A(n, m)$. For a fixed N , there are compact exact state sets $\mathcal{S}_n \subset \mathbb{R}^{d_A(n, N-n)}$ and continuous-in-command causal updates attaining this dimension at every n . Their state represents the posterior-dependent future law, not an uncharged past-dependent task.*

Proof. Compose a sufficient continuous encoding with the section in Theorem 3.3. Different points in the projected ball have different predictions, so this composition is injective. A continuous injection of an open d -ball into $\mathbb{R}^k$, $k < d$, is impossible: embedding $\mathbb{R}^k$ as a coordinate plane of $\mathbb{R}^d$ would contradict invariance of domain. The decoder need not be continuous for this argument.

For the global upper bound, normalize each observed factor by its own prior integral, $\tilde{f}_i = f_i / \mu f_i$. Each normalized factor is in the affine hyperplane $\mu f = 1$ of W_A , with $r - 1$ coordinates. The chronologically ordered n factors determine the posterior product and use $n(r - 1)$ real coordinates. This representation may distinguish permutations; sufficiency does not forbid that. Alternatively, retain the $h_A(m) - 1$ nonconstant posterior moments in mA . Proposition 2.3 proves this alternative sufficient on every history. Use whichever representation is smaller. Both are continuous on each fixed-report stratum. Finite unions of compact continuous images of the command cube give compact state sets. The cases $n = 0$ or $m = 0$ have a single class and zero coordinates.

For causal implementation retain normalized factors while $n(r - 1) \leq h_A(N - n) - 1$. The left side increases and the right side nonincreases, so there is at most one changeover. When the inequality first fails, multiply the retained factors and compute the required moments. Afterward, if $m = N - n$ and the new report factor is $f(t) = \sum_{a \in A} f_a t^a$, update by

$$(11) \quad M'_s = \frac{\sum_{a \in A} f_a M_{s+a}}{\sum_{a \in A} f_a M_a}, \quad s \in (m - 1)A \setminus \{0\}, \quad M_0 = 1.$$

All indices $s + a$ belong to mA . The denominator is the report probability and is at least κ . At the changeover the integrands are products of n factor monomials and a remaining-test monomial, so their exponents are in $nA + (N - n)A = NA$. Normalizing each factor requires only moments in $A \subset NA$. Thus a finite read-only table $(\mu(t^s))_{s \in NA}$ suffices for all updates. Formula (11) then gives the exact shrinking state to time N . $\square$

For arbitrary nonmonomial spaces W , the starting identity remains (10), with the true product tangent and W^m . What is special here is the explicit attainable tangent (8) and the

strictly positive mixed pairing. No dimension formula depending only on $\dim W$ is inferred outside this class.

4. INTERPOLATION AND THE ANALYTIC INPUTS

The interpolation identities in this section are classical. They fix the normalization used in the geometric argument; they do not assert statistical attainment. The independent repository derivation of the finite spectral comparison is acknowledged in [23].

Lemma 4.1 (Arbitrary-prefix Hermite data). *Let $y_1, \dots, y_p$ be an ordered multiset of real numbers, and let q_y be the multiplicity of y in the full multiset. The functionals*

$$f \mapsto [y_1, \dots, y_j]f, \quad 1 \leq j \leq p,$$

form a basis of the space generated by $f \mapsto f^{(k)}(y)$, $0 \leq k < q_y$. Repetitions need not be adjacent. For positive nodes, application to $f(z) = t^{Hz}$ gives exactly the complete blocks $t^{Hy}(\log t)^k$, $0 \leq k < q_y$.

Proof. A prefix divided difference depends only on the Hermite data at that prefix, so belongs to the stated p -dimensional space. On polynomials $1, z, \dots, z^{p-1}$ its values vanish in degrees below $j - 1$ and equal one in degree $j - 1$. The resulting matrix is triangular with unit diagonal, so these p functionals are independent. Substitution of t^{Hz} transforms differentiation at a node into multiplication by the nonzero factor H^k and the indicated logarithmic function. This is the Newton–Hermite prefix principle; see [12, Proposition 7]. It does not assert attainable posterior rank. $\square$

Proposition 4.2 (Exterior spectral form of determinant volumes). *Fix q and nodes $x_1, \dots, x_q \in [0, 1]$, retaining repetitions. Let $W = (x_i^{j-1})_{i,j=1}^q$ and let $\sigma_1 \geq \dots \geq \sigma_q \geq 0$ be its singular values. Define $\mathcal{V}_\ell$ as the maximal absolute Vandermonde product over ℓ labels, with $\mathcal{V}_1 = 1$. Then*

$$c_q \mathcal{V}_\ell \leq \prod_{j=1}^{\ell} \sigma_j(W) \leq C_q \mathcal{V}_\ell, \quad 1 \leq \ell \leq q.$$

The constants are independent of all gaps, and the assertion includes coincident nodes. Equivalently the middle expression is the operator norm of $\bigwedge^\ell W$.

Proof. Use the finite Leja ordering of Lemma 5.1, and let d_j be its nonincreasing pivots. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$, and let T have as its j th column the coefficients of P_{j-1} in the power basis. It is upper triangular with diagonal one. Elementary symmetric polynomial bounds on nodes in $[0, 1]$ bound T uniformly in q . Writing $T = I + U$ with $U^q = 0$ gives $T^{-1} = \sum_{k=0}^{q-1} (-U)^k$, hence a uniform bound on its inverse.

Evaluation gives $WT = LD$, $D = \text{diag}(d_j)$. If s is the number of distinct sites, the active leading s -block of L is lower triangular with diagonal ± 1 , bounded entries, and bounded inverse. Replace each inactive column by the corresponding coordinate vector. The resulting matrix $\tilde{L}$ and its inverse are uniformly bounded: it is block lower triangular, with the active block and an identity block on its diagonal. Because inactive entries of D are zero, $W = \tilde{L}DT^{-1}$ still holds. Multiplication by these uniformly bounded invertible matrices compares the ordered singular values of W with the ordered d_j , including zero singular values. Finally Lemma 5.1 gives

$$\prod_{j=1}^{\ell} d_j \leq \mathcal{V}_\ell \leq \ell! \prod_{j=1}^{\ell} d_j.$$

This proves the proposition without dividing by a vanishing pivot. The Leja lemma is a finite interpolation calculation proved below before its use in the statistical theorem, so this comparison introduces no circular dependence on attainment or memory. $\square$

4.1. Positivity and global covering. The next two lemmas are the analytic inputs to the intrinsic theorem. Their complete proofs are placed here, before the application, while the special-case arguments using them are collected in the appendices.

Lemma 4.3 (Strict confluent mixed pairing). *Let $\alpha_1 < \dots < \alpha_s$ be nonnegative, and let λ run through distinct nonnegative numbers with multiplicities q_λ summing to s . Include the columns $t^\lambda (\log t)^k / k!$, $0 \leq k < q_\lambda$, in this order within each increasing λ cluster. Assume $q_0 = 1$ if zero occurs. The matrix of their integrals against $t^{\alpha_i} \mu(dt)$ has strictly positive determinant. Consequently the corresponding rectangular pairing has full column rank whenever there are at least as many distinct row exponents as columns.*

Proof. Put $t = e^x$. The functions become $e^{\lambda x} x^k / k!$. A nonzero linear combination with total multiplicity s has at most $s - 1$ distinct zeros. Indeed divide by the first exponential and differentiate q_{λ_1} times. Rolle's theorem reduces a hypothetical s zeros to at least $s - q_{\lambda_1}$ zeros. The remaining terms are exponential polynomials with distinct nonzero exponents and the same remaining polynomial degrees: the operator $(D + \lambda - \lambda_1)^{q_{\lambda_1}}$ is invertible on polynomials of each fixed degree bound. Induction gives a contradiction. If no remaining term occurs, the original function is an exponential times a polynomial of degree less than s , and the assertion is immediate.

Thus the evaluation determinant at distinct ordered points never vanishes. Its sign is positive. One can determine the sign by letting ordered simple exponents approach the prescribed repeated exponents and dividing by their positive within-cluster Vandermonde factors. Equivalently, the Wronskian of the displayed exponential-polynomial basis has sign

$$\prod_{\lambda < \lambda'} (\lambda' - \lambda)^{q_\lambda q_{\lambda'}} > 0;$$

the exponential factor and the factorial normalizations are positive. Nonvanishing fixes this sign throughout the connected ordered-point domain.

Use the determinant integration identity from Lemma 3.1, with one determinant replaced by this confluent evaluation determinant. Both determinants have the same positive sign on ordered distinct positive points. The integrand is nonnegative elsewhere by symmetry and continuity. Separated ordered subintervals have positive prior mass, so the integral is strictly positive. The logarithmic functions are bounded at zero under the stated hypotheses, which also justifies Fubini. Select any s rows for the rectangular assertion. $\square$

A bounded semialgebraic format means a bound on the number of variables, polynomial relations and their degrees in a defining formula; arbitrary real coefficients are allowed. Quantified formulas of fixed format can also be used, since real quantifier elimination gives a bound depending only on this format. Neither a coefficient norm nor a lower bound on a nonzero coefficient is part of the definition.

Lemma 4.4 (Bounded-format sets in a thin rectangle). *Let S be a nonempty compact semialgebraic subset of $\prod_{i=1}^q [-a_i, a_i]$, where $a_1 \geq \dots \geq a_q \geq 0$. Suppose its format is bounded and its dimension is at most p , with $1 \leq p \leq q$. There is a constant C depending only on the format and q such that, for every $\varepsilon > 0$,*

$$(12) \quad \mathcal{N}(S, \varepsilon) \leq C \sum_{j=0}^p \varepsilon^{-j} \prod_{i=1}^j a_i.$$

The empty product is one. Consequently S has a covering by at most M representatives belonging to S , of radius at most

$$(13) \quad C' \max_{1 \leq j \leq p} \left(M^{-1} \prod_{i=1}^j a_i \right)^{1/j}.$$

If all a_i vanish, one representative suffices with radius zero.

Proof. We use two classical real-geometric facts. First, intersections of a bounded-format semialgebraic set with arbitrary affine subspaces have a uniform bound on their number of connected components. The bound is independent of the coefficients; one formulation is the regularity statement in [18, Lemma 2.18], together with real quantifier elimination. Second, the real Vitushkin–Ivanov entropy inequality is

$$(14) \quad \mathcal{N}(S, \varepsilon) \leq C_q \sum_{j=0}^q \varepsilon^{-j} V_j(S), \quad V_j(S) = c_{q,j} \int b_0(S \cap L) \, dL.$$

For $j > 0$ the integral is over affine subspaces of codimension j , with normalized rotational measure and Lebesgue translation measure; $V_0(S) = b_0(S)$. This is the classical real inequality in [16, Theorem 3.5], also recalled as equations (4)–(5) in [17]. No nonarchimedean result is used here.

For $j > p$, almost every such affine section is empty: its translation parameter lies outside a projection of dimension at most $p < j$. Thus $V_j(S) = 0$. For $1 \leq j \leq p$, write an affine section as $L = u + E^\perp$, $u \in E$, for a j -plane E . If it meets S , then u belongs to the projection onto E of the containing rectangle. The component bound, denoted B , therefore gives

$$V_j(S) \leq c_{q,j} B \int_{G(q,j)} \text{vol}_j \left(\text{proj}_E \prod_i [-a_i, a_i] \right) \, dE.$$

This projected rectangle is a zonotope. Its volume is the sum, over j -element subsets of the coordinate directions, of their parallelepiped volumes. Each corresponding orthogonal-projection minor has absolute value at most one. Hence its volume is at most $2^j \binom{q}{j} a_1 \cdots a_j$. Also $V_0(S) \leq B$. Substitution into (14) proves (12), including zero-width axes. This argument explains why the products stop at the dimension of S , not at the number of sides of its ambient rectangle.

To respect the integer budget, let e be the maximum in (13) without its constant. If $a_1 > 0$, then $e > 0$ and $a_1 \cdots a_j \leq M e^j$ for every $j \leq p$. Choose $L \geq 2$ with $C \sum_{j=1}^p L^{-j} \leq 1/2$. For $M \geq 2C$, (12) at radius Le is at most $C + M/2 \leq M$. Discard empty balls and replace each center by a point of S in its ball; the radius at most doubles. For $M < 2C$, a single point of S covers it at radius $2\sqrt{q} a_1$, while $e \geq a_1/M \geq a_1/(2C)$. Increasing the constant proves the assertion for these budgets as well. A nearest-representative map, with a fixed tie convention, is Borel. If $a_1 = 0$, the set is a singleton. $\square$

5. INTRINSIC RESOLUTION AT ARBITRARY EXPONENT COLLISIONS

The affine hypothesis can be removed altogether. The appropriate invariant is a family of maximal Vandermonde products on the future exponent set. It is defined at a collision itself, does not require a chosen path into that collision, and determines both attainable checkpoint resolution and causal memory. We first establish a uniformly conditioned Newton flag, then prove its attainable transversality. The covering and causal arguments are given separately, so that a local flag is never substituted for a global description of the reachable set.

5.1. Calibration chamber and determinant volumes. Fix $r \geq 2$ and a compact subset K of the open chamber

$$\{(a_1, \dots, a_{r-1}) : 0 < a_1 < \dots < a_{r-1}\}, \quad a_0 = 0.$$

The parameter of the experiment is $t \in [0, 1]$, with one fixed full-support prior μ . The known calibration is $a \in K$. One fixed matrix $C = (c_{ji})$ of column rank r defines the cells

$$k_j^a(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i}, \quad \sum_j k_j^a(t) = 1, \quad k_j^a(t) \geq k_* > 0.$$

The command cube is $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$, as before. These assumptions are imposed on all of K ; no regularity or semialgebraicity of K is required. Compactness inside the chamber separates the one-step exponents, but *does not* separate their additive sums. The finitely many report likelihoods have a common positive lower bound κ .

Fix $N \geq 2$, and choose once and for all $H \geq \max\{1, N \max_{a \in K} a_{r-1}\}$. For $m \geq 1$ write

$$\Lambda_m = \{\alpha \in \mathbb{N}_0^r : |\alpha| = m\}, \quad \Lambda_m^+ = \Lambda_m \setminus \{me_0\}, \quad q_m = |\Lambda_m^+|, \quad x_\alpha(a) = H^{-1} \sum_i \alpha_i a_i.$$

Here $\mathbb{N}_0$ includes zero. Formal labels are retained even when their values coincide. All x_α , $\alpha \in \Lambda_m^+$, lie in $[a_*/H, 1]$, where $a_* = \min_{a \in K} a_1 > 0$. For $1 \leq \ell \leq q_m$ define

$$(15) \quad \mathcal{V}_{m,\ell}(a) = \max_{\substack{J \subset \Lambda_m^+ \\ |J|=\ell}} \prod_{\substack{\alpha, \beta \in J \\ \alpha < \beta}} |x_\alpha(a) - x_\beta(a)|, \quad \mathcal{V}_{m,1} = 1.$$

The ordering inside the product is immaterial. The maximum is zero if fewer than ℓ distinct positive future exponents remain. Put

$$(16) \quad \begin{aligned} p_{n,m} &= \min\{n(r-1), q_m\}, \\ \Psi_{n,m}(M, a) &= \max_{1 \leq \ell \leq p_{n,m}} \left(\frac{\mathcal{V}_{m,\ell}(a)}{M} \right)^{2/\ell}, \\ \Xi_N(M, a) &= \max_{1 \leq n < N} \Psi_{n,N-n}(M, a). \end{aligned}$$

Set $\Psi_{0,m} = \Psi_{n,0} = 0$. Changing H to any other fixed admissible value changes only comparison constants. In particular the invariant does not depend on a convention for measuring exponents.

Use the failure-factor basis $1/2$ and $1/2 + \delta t^{a_i}$, $1 \leq i < r$, realized by the fixed right inverse of C for one sufficiently small $\delta > 0$. Its ordered m -fold products are the query menu, sampled uniformly after the memory index is formed. Expansion in the symbolic variables $X_0, \dots, X_{r-1}$ gives a fixed matrix G_m of column rank $|\Lambda_m|$: an invertible linear change of variables on homogeneous polynomials of degree m is invertible. Consequently the physical query norm and the norm of the formal raw moment vector are uniformly equivalent. Coincident exponent values impose relations on attainable vectors, not a loss of column rank of this *formal* coefficient matrix.

5.2. A uniformly conditioned flag. Order the positive nodes greedily. Choose the first label by a fixed rule; for example take its least node, breaking ties by a fixed label order. Having selected $x_1, \dots, x_{j-1}$, select a remaining label maximizing

$$\prod_{i < j} |x - x_i|, \quad d_1 = 1, \quad d_j = \prod_{i < j} |x_j - x_i|.$$

Break every tie by the same label rule. After all distinct values have been selected, continue through the remaining labels; their d_j are zero. This is finite Leja ordering. The ordering and

its use in Newton interpolation are classical [20, 21]; the estimates needed here are proved below, including their zero-pivot form.

Lemma 5.1 (Newton scales and maximal determinant volumes). *For every calibration, $1 = d_1 \geq d_2 \geq \dots \geq d_{q_m} \geq 0$. There is a matrix L_a , whose entries have absolute value at most one, such that*

$$(17) \quad (t^{Hx_i})_{i=1}^{q_m} = L_a \operatorname{diag}(d_j)(g_j^a(t))_{j=1}^{q_m}, \quad g_j^a(t) = [x_1, \dots, x_j](z \mapsto t^{Hz}).$$

Divided differences at repeated nodes have their Hermite values. If s is the number of distinct positive nodes, the first s rows and columns of L_a form a lower triangular matrix with diagonal entries ± 1 ; its inverse has a bound depending only on q_m . Thus, for any vector v ,

$$(18) \quad c_m \|\operatorname{diag}(d_j)v\| \leq \|L_a \operatorname{diag}(d_j)v\| \leq C_m \|\operatorname{diag}(d_j)v\|.$$

Moreover, for every $1 \leq \ell \leq q_m$,

$$(19) \quad \prod_{j=1}^{\ell} d_j \leq \mathcal{V}_{m,\ell}(a) \leq \ell! \prod_{j=1}^{\ell} d_j.$$

All these assertions hold at exact coincidences.

Proof. Every distance is at most one. The maximization defining d_j and the removal of a candidate imply $d_{j+1} \leq d_j$. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$. For $d_j > 0$ set $L_{ij} = P_{j-1}(x_i)/d_j$; for $d_j = 0$ set that column to zero. A previously selected node gives zero when $i < j$, and the greedy choice gives $|L_{ij}| \leq 1$ on all remaining nodes. The leading nonzero block is lower triangular with diagonal ± 1 . Back substitution bounds its inverse in a fixed dimension. If $d_j = 0$, all remaining node values already occur among the earlier ones, so P_{j-1} vanishes on every node. Newton's interpolation identity, evaluated at the nodes, is therefore exactly (17), also when there are repeats. Only the first s scaled coordinates can be nonzero, proving (18).

For $\ell \leq s$, evaluation of the monic polynomials $P_0, \dots, P_{\ell-1}$ on any ℓ chosen nodes has determinant equal to their Vandermonde determinant. Its matrix is the selected rows of $L_{:,1:\ell}$ times $\operatorname{diag}(d_1, \dots, d_\ell)$. The Leibniz formula bounds the first determinant by $\ell!$. For the first ℓ selected nodes its absolute value is one, giving both bounds in (19). For $\ell > s$, both sides vanish. No division by a zero pivot has been made. $\square$

The next assertion is the attainable part of the argument. Bounds for a Vandermonde matrix alone would not prove it.

Lemma 5.2 (Uniform attainable Newton flags). *For every $n, m \geq 1$ with $n + m \leq N$, let $z_a(h) = (\mathbb{E}[g_j^a(t) \mid h])_{j=1}^{q_m}$ and $p = p_{n,m}$. At one common interior all-failure command tuple, the first p coordinates of z_a have a surjective differential. Its least row singular value is bounded below uniformly over $a \in K$ and every ordering of the formal nodes, including limiting orderings at collisions. Under independent uniform acquisition commands, their all-failure subprobability law dominates β times uniform probability on a translated p -cube of side 2ρ , where $\beta, \rho > 0$ are uniform.*

Proof. We first justify continuity at all repeated-node configurations, without an assumption on how a collision is approached. The Hermite–Genocchi formula [12] expresses $g_j^a(t)$ as the integral of $(H \log t)^{j-1} t^{Hz}$ over the simplex of convex combinations of $x_1, \dots, x_j$; the simplex has volume $1/(j-1)!$. Since $Hz \geq a_* > 0$,

$$|g_j^a(t)| \leq \frac{H^{j-1}}{(j-1)!} t^{a_*} |\log t|^{j-1}, \quad 0 < t \leq 1.$$

The right side is bounded and tends to zero at $t = 0$. The same argument with higher logarithmic powers gives continuity, uniform in t , as the nodes vary in their compact range. Define $g_j^a(0) = 0$.

For any initial multiset of p nodes, the p divided-difference functionals are independent on polynomials of degree at most $p - 1$: their values on $1, z, \dots, z^{p-1}$ form an upper triangular matrix with diagonal one. They span the Hermite evaluation functionals at the distinct nodes, with all derivative orders below the corresponding multiplicities. Applied to t^{Hz} , their span is therefore the complete exponential-polynomial space

$$\text{span}\{t^{H\lambda}(\log t)^k : \lambda \text{ in the multiset, } 0 \leq k < m_\lambda\}.$$

This argument permits repeated nodes at nonadjacent positions. Adjoin the constant function. The resulting $p + 1$ tests form a complete confluent Chebyshev space, not a selection of isolated high derivatives.

Choose the common binomial factors $F_i = (1 + c_i t^{a_{r-1}})/2$, with distinct sufficiently small positive c_i . Their command coefficients are independent of a . By Lemma 3.2 their product tangent has $n(r - 1) + 1$ distinct monomials. The mixed and confluent pairing of Lemmas 3.1 and 4.3 has rank $p + 1$ against the selected tests, at every node configuration. In the normalized differential

$$Q \mapsto (\mu F)^{-1}(LQ - v e_0 LQ), \quad v = LF/\mu F,$$

the product belongs to the tangent and $e_0 v = 1$. Thus exactly the constant direction is lost and the differential onto the other p coordinates is surjective.

For each fixed permutation of formal labels this derivative is continuous on K , including coincidences, by the uniform bounds just proved and positivity of evidence. There are finitely many permutations. Compactness therefore supplies a common positive least row singular value. First and second derivatives in command coordinates are uniformly bounded because the numerator is a polynomial and the evidence is at least κ^n .

To record the probability step explicitly, complete the derivative's row space by orthonormal kernel coordinates. A ball of uniform radius in command space gives a square map whose derivative differs from its invertible central derivative by at most half its least singular value. The contraction proof of the inverse theorem gives a fixed-size product box in output and kernel coordinates, with its inverse inside the command cube. The inverse Jacobian determinant is bounded below by the reciprocal of a uniform power of the forward derivative bound. The command density is constant, and the joint density of those commands and the all-failure word is at least that density times η^n . Integrating over the kernel box proves the stated subprobability minorization. No prior density is involved and no failure evidence is discarded. $\square$

5.3. The global attainable law.

Theorem 5.3 (Collision-uniform checkpoint classification). *Under the assumptions of this section there are $0 < c \leq C < \infty$ such that, for every $a \in K$ and integer $M \geq 1$,*

$$(20) \quad c\Psi_{n,m}(M, a) \leq \bar{R}_{M;n,m}^a \leq R_{M;n,m}^{a,\max} \leq C\Psi_{n,m}(M, a).$$

The unconditional lower law uses independent uniform acquisition commands. The upper code is deterministic, has at most M reachable representatives, and works on every admitted history. The lower law permits the independent coding randomization of Definition 2.4. Constants may depend on K, C, η, μ, n, m , but not on a or M .

Proof. Let $D_a = \text{diag}(d_j)$. For one report word each coordinate of $z_a(h)$ has the form $N_j(u)/Z(u)$, with numerator and evidence polynomials of degree at most n in the command variables and $Z(u) \geq \kappa^n$. Thus

$$S_a = \{D_a z_a(h) : |h| = n\}$$

is compact and has coefficient-independent bounded semialgebraic format: it is a finite union of projections of the formulas $Z(u)y_j = d_j N_j(u)$ on the command cube. Prior integrals and node values are real coefficients in these formulas. They need not depend semialgebraically on a .

Individually normalizing the n report factors puts them in affine hyperplanes of dimension $r - 1$. The moment map is rational in those coordinates. Hence $\dim S_a \leq p$. Lemma 5.2 also gives $|z_{a,j}| \leq B$, so S_a lies in the rectangle with half-widths $Bd_1 \geq \dots \geq Bd_{q_m}$. By Lemma 4.4, it has at most M reachable representatives with covering radius bounded by

$$C \max_{\ell \leq p} \left(M^{-1} \prod_{j \leq \ell} d_j \right)^{1/\ell}.$$

The physical metric equivalence follows from (18) and the full-column-rank product-probe matrix G_m . A fixed tie rule makes nearest-representative encoding Borel. Squaring and applying (19) gives the upper bound in (20), with the integer budget already enforced by Lemma 4.4.

For the lower bound, first average any randomized decoder conditional on its label; squared loss cannot increase. Orthogonally project its query centers onto the affine physical query span, apply a fixed left inverse of G_m , and then the inverse of the leading nonzero block of L_a . These maps have uniformly bounded norms. Projection onto any first $\ell \leq \min(p, s)$ coordinates reduces the squared-error problem to the marginal of the cube in Lemma 5.2, scaled by $d_1, \dots, d_\ell$. The probability that M radius- b balls cover this uniform rectangle is at most

$$C_\ell M b^\ell / (d_1 \cdots d_\ell).$$

Choose $b = c_\ell (d_1 \cdots d_\ell / M)^{1/\ell}$ with the constant small enough that at least half the minorized mass remains. Its mean squared error is at least a fixed multiple of b^2 . An encoder's randomized assignment cannot improve upon the distance to the nearest of its centers. Independent public coding randomness can be fixed and then averaged, since it does not generate the acquisition law. Maximize over ℓ . For $\ell > s$ the determinant volume is zero, so no additional term is needed. Equation (19) now gives the lower bound. The unconditional optimum is bounded above by the minimax optimum, completing the proof. $\square$

Theorem 5.4 (Causal classification across all collision strata). *For the same fixed experiment and horizon, there exist $c, C > 0$ such that for every known $a \in K$ and every integer $M \geq 1$ there exists a single, stage-compatible deterministic M -state transducer with maximum-history, maximum-checkpoint regret at most $C\Xi_N(M, a)$. Every M -state transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under one common independent uniform exploration law. Equivalently,*

$$(21) \quad c\Xi_N(M, a) \leq \mathfrak{R}_{M,N}^{a,\text{av}} \leq \mathfrak{R}_{M,N}^{a,\text{max}} \leq C\Xi_N(M, a).$$

The transducer may depend on a and M . Constants are uniform in both; codebooks are not asserted to be calibration-blind. It stores only its representative index. The clock, known calibration and fixed real-valued program are read-only, as in Definition 2.4.

Proof. At stage n let $m = N - n$ and store a representative in the reachable raw moment image indexed by Λ_m , including the constant coordinate $v_{me_0} = 1$. For a new factor $f(t) = \sum_i f_i(g, x)t^{a_i}$, the exact update is

$$(22) \quad v'_\alpha = \frac{\sum_i f_i(g, x)v_{\alpha+e_i}}{\sum_i f_i(g, x)v_{e_i+(m-1)e_0}}, \quad \alpha \in \Lambda_{m-1}.$$

Every index on the right belongs to Λ_m . Coincident labels have equal moment values and cause no ambiguity. On the segment joining any two reachable moment vectors, the denominator

is a report probability under a posterior mixture and is at least κ . Coefficients are uniformly bounded and raw moments belong to $[0, 1]$. Differentiating the quotient gives a Lipschitz constant independent of all exponent gaps.

At every stage Theorem 5.3 provides a raw reachable codebook of cardinality at most M and covering radius at most $C_n \Psi_{n,N-n}(M, a)^{1/2}$. Update a representative by (22), then quantize in the next reachable codebook. Updating a reachable representative produces a reachable next state, so this quantizer is always evaluated on its domain. If e_n is the raw error, then

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_{n+1} \Psi_{n+1,N-n-1}(M, a)^{1/2}.$$

The finite recurrence bounds every e_n by $C_N \Xi_N(M, a)^{1/2}$. The query matrix converts this to the asserted squared-regret bound. The exact prefix appears only in this error analysis, not in the transducer. No divided-difference coordinate, reciprocal pivot or stored command tape is used in a transition.

Conversely, at each checkpoint the streaming index is an M -message encoder of the prefix. All command laws used in Theorem 5.3 are prefixes of one common exploration law. Each transducer therefore satisfies every checkpoint lower bound; taking their maximum and then the infimum proves the lower bound in (21). Finitely many checkpoint constants can be replaced by their positive minimum. One checkpoint and its remaining query trials are executed in a run, as before. $\square$

Corollary 5.5 (Intrinsic bit law). *Let $B_*(\varepsilon, a)$ be the least integer persistent bit budget for maximum-history streaming regret at most ε in this experiment. Uniformly for $0 < \varepsilon \leq 1$ and $a \in K$,*

$$(23) \quad B_*(\varepsilon, a) = \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n,N-n}}} \left[\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} + \log_2 \mathcal{V}_{N-n,\ell}(a) \right]_+ + O(1).$$

A zero determinant volume contributes zero to this maximum, by interpreting its logarithm as $-\infty$ before taking the positive part. The same law holds for the unconditional maximum-checkpoint criterion. At each fixed calibration the small-error slope is $D_A(N)/2$.

Proof. With $M = 2^B$, solving each inequality $(\mathcal{V}_{m,\ell}/M)^{2/\ell} \leq \varepsilon$ gives the displayed lower bound on B . Constants in (21) change it by at most a bounded amount, since ℓ is bounded. Integer rounding costs at most one further bit. For fixed a , the largest nonzero index at checkpoint (n, m) is $\min\{n(r-1), |mA| - 1\} = d_A(n, m)$, proving the slope assertion. $\square$

5.4. Collision orders and a finite tree formula. The determinant form has a consequence not requiring interpolation coordinates: it classifies arbitrary orders of tangency to a collision. It also gives a finite optimization rule for all resolution phases.

Theorem 5.6 (Collision-tree determination of memory phases). *Let $a(\theta) \in K$, $0 \leq \theta \leq \theta_0$, be a known path. For a fixed m , identify formal sums that are identical along the path. Suppose that for every remaining pair of labels*

$$|x_\alpha(\theta) - x_\beta(\theta)| \asymp \theta^{w_{\alpha\beta}}, \quad w_{\alpha\beta} \in [0, \infty), \quad 0 < \theta \leq \theta_0.$$

After decreasing θ_0 this hypothesis holds for every real analytic path whose nonidentical sums are analytic at zero. Define

$$(24) \quad \beta_{m,\ell} = \min_{\substack{J \text{ a set of} \\ |J|=\ell}} \min_{\text{remaining labels}} \sum_{\{\alpha,\beta\} \subset J} w_{\alpha\beta}, \quad \beta_{m,1} = 0.$$

Then $\mathcal{V}_{m,\ell}(a(\theta)) \asymp \theta^{\beta_{m,\ell}}$. For unavailable cardinalities the volume is zero. Consequently the checkpoint law is

$$(25) \quad \overline{R}_{M;n,m}^{a(\theta)} \asymp \max_{\ell \leq p_{n,m}} \theta^{2\beta_{m,\ell}/\ell} M^{-2/\ell},$$

with zero-volume terms omitted; the streaming law is its maximum over checkpoints. The exponents $\beta_{m,\ell}$ depend only on the pairwise collision orders, and can be computed by a finite tree allocation rule.

Proof. Each fixed subset product has order equal to the sum of its pair orders. There are finitely many subsets, so their maximum has the smallest such order. This proves the determinant assertion and then (25) by Theorems 5.3 and 5.4. Constants may depend on the path's nonzero leading coefficients; the intrinsic determinant theorem, in contrast, is already uniform on K .

For completeness we give the allocation rule and its justification. The triangle inequality for node differences implies

$$w_{\alpha\gamma} \geq \min\{w_{\alpha\beta}, w_{\beta\gamma}\}.$$

Indeed a smaller order on the left would contradict the upper bound by the sum of the two differences on the right as $\theta \downarrow 0$. Thus thresholds on w define nested equivalence classes. Make a rooted tree from these classes, with root height zero; an internal cluster C has height $h_C = \min_{\alpha \neq \beta \in C} w_{\alpha\beta}$, and its children are the equivalence classes for orders strictly greater than h_C . Split the root at orders greater than zero. Singleton children are leaves. Redundant equal-height vertices can be omitted. For an internal nonroot vertex put $\Delta_C = h_C - h_{\text{parent}(C)}$. The sum of these increments on the common ancestor chain of two leaves is exactly their pair order. Hence, for a selected leaf set J ,

$$(26) \quad \sum_{\{\alpha,\beta\} \subset J} w_{\alpha\beta} = \sum_{C \text{ internal, } C \neq \text{root}} \Delta_C \binom{|J \cap C|}{2}.$$

Set $F_C(0) = F_C(1) = 0$ for a leaf and $F_C(k) = +\infty$ for $k > 1$. For an internal vertex, including the root with $\Delta_{\text{root}} = 0$, compute

$$(27) \quad F_C(k) = \Delta_C \binom{k}{2} + \min_{\sum_D \text{ child of } C} \sum_{k_D=k} F_D(k_D).$$

Equation (26) proves this recurrence by induction. It gives $F_{\text{root}}(\ell) = \beta_{m,\ell}$. Successive finite min-convolutions compute the table; no optimization over measures, posterior encoders or codebooks is involved. For analytic paths the pair orders are the finite orders of their first nonzero Taylor coefficients, after identical sums have been identified. This proves the assertion. $\square$

For the affine family of Section B, different limiting clusters have pair order zero, and distinct members of a cluster have pair order one. Thus a choice of k nodes from that cluster costs $\binom{k}{2}$; its successive costs are $0, 1, \dots, s-1$. Minimizing their sum picks the smallest costs from all clusters. Therefore $\beta_{m,\ell} = \nu_{m,1} + \dots + \nu_{m,\ell}$. This recovers Theorem D.1, while its separate proof remains available. Higher orders of contact between distinct branches require the pair orders above and are not encoded by the affine multiplicity list.

5.5. Intersecting collision loci and arbitrary tangency order. Here is a two-parameter consequence with different ranks on intersecting loci and arbitrarily deep resolution phases along tangent approaches. It illustrates the geometry of the determinant theorem rather than requiring a separate asymptotic analysis for each path.

Corollary 5.7 (A two-parameter collision arrangement). *Let $A_{u,v} = \{0, 1, 2 + u, 3 + v\}$, with $|u|, |v| \leq 1/16$, and take the four cells*

$$k_0(t) = \frac{1}{4} + \frac{1}{16}(t + t^{2+u} + t^{3+v}), \quad k_i(t) = \frac{1}{4} - \frac{1}{16}t^{a_i}, \quad 1 \leq i \leq 3.$$

Use any fixed admitted command cube and a fixed full-support prior. For total horizon five set $\rho = \max\{|u|, |v|\}$ and $\tau = \min\{|u|, |v - u|, |v - 2u|\}$. Then, uniformly on the square,

$$(28) \quad \mathfrak{R}_{M,5}^{u,v} \asymp \max \left\{ M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau)^{2/9} M^{-2/9} \right\}$$

for either streaming criterion. The exact peak dimension is nine off the three collision lines, eight on any such line away from their intersection, and six at $(0, 0)$.

Along $u = \theta$, $v = \theta + \theta^k$ with an integer $k \geq 2$, the three regimes have respective orders

$$M^{-1/3}, \quad \theta^{1/2} M^{-1/4}, \quad \theta^{(4+2k)/9} M^{-2/9}.$$

Their crossover budgets have orders θ^{-6} and $\theta^{-(8k-2)}$, with regrets of orders θ^2 and θ^{2k} , respectively. These are comparison orders, not exact integer thresholds.

Proof. The cells sum to one, lie between $3/16$ and $7/16$, and their coefficient matrix has rank four: its last three rows separate the three nonconstant coefficients and the sum of its rows gives the constant. They therefore satisfy the compact-chamber hypotheses.

At checkpoint $n = 3, m = 2$ the nine positive formal exponents are

$$1, \quad 2, \quad 2 + u, \quad 3 + u, \quad 3 + v, \quad 4 + 2u, \quad 4 + v, \quad 5 + u + v, \quad 6 + 2v.$$

They form six uniformly separated groups, with three within-group gaps $|u|, |v - u|, |v - 2u|$. Denote these gaps in decreasing order by $g_1 \geq g_2 \geq g_3$. For $\ell \leq 6$ the determinant volumes are comparable to one. For $\ell = 6 + j$, $1 \leq j \leq 3$, they are comparable to $g_1 \cdots g_j$: choose one exponent from every group and a second from the j groups with largest gaps for the lower bound. Any selection of $6 + j$ exponents must double at least j groups; all other factors are bounded, proving the upper bound. Fixed powers of H only change constants. Since $p_{3,2} = 9$, its profile is

$$\max \{ M^{-1/3}, g_1^{2/7} M^{-2/7}, (g_1 g_2)^{1/4} M^{-1/4}, (g_1 g_2 g_3)^{2/9} M^{-2/9} \}.$$

The signed differences satisfy $(v - u) - (v - 2u) = u$, so the largest absolute gap is at most twice the second largest. Moreover $g_1 \asymp \rho$, hence $g_2 \asymp \rho$ and $g_3 = \tau$. The seven-dimensional term is the geometric interpolation, with weights $3/7$ and $4/7$, of the six- and eight-dimensional terms, up to uniform constants. This proves the expression in (28) for the checkpoint, also on the collision lines.

At the other checkpoints, the attainable dimensions are bounded by three, six and three, respectively. Since every normalized gap is at most one, their squared covering errors are at most $CM^{-1/3}$. The causal classification therefore gives the same entire-horizon profile and the checkpoint-three lower bound supplies its converse. Counting distinct two-fold exponents gives the three exact peak dimensions asserted.

Finally, on the stated path $\rho \asymp \theta$ and $\tau \asymp \theta^k$. Substitution gives the three displayed terms. Equality in order of the first two gives $M = \theta^{-6}$, at regret θ^2 ; equality in order of the last two gives $M = \theta^{-(8k-2)}$, at regret θ^{2k} . As $k > 1$, these budgets are strictly separated as $\theta \downarrow 0$ and all three regimes occur. The exponent k records contact with a collision locus, not an additional assumed observable jet. $\square$

6. A STRUCTURAL CLASSIFICATION OF POSITIVE POLYNOMIAL EXPERIMENTS

We first treat finite detectors without an ordered latent space or a prescribed resolution flag. Polynomial dependence below is on the commands, not on the latent variable. In particular, the class includes every finite detector with affine retention commands and arbitrary continuous cells.

6.1. Experiments, queries, and the augmented moment matrix. Let X be a compact metric space and let μ be a probability on X . Fix $2 \leq N < \infty$. At time i , a command belongs to $U_i = [-1, 1]^{b_i}$, where $b_i \geq 1$, and the report belongs to a finite set $\mathcal{Y}_i$. The likelihoods have the form

$$(29) \quad L_{i,y}(u, x) = \sum_{\alpha \in F_{i,y}} u^\alpha c_{i,y,\alpha}(x), \quad c_{i,y,\alpha} \in C(X), \quad L_{i,y} \geq \kappa > 0, \quad \sum_y L_{i,y} = 1.$$

Each $F_{i,y}$ is finite. The commands in the exploration experiment are independent uniform random variables, independent of the latent variable and of coding randomness. A command is read once. The persistent resource is one of M labels; the clock and the known experiment are read-only. The meaning of this resource is exactly that of Definition 2.4. In particular, a command-generation seed that remembers the prefix is not free side information.

For $0 \leq n < N$, let $\mathcal{V}_n$ be the span in $C(X)$ of the likelihoods of all specified report words in the remaining times $n+1, \dots, N$, with arbitrary fixed future commands. It is finite dimensional, by (29), and contains 1: summing over all future report words at any fixed future commands gives 1. Choose a finite list of such physical queries $H_{n,1}, \dots, H_{n,q_n}$ spanning $\mathcal{V}_n$, and fixed weights $w_{n,j} > 0$ with sum one. A query is chosen independently after the checkpoint label is formed. Only that query and its remaining trials are executed. Its binary outcome is the indicator of its specified report word. Thus squared-loss excess is the squared Euclidean error in

$$p_n(h) = \left(\sqrt{w_{n,j}} \frac{\mu(P_h H_{n,j})}{\mu P_h} \right)_{j=1}^{q_n}.$$

Open-loop word queries suffice for this span; an adaptive future tree only forms finite sums of the same products.

For a fixed past report word $a = (a_1, \dots, a_n)$ set

$$P_a(\mathbf{u}, x) = \prod_{i=1}^n L_{i,a_i}(u_i, x), \quad Z_a = \mu P_a, \quad D_n = \sum_{i=1}^n b_i,$$

and form the polynomial vector and its augmented matrix

$$(30) \quad \begin{aligned} Y_a(\mathbf{u}) &= \left(Z_a, (\sqrt{w_{n,j}} \mu(P_a H_{n,j}))_{j=1}^{q_n} \right)^t, \\ \mathcal{A}_a(\mathbf{u}) &= [Y_a \mid \partial_1 Y_a \mid \dots \mid \partial_{D_n} Y_a], \\ d_n(\mu) &= \max_{a \in \prod_{i \leq n} \mathcal{Y}_i} \text{rank}_{\mathbb{R}(\mathbf{u})} \mathcal{A}_a - 1. \end{aligned}$$

The rank is over the rational-function field with the actual prior moments as real coefficients. No genericity of the prior is implicit. The initial state is a singleton; set $d_0 = 0$.

Lemma 6.1 (Augmentation and normalization). *Let $Y = (Z, Zp)^t$ be a differentiable vector with $Z > 0$. Then*

$$(31) \quad \text{rank}[Y \mid DY] = 1 + \text{rank } Dp.$$

Consequently $d_n(\mu)$ is the maximum rank of the actual checkpoint prediction map over interior command tuples and report words. It is independent of the chosen spanning query coordinates, provided their weights are positive.

Proof. Subtract p_j times the first row from row $j+1$. The first column becomes $(Z, 0)^t$, and each derivative column becomes $(\partial_i Z, Z \partial_i p)^t$. Subtracting $(\partial_i Z/Z)$ times the first column removes its top entry. These invertible operations prove (31). A polynomial minor is either identically zero or nonzero on a dense open subset of the interior command cube. Thus rational-function rank is the maximum attained rank there. A different spanning query list gives linear maps in both directions on the vector augmented by its constant coordinate. After differentiation these maps identify the ranks. Notice that no assertion $P \in \text{im } DP$ was needed: the added evidence column is removed by normalization, not treated as an additional acquired direction. $\square$

Write $R_{M,n}^{\text{av}}$ for the optimal unconditional checkpoint excess when an arbitrary encoder of the complete prefix may retain M labels, and $R_{M,n}^{\text{max}}$ for its worst-history analogue. Write $\mathfrak{R}_{M,N}^{\text{av}}$ and $\mathfrak{R}_{M,N}^{\text{max}}$ for the infima of the corresponding maximum over checkpoints when *one* causal filter is used. Independent encoder and decoder randomness is allowed in all infima.

Theorem 6.2 (Product-rank classification). *In every experiment (29), for every fixed prior μ , the whole reachable prediction set at checkpoint n is compact semialgebraic of dimension $d_n(\mu)$. If $d_n(\mu) > 0$, then for all integers $M \geq 1$*

$$(32) \quad c_n M^{-2/d_n} \leq R_{M,n}^{\text{av}} \leq R_{M,n}^{\text{max}} \leq C_n M^{-2/d_n}.$$

If $d_n = 0$, the reachable set is finite, with at most $\prod_{i \leq n} |\mathcal{Y}_i|$ points; its checkpoint risk is zero once that many labels are available.

Set $d_ = \max_{1 \leq n < N} d_n$. If $d_* > 0$, then*

$$(33) \quad c M^{-2/d_*} \leq \mathfrak{R}_{M,N}^{\text{av}} \leq \mathfrak{R}_{M,N}^{\text{max}} \leq C M^{-2/d_*}.$$

The upper bound is attained by one deterministic stage-compatible filter with reachable representatives. If $d_ = 0$, one finite filter is exact at every checkpoint. These conclusions require no full-support assumption on μ , no Chebyshev system, and no nondegeneracy hypothesis besides the rank computed in (30). For fixed detector and query lists, all positive-rank comparison constants can be chosen uniformly on a compact set of prior moment vectors on which the relevant ranks are constant.*

6.2. Whole images and actual history mass.

Lemma 6.3 (Image dimension and a minorized chart). *For a fixed checkpoint, let S_n be the union of all reachable vectors $p_n(h)$. Its semialgebraic format is bounded in terms of the finite polynomial data and horizon. Its dimension is d_n . If $d_n > 0$, some coordinate projection $\pi : \mathbb{R}^{q_n} \rightarrow \mathbb{R}^{d_n}$ and some cube Q of positive side length satisfy*

$$(34) \quad \pi_* \text{Law}(p_n(h)) \geq \beta \text{Unif}(Q), \quad \beta > 0,$$

under the stated exploration law, with reports included rather than conditioned away.

Proof. For each word, the graph is specified by its command cube and the polynomial equations $Z_a p_j = Y_{a,j+1}$. Here $Z_a \geq \kappa^n$ on that cube. Projection gives a semialgebraic set of bounded format. Its compactness follows from the continuous rational map on a compact cube. The finite union over words has the same properties. Prior integrals are simply real coefficients in these equations; neither X nor the prior has been assumed semialgebraic.

For clarity, the real-geometric dimension fact used here is the following standard consequence of semialgebraic constant-rank stratification [31]: a rational map with nonvanishing denominator

on a cube has image dimension equal to its largest interior differential rank. Indeed, stratify its domain, including the boundary faces, into finitely many smooth semialgebraic pieces on which the restricted map has constant rank. The constant-rank theorem bounds each image dimension by that rank. Every restricted rank is at most the maximum ambient rank, which is also the maximum interior rank: a nonzero minor at a boundary point stays nonzero at nearby interior points. Conversely, a point of maximum interior rank has a neighborhood whose image contains a submanifold of that dimension. Applying this fact and Lemma 6.1 proves $\dim S_n = d_n$.

Choose a word and an interior tuple u_* where the rank is d_n , and choose d_n output rows with independent differentials. Put $z = \pi p_n$ and $J = Dz(u_*)$. Let the rows of K be an orthonormal basis of $\ker J$. The square map

$$F(u) = (z(u), K(u - u_*))$$

has an invertible derivative at u_* . The inverse-function theorem provides a diffeomorphism over a product of a d_n -cube and a $(D_n - d_n)$ -cube. Shrinking these cubes makes the forward Jacobian bounded, say by B in operator norm. Thus its inverse determinant is at least B^{-D_n} . In command coordinates the subprobability density of this very report word is $Z_a / \text{vol}(\prod U_i)$, bounded below by $\kappa^n / \text{vol}(\prod U_i)$. Changing variables and integrating over the entire kernel-coordinate cube gives a positive lower density on the first cube. This proves (34). Restricting to a local section instead would discard its ambient command mass; we have not done so. $\square$

Proof of Theorem 6.2. First let $d = d_n > 0$. The vector p_n lies in a fixed unit box. Lemma 4.4, applied to the whole set of Lemma 6.3, gives $\mathcal{N}(S_n, \varepsilon) \leq A\varepsilon^{-d}$ for $0 < \varepsilon \leq 1$. The constant depends only on its bounded format and ambient dimension. Replacing every nonempty covering center by a point of S_n at most doubles its radius. Consequently M reachable representatives give error at most $C_n M^{-1/d}$; for the finitely many small budgets in this deduction increase C_n and use one point. Encoding a nearest representative proves the checkpoint upper bound.

For the lower bound, project the M arbitrary prediction centers to the coordinates in (34). If the cube has volume v and ω_d is the volume of the unit d -ball, the union of their radius- r balls occupies at most $M\omega_d r^d$ of the cube. Take $r = (v/(2M\omega_d))^{1/d}$. At least half the uniform mass remains at distance at least r , so the unconditional squared error is at least $\beta r^2/2$. Conditional averaging first makes each randomized decoder a deterministic vector for each label. A randomized assignment to these vectors cannot improve their pointwise minimum distance. An independent public coding seed can be fixed and then averaged; its conditioning does not change the exploration law. This proves the lower bound with exactly the permitted randomization. Worst-history risk is at least unconditional risk.

When $d_n = 0$, every fixed-word prediction map has derivative zero on the connected interior cube, and is therefore constant there and on its closure. There is at most one point per word. This conclusion allows several distinct points; zero differential rank does not assert that one label suffices in a general finite-report experiment.

We next construct one causal filter. For fixed u, y , multiplication by $L_{n+1,y}(u, \cdot)$ maps $\mathcal{V}_{n+1}$ into $\mathcal{V}_n$. Since $1 \in \mathcal{V}_{n+1}$, it also puts the next likelihood in $\mathcal{V}_n$. Choose expansions in the fixed spanning list at time n . Their coefficients are polynomial in u and bounded on U_{n+1} : this follows by choosing a fixed linear right inverse to that spanning list on the finite-dimensional coefficient space. Hence the next prediction vector is a well-defined rational function

$$(35) \quad T_n(p, u, y)_j = \sqrt{w_{n+1,j}} \frac{\nu_p(L_{n+1,y}(u, \cdot) H_{n+1,j})}{\nu_p L_{n+1,y}(u, \cdot)}.$$

Here ν_p denotes any posterior realizing p ; its action on $\mathcal{V}_n$ is determined by p and $\nu_p 1 = 1$, so the expression is independent of the realization.

The denominator in (35) is at least κ , not only on S_n but also on line segments between its points: such a segment is represented on $\mathcal{V}_n$ by the corresponding mixture of posteriors. Bounded coefficients and the quotient rule therefore give

$$\|T_n(p, u, y) - T_n(p', u, y)\| \leq L_n \|p - p'\| \quad (p, p' \in S_n),$$

with $L_n < \infty$ independent of histories and commands. If p is reachable, so is $T_n(p, u, y)$, since the appended report has strictly positive probability.

For M large enough, choose at stage n a reachable quantizer Q_n of radius at most $A_n M^{-1/d_n}$ when $d_n > 0$, and store every point when $d_n = 0$. With $d_* > 0$ all these radii are at most AM^{-1/d_*} . Store only the representative index and use

$$\hat{p}_{n+1} = Q_{n+1}(T_n(\hat{p}_n, u_{n+1}, y_{n+1})).$$

This is an admitted transition using only the preceding index and the current input. Starting with the known exact initial vector, its errors satisfy

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + AM^{-1/d_*}.$$

Finite iteration gives $e_n \leq B_N M^{-1/d_*}$ at every checkpoint. For the finitely many excluded budgets, the squared prediction loss is at most one, so increasing the constant supplies the same assertion. The decoder uses the actual probabilities of the representative. This proves the simultaneous upper bound without retaining or reconstructing an exact discarded prefix. Every causal filter is in particular an M -message prefix encoder at a checkpoint attaining d_* ; the lower bound already proved there gives (33). If every $d_n = 0$, retain all finitely many reachable states at each stage instead; the same exact updates give a finite exact filter.

Finally fix the detector and queries and vary their finite prior moment vector in a compact constant-rank set. All first and second command derivatives of the normalized maps are uniformly bounded, because evidence is bounded below by κ^n . At each parameter choose a nonzero d_n -minor and an interior tuple. Continuity preserves a positive minor on a parameter neighborhood. A finite subcover gives uniform chart radii and lower mass by the preceding inverse-function argument. The whole-image format bound was already coefficient independent. The update coefficients are fixed by the detector and queries, and the denominator is still at least κ . These observations give all asserted uniform constants. $\square$

The theorem identifies the exponent directly from the experiment, not from a postulated scale representation of its risk. It makes no claim that the integer d_n determines a uniform profile when ranks change. The covariance classification in Section 7, and the collision and contrast classifications later, supply that additional information in their respective classes. The results also distinguish semialgebraic image dimension from the dimension of a Euclidean space into which the entire exact state set might embed.

6.3. Algebraic rank strata in the moment body. Choose a basis $1, g_1, \dots, g_s$ of the finite-dimensional space generated by all coefficient functions needed to form all vectors Y_a at the fixed horizon. Write

$$m(\mu) = (\mu g_1, \dots, \mu g_s), \quad K = \text{conv}\{(g_1(x), \dots, g_s(x)) : x \in X\}.$$

This convex hull is compact. Linear independence with 1 means that K has nonempty interior in $\mathbb{R}^s$; the case $s = 0$ is read in its zero-dimensional affine space. Entries of every $\mathcal{A}_a$ are polynomials in commands whose coefficients are affine functions of m .

Theorem 6.4 (Exact moment strata and generic maximal acquisition). *For every checkpoint n and integer $k \geq 0$ there is an explicit finite list of polynomials $\mathcal{F}_{n,k}$ in m such that*

$$(36) \quad d_n(\mu) \leq k \iff f(m(\mu)) = 0 \text{ for every } f \in \mathcal{F}_{n,k}.$$

The list consists of the command coefficients of all $(k+2)$ -minors of the matrices (30), over all report words. Thus the complete checkpoint and causal memory exponents are determined by finitely many polynomial equalities in the actual prior moments.

The moments of full-support probabilities are precisely $\text{int } K$. On this set there is a relatively open dense complement of a proper algebraic set on which all checkpoint ranks simultaneously take their maximum values. These generic ranks can be reached by arbitrarily small bounded-density perturbations of any full-support prior. Lower-rank loci, when present, are exactly the algebraic loci in (36), intersected with the moment body.

Proof. The equivalence is the polynomial rank criterion: a command polynomial vanishes on the interior cube exactly when every one of its coefficients vanishes. The normalization rank is one smaller by Lemma 6.1. An empty list of minors has its usual vacuous meaning when k exceeds the possible rank. This gives a finite algebraic test using exact moments, not an equality test from inexact numerical moment names.

Suppose μ has full support and $m(\mu)$ were on a proper supporting hyperplane of K . Its defining affine function would give a continuous nonnegative function on X , with integral zero under μ , but not identically zero. Positivity on a nonempty open set contradicts full support. Hence $m(\mu) \in \text{int } K$. Conversely, take an interior m and any fixed full-support μ_0 . Such a probability exists on a compact metric space, for example by assigning positive summable masses to a countable dense set. For sufficiently small $\epsilon > 0$,

$$m' = \frac{m - \epsilon m(\mu_0)}{1 - \epsilon} \in K.$$

By the definition of K and finite-dimensional convexity, m' is a finite convex combination of points $g(x)$ and hence is the moment of a probability ν . The probability $\epsilon\mu_0 + (1 - \epsilon)\nu$ has full support and moment m .

For each checkpoint of positive maximal rank choose one nonzero coefficient of a maximal nonzero minor. Its polynomial is not identically zero on $\mathbb{R}^s$: K has interior. The product of the finitely many chosen polynomials is nonzero. Away from its zero set all maximal ranks are attained simultaneously. Checkpoints whose maximal rank is zero impose no condition. A nonzero real polynomial has a zero set of empty interior and Lebesgue measure zero, giving the claimed dense open set. No semialgebraicity of K itself is asserted.

For the final assertion fix a full-support μ and put $m = m(\mu)$. The covariance matrix $G = \mu[(g - m)(g - m)^t]$ is positive definite: zero variance of a linear combination would make it constant on X , contrary to the independence of $1, g_1, \dots, g_s$. For sufficiently small $t \in \mathbb{R}^s$ the probability

$$d\mu_t = (1 + t^t(g - m)) d\mu$$

is positive, has total mass one, has full support, and has moment $m + Gt$. These moments fill a neighborhood of m , while the densities tend uniformly to one. Every such neighborhood meets the generic maximal-rank set just constructed. $\square$

Together, Theorems 6.2 and 6.4 give a necessity-and-sufficiency description of the memory exponent throughout this class, including exceptional priors. The algebraic test does not presuppose a Chebyshev minorization, a rectangular risk curve, or a known attainable geometry. What it does not encode is a sharp anisotropic profile near an exceptional locus. We now obtain such a profile, uniformly through all rank losses, for a whole class of affine acquisitions.

7. THE COMPLETE RESOLUTION PROFILE OF POSITIVE AFFINE ACQUISITION

The rank classification does not prescribe how acquired directions vanish. For a general affine acquisition, that question has an exact answer in terms of a cross-covariance operator.

Unlike a future-test spectrum alone, this operator contains both acquisition and observation. No assumption on the dimension, ordering, or density of the latent space is required.

7.1. The experiment-level invariant. Let $b, q \geq 1$, let $f = (f_1, \dots, f_b)^t \in C(X)^b$, and suppose

$$(37) \quad \sup_{x \in X} \sum_{i=1}^b |f_i(x)| \leq a < 1.$$

The acquisition command is uniform on $U = [-1, 1]^b$. The two report likelihoods are

$$(38) \quad L_y(u, x) = \frac{1}{2}(1 + y u^t f(x)), \quad y \in \{-1, 1\}.$$

Both are at least $(1 - a)/2$. Let $H_1, \dots, H_q \in C(X)$ be the success probabilities of any fixed finite list of subsequent physical queries, with $0 \leq H_j \leq 1$, and let $w_j > 0$, $\sum_j w_j = 1$. The chosen query is revealed independently after the label is stored. Put

$$(39) \quad h = (\sqrt{w_j} H_j)_{j=1}^q, \quad \bar{h} = \mu h, \quad m = \mu f, \quad C_\mu = \mu[(h - \bar{h})(f - m)^t] : \mathbb{R}^b \rightarrow \mathbb{R}^q.$$

The command cube fixes the units of f . We do not whiten either side of this matrix by a possibly singular covariance. Such a whitening would change the physical scale being classified.

Let $d = \min(b, q)$, let $s_1 \geq \dots \geq s_d \geq 0$ be the singular values of C_μ , and set

$$(40) \quad V_\ell(C_\mu) = \prod_{i=1}^\ell s_i = \|\bigwedge^\ell C_\mu\|_{\text{op}}, \quad E_{C_\mu}(M) = \max_{1 \leq \ell \leq d} \left(\frac{V_\ell(C_\mu)}{M} \right)^{2/\ell}.$$

Zero products remain zero. Write R_M^{av} and R_M^{max} for the unconditional and worst-history checkpoint risks, with the same finite-label and randomization conventions as above.

Theorem 7.1 (Uniform affine covariance classification). *There are $0 < c \leq C < \infty$, depending only on a, b, q , such that every experiment (38), every prior μ , every query list as above, and every integer $M \geq 1$ satisfy*

$$(41) \quad cE_{C_\mu}(M) \leq R_M^{\text{av}} \leq R_M^{\text{max}} \leq CE_{C_\mu}(M).$$

The upper code can use only reachable prediction vectors. The constants are independent of the nonzero singular values, their multiplicities, the prior, and the query weights. In particular, the assertion is uniform through every rank loss in C_μ . If $C_\mu = 0$, a single label is exact for the prescribed query task.

7.2. Exact acquisition geometry, including evidence.

Lemma 7.2 (Projective acquisition coordinates). *Set $v = yu$ and $\theta = v/(1 + m^t v)$. The acquired prediction vector is exactly*

$$(42) \quad p(u, y) = \bar{h} + C_\mu \theta.$$

The law of v has density $(1 + m^t v)/2^b$ on U . The image $\Theta = \{v/(1 + m^t v) : v \in U\}$ satisfies

$$(43) \quad rB_b \subset \Theta \subset RB_b, \quad r = \frac{1}{2(1+a)}, \quad R = \frac{\sqrt{b}}{1-a},$$

where B_b is the Euclidean unit ball. On Θ the density of θ is

$$(44) \quad \frac{(1 - m^t \theta)^{-(b+2)}}{2^b},$$

and lies between $(1 - a)^{b+2}/2^b$ and $(1 + a)^{b+2}/2^b$.

Proof. Bayes' formula and the definition of covariance give

$$p(u, y) = \frac{\bar{h} + \mu(hf^t)v}{1 + m^t v} = \bar{h} + \frac{C_\mu v}{1 + m^t v}.$$

For each sign y , the change $u \mapsto v = yu$ preserves volume and U . The joint density of this branch is $(1 + m^t v)/(2 \cdot 2^b)$. Adding the two branches gives the stated density of v . This calculation includes the probability of the observed report; it is not a distribution obtained by conditioning away evidence.

Write $F(v) = v/(1 + m^t v)$. Since $|m^t v| \leq a$ on U , its inverse is $F^{-1}(\theta) = \theta/(1 - m^t \theta)$ on its image. Direct differentiation and the rank-one determinant identity give

$$DF(v) = \frac{I}{1 + m^t v} - \frac{vm^t}{(1 + m^t v)^2}, \quad \det DF(v) = (1 + m^t v)^{-(b+1)}.$$

Multiplying the density of v by the inverse Jacobian gives (44). Its bounds follow from $1 - a \leq 1 + m^t v \leq 1 + a$.

The outer inclusion in (43) follows from $\|v\| \leq \sqrt{b}$. For the inner inclusion, $\|m\| \leq \|m\|_1 \leq a$. If $\|\theta\| \leq r$, then $|m^t \theta| < 1/2$ and $\|\theta/(1 - m^t \theta)\| \leq 2r \leq 1$. This inverse point belongs to $B_b \subset U$, and substitution verifies that its image is θ . Thus the full ball is contained in Θ . $\square$

Lemma 7.3 (Quantization of a linearly observed thick set). *Let $A : \mathbb{R}^b \rightarrow \mathbb{R}^q$ be linear. Suppose a random vector θ is supported in RB_b , and its law dominates γ times Lebesgue measure on rB_b , where $r, R, \gamma > 0$. Let $s_1 \geq \dots \geq s_d \geq 0$ be the singular values of A , $d = \min(b, q)$. For all $M \geq 1$, the optimal mean squared M -center error is bounded below by a constant times*

$$\max_{1 \leq \ell \leq d} (s_1 \cdots s_\ell / M)^{2/\ell}.$$

The whole image of the support has an M -center covering of squared radius bounded above by another constant times the same quantity. If that image is compact, its centers can be chosen in the image. The constants depend only on b, q, r, R, γ and not on A .

Proof. Choose orthogonal singular coordinates. Fix an index ℓ with $s_\ell > 0$. Project the first ℓ input coordinates. For every $z \in (r/2)B_\ell$, the fiber of rB_b has $(b - \ell)$ -dimensional volume at least

$$\omega_{b-\ell}(\sqrt{3}r/2)^{b-\ell},$$

where the zero-dimensional volume is one. Consequently the law of the first ℓ coordinates dominates a positive constant times Lebesgue measure on $(r/2)B_\ell$. After multiplication by $\text{diag}(s_1, \dots, s_\ell)$ it dominates a fixed positive mass times the uniform measure of an ellipsoid of volume $\omega_\ell(r/2)^\ell s_1 \cdots s_\ell$. The union of M projected radius- t balls has volume at most $M\omega_\ell t^\ell$. Taking $t = (r/2)(s_1 \cdots s_\ell/(2M))^{1/\ell}$ leaves at least half that uniform mass outside. This proves the lower bound for the chosen index and hence for their maximum. It remains valid when t exceeds a short ellipsoid axis: the ball-volume estimate is an upper bound on the volume of its intersection with the ellipsoid, not an assumption that the ball fits inside. The argument also allows arbitrary centers and randomized encoders and decoders by conditional averaging.

For the upper bound, the image lies in the rectangle with active half-sides Rs_i . A grid of mesh $\varepsilon/\sqrt{d}$ on each active axis covers that rectangle by at most

$$\prod_{i=1}^d (1 + B_d s_i / \varepsilon) \leq A_d \sum_{j=0}^d \varepsilon^{-j} s_1 \cdots s_j$$

balls of radius ε , after increasing constants depending on d, R . The inequality follows by expanding the product: each j -fold product is at most the product of the largest j singular

values. Zero axes need a single coordinate and cause no division. Let $e = \max_j (s_1 \cdots s_j / M)^{1/j}$. If $e > 0$, then $s_1 \cdots s_j \leq M e^j$. Choose L large enough that $A_d \sum_{j=1}^d L^{-j} \leq 1/2$. For $M \geq 2A_d$, radius Le therefore suffices with at most M centers. Empty balls are discarded; replacing a remaining center by an image point in that ball at most doubles the radius. For $M < 2A_d$, one image point has covering radius at most $2Rs_1$, while $e \geq s_1/M$. Increasing the constant handles these budgets. If $A = 0$, the image is a singleton and every assertion has zero error. No rank-dependent lower bound on a singular value entered either argument. $\square$

Proof of Theorem 7.1. Apply Lemma 7.3 to $A = C_\mu$ and the exact law in Lemma 7.2. The constants r, R, γ in that application depend only on a, b . Translation by $\bar{h}$ changes neither coding nor error. Weighted query probabilities identify physical squared-loss excess with the Euclidean norm used in those lemmas, so no inverse query weight is paid in the constants. Every representative belongs to $\bar{h} + C_\mu \Theta$ and hence is a posterior prediction vector of an admitted history. Its unweighted entries are valid query probabilities. This proves the upper bound for every history, while the exact unconditional law proves the lower bound. If $C_\mu = 0$, (42) is independent of the history and its known constant prediction requires one label. $\square$

7.3. Operational completeness and an interior degeneration.

Corollary 7.4 (Complete invariant in the affine class). *In any family with fixed bounds on $a < 1$, b , and q , the products $V_\ell(C_\mu)$ characterize uniform comparison of entire integer-budget checkpoint-risk curves. More precisely, after padding to a fixed maximum dimension, two such families have uniformly comparable curves if and only if their corresponding products are uniformly comparable, including matching zero products. Moreover*

$$(45) \quad \inf_{M \in \mathbb{N}, M \geq 1} M(R_M^{\text{av}})^{\ell/2} \asymp V_\ell(C_\mu).$$

For positive rank r_μ , the eventual logarithmic slope is $-2/r_\mu$.

Proof. The singular values are ordered and Theorem 7.1 provides the forward representation uniformly for this entire experiment class. Lemma 14.1 gives, for $e(M) = E_{C_\mu}(M)^{1/2}$,

$$V_\ell \leq \inf_{M \geq 1} M e(M)^\ell \leq 2V_\ell.$$

Combining it with (41) proves (45). Comparison of risks therefore implies comparison of products; conversely compare each term in (40) and take their finite maximum. The argument includes zero products by the same envelope identity. For fixed positive rank, the last positive branch dominates for all sufficiently large budgets and yields the slope. The all-zero case is handled by the one-label assertion of the theorem, not by taking a logarithm of zero. $\square$

This is an experiment-level classification: the covariance is computed from likelihood features and physical future tests before the coding problem is solved. It does not posit an ordered-envelope law and then use that law to discover the forward geometry. The inverse still uses the full unbounded sequence of integer budgets; it is not reconstruction of an unknown experiment from a finite sample of observed errors.

Proposition 7.5 (Rank change among full-support priors). *Let $X = [-1, 1]$, use $f(x) = \alpha x$ with $0 < \alpha < 1$, and let the future query be a Bernoulli experiment with success probability $H(x) = 1/2 + \beta x^2$, where $0 < \beta < 1/2$. For the full-support priors*

$$d\mu_t(x) = (1 + tx) dx/2, \quad |t| \leq t_0 < 1,$$

the sharp acquisition-checkpoint law is

$$(46) \quad R_M^{\text{av}} \asymp R_M^{\text{max}} \asymp (\alpha\beta)^2 t^2 M^{-2}, \quad M \geq 1,$$

uniformly in t , including zero. At $t = 0$ one label is exact for the specified future query; at every $t \neq 0$ the attainable rank is one.

Proof. Direct integration gives

$$\mu_t x = t/3, \quad \mu_t x^2 = 1/3, \quad \mu_t x^3 = t/5, \quad \text{Cov}_{\mu_t}(x^2, x) = 4t/45.$$

Thus the sole singular value of (39) is $4\alpha\beta|t|/45$. Theorem 7.1 proves (46). Equivalently, with $v = yu$, the posterior query probability is exactly

$$\frac{1}{2} + \beta \frac{1/3 + \alpha vt/5}{1 + \alpha vt/3},$$

whose derivative in v is $4\alpha\beta t/[45(1 + \alpha vt/3)^2]$. It is zero throughout the command interval precisely at $t = 0$. Both acquisition reports and both future outcomes have uniformly positive likelihoods. Thus the degeneration occurs inside a dominated full-support prior family; it is not loss of support or vanishing evidence. The centered moment equation $\mu x^3 - \mu x \mu x^2 = 0$ is exactly its rank stratum. $\square$

The example exhibits a degeneration governed by the interaction of acquisition and observation rather than a collision of future exponents. Its role is to make the structural invariant concrete; the uniform classification applies to arbitrary dimensions and arbitrary latent spaces in (37), not only to this example. The multi-step collision and contrast laws are retained below and are not replaced by a one-step assertion.

8. ATTAINMENT UNDER EVERY FULL-SUPPORT PRIOR

The rank formula uses the specified prior. We now characterize when a square product pairing has full normalized rank for *every* full-support prior. This replaces separate Chebyshev assumptions by an exact sign condition on the paired evaluation determinants. The condition is intrinsic to the two function spaces and does not require an ordering of X .

8.1. Positive histories and their augmented tangents. Fix an interior command tuple u_* for a positive history product $P(u, \cdot) \in C(X)$, twice continuously differentiable in u . Write $P = P(u_*, \cdot) > 0$, and let

$$(47) \quad \tilde{T} = \text{span}\{P\} + \text{im } DP(u_*).$$

Let $p \geq 1$ and $\Phi = \text{span}\{1, \phi_1, \dots, \phi_p\}$, where the functions are real and continuous, and suppose

$$(48) \quad \dim \tilde{T} = \dim \Phi = p + 1 =: d.$$

This is the square pairing regime; no claim that every rectangular pairing reduces to it is made. For a basis $v_0, \dots, v_p$ of $\tilde{T}$, with $\phi_0 = 1$, define the continuous symmetric function on X^d

$$(49) \quad \Delta_{v, \phi}(x_0, \dots, x_p) = \det[v_i(x_k)]_{i,k=0}^p \det[\phi_j(x_k)]_{j,k=0}^p.$$

It is symmetric because a permutation changes both determinants by the same sign. Changing either basis multiplies it by a nonzero constant; being of one weak sign and nonzero somewhere is therefore a property of the pair $(\tilde{T}, \Phi)$, not of its oriented bases.

Theorem 8.1 (Universal square-pairing criterion). *Under (48), the following assertions are equivalent.*

(i) *For every full-support probability μ on X , the actual command map*

$$u \longmapsto (\mu(P(u)\phi_j)/\mu P(u))_{j=1}^p$$

has rank p at u_* .

(ii) The function $\Delta_{v,\phi}$ has one weak sign on X^d and is nonzero at at least one tuple.

Under these equivalent conditions, independent commands with a density bounded below near u_* give a positive subprobability minorization of a p -dimensional prediction cube for every such prior. If the positive history products, tests, interior tuples, and command derivatives vary continuously in a compact family, the dimensions in (48) remain fixed, and (ii) holds at every parameter, the minorization is uniform over that family and all probabilities $\mu \geq c\mu_0$, where $c > 0$ and μ_0 has full support. A common positive lower bound for the likelihoods, a common command neighborhood, and a common lower command-density bound are understood in this uniform assertion.

Proof. For each μ define $L_\mu Q = (\mu Q, \mu(Q\phi_1), \dots, \mu(Q\phi_p))^t$ and $Z = \mu P$. The normalized differential is the map

$$Q \mapsto Z^{-1} \left(L_\mu Q - \frac{L_\mu P}{Z} \mu Q \right)$$

on actual product derivatives, with its constant coordinate omitted. Adding the direction P does not change its image, because that direction maps to zero. On $L_\mu(\tilde{T})$ the kernel of the projection in parentheses is exactly the span of $L_\mu P$, whose constant coordinate is $Z > 0$. Therefore the normalized command rank is $\text{rank}(L_\mu|\tilde{T}) - 1$. In the square case it equals p if and only if

$$D(\mu) := \det[\mu(v_i\phi_j)]_{i,j=0}^p \neq 0.$$

This argument explains why adjoining the evidence direction in (47) is legitimate even when P is not an actual command derivative.

Determinant expansion, Fubini, and symmetrization give the classical integration identity

$$(50) \quad D(\mu) = \frac{1}{d!} \int_{X^d} \Delta_{v,\phi}(x_0, \dots, x_p) \prod_{k=0}^p \mu(dx_k).$$

For completeness, expand both determinants on the right into their permutation sums. Integrating a term gives a product of d scalar integrals. Reindexing one permutation relative to the other produces the determinant on the left exactly $d!$ times. All functions are bounded, so the expansions and integrations are legitimate. This is Andréief's identity, not a new integration formula; see [32] for its source and attribution.

Suppose (ii) holds and orient a basis so that $\Delta \geq 0$. Continuity and a strictly positive value give a product of nonempty open sets on which $\Delta > 0$. Every full-support probability gives this product positive mass. Equation (50) therefore gives $D(\mu) > 0$, proving (i).

Conversely, the full-support probabilities form a convex set. If (i) holds, continuity along every line segment of this set shows that $D(\mu)$ has the same nonzero sign throughout it. Orient a basis so that the sign is positive and fix one full-support μ_0 . For an arbitrary tuple $(x_0, \dots, x_p)$ put $\nu = d^{-1} \sum_k \delta_{x_k}$. The probabilities $(1 - \epsilon)\nu + \epsilon\mu_0$ have full support for every $0 < \epsilon < 1$. Letting $\epsilon \downarrow 0$ gives $D(\nu) \geq 0$. Matrix multiplication at this atomic probability gives, also for repeated points,

$$D(\nu) = d^{-d} \det[v_i(x_k)] \det[\phi_j(x_k)] = d^{-d} \Delta_{v,\phi}(x_0, \dots, x_p).$$

Hence $\Delta \geq 0$ everywhere. Since $D(\mu_0) > 0$ in (50), Δ is not identically zero. This proves (ii) and its necessity without assuming either evaluation determinant separately has a fixed sign.

The minorization follows by augmenting the onto normalized derivative with orthonormal kernel coordinates, applying the inverse-function theorem, and integrating the entire kernel-coordinate cube, exactly as in the proof of Lemma 6.3. The density of the specified report word contains its evidence, bounded below by the product of the likelihood lower bounds.

For uniformity, the set of probabilities dominating $c\mu_0$ is weakly compact: it is a closed subset of the probabilities on compact X , as is seen by testing the domination inequality against continuous nonnegative functions. Every limiting probability still has full support. The normalized command Jacobian is jointly continuous in the parameter and this weakly varying probability; this follows from the uniform-norm continuity of the finitely many integrands and the positive evidence bound. It is onto everywhere by the equivalence just proved. Compactness gives a positive lower bound for its least row singular value. Uniform second derivative bounds and uniformly interior tuples then give uniform inverse-chart radii, while the specified command and evidence lower bounds give uniform minorization mass. No conclusion about a later observation operator's small singular values is used. $\square$

Corollary 8.2 (Matched score and query spaces on arbitrary latent spaces). *If $\tilde{T} = P\Phi$ and $\dim \Phi = p + 1$, then the normalized history map has rank p for every full-support prior. More generally, the same conclusion holds whenever the paired evaluation determinants have one weak sign and are nonzero somewhere, even if both individual determinants change sign.*

Proof. Choose $v_i = P\phi_i$. Then

$$\Delta_{v,\phi}(x_0, \dots, x_p) = \left(\prod_{k=0}^p P(x_k) \right) \det[\phi_i(x_k)]_{i,k=0}^p{}^2 \geq 0.$$

The evaluation determinant is nonzero somewhere. Indeed, if the vectors $(\phi_i(x))_{i=0}^p$ did not span $\mathbb{R}^{p+1}$ as x varies, a nonzero linear functional annihilating all of them would contradict the independence of the functions. Selecting a spanning set of $p + 1$ evaluation vectors gives the desired tuple. Apply Theorem 8.1. Its same equivalence proves the more general assertion. $\square$

8.2. Relation to positive products and to rank strata. For the positive products of Section 11, the original normalized-product criterion remains valid. The present criterion makes two further distinctions. It uses the actual augmented tangent, so it does not require a physically available radial command. It also characterizes, in the square regime, the full-support-prior quantifier itself. Two separately strict Chebyshev systems with compatible orientations make (49) positive on distinct ordered tuples and are a sufficient instance. They are not necessary: the matched-space corollary works on compact spaces of any dimension.

For example, at $u = 0$ in (38), the augmented tangent is $\text{span}\{1, f_1, \dots, f_b\}$. When the future tests span that same space and these functions are independent, the corollary proves maximal acquisition for every full-support prior. In contrast, for $v = (1, x)$ and $\phi = (1, x^2)$ on $[-1, 1]$, the paired evaluation determinant is

$$(x_1 - x_0)^2(x_0 + x_1),$$

which takes both signs. The normalized rank can consequently depend on the full-support prior. Proposition 7.5 computes that dependence and its entire finite-label profile exactly. These observations link the universal criterion, the algebraic moment strata, and the covariance classification without substituting a local pairing argument for either a global cover or a causal construction.

9. RECTANGULAR PAIRINGS AND ATTAINABLE RANK

The square criterion has a rectangular counterpart, but the counterpart is not a choice of one square subpairing. The appropriate object is the cone of products of functions in the two spaces. We give both a full-rank criterion and a formula for the smallest rank over all full-support priors. The latter produces priors whose kernels contain the permitted subspaces and realizes the minimum rank. Exact realization of a prescribed subspace is treated in Section 10.

Throughout this section, X is a nonempty compact metric space and $E, F \subset C(X; \mathbb{R})$ have dimensions d, k , with $1 \leq k \leq d$. Choose bases $e_1, \dots, e_d$ and $f_1, \dots, f_k$, and set

$$e(x) = (e_1(x), \dots, e_d(x))^t, \quad H_\mu = (\mu(e_a f_i))_{a,i} \in \mathbb{R}^{d \times k}.$$

The kernel is regarded as a subspace of F , independently of the chosen bases. Relative interiors below are taken in the affine hull of the indicated convex body.

9.1. Full-support moments.

Lemma 9.1 (The full-support moment image). *For a continuous map $G : X \rightarrow \mathbb{R}^s$, put $K_G = \text{conv } G(X)$. Then*

$$(51) \quad \{\mu G : \mu \in \mathcal{P}(X), \text{ supp } \mu = X\} = \text{ri } K_G.$$

More precisely, fix any full-support probability μ_0 . Each $b \in \text{ri } K_G$ is realized by $\mu = \varepsilon \mu_0 + (1 - \varepsilon) \nu$, where $0 < \varepsilon < 1$ and ν has at most $1 + \dim \text{aff } K_G$ atoms. The number ε may depend on b .

Proof. A compact metric space is separable. Assigning positive summable masses to a countable dense set supplies a full-support probability, so the class in (51) is nonempty. The convex hull K_G is compact: by the finite-dimensional convex-hull theorem, its points are convex combinations of at most $s + 1$ points of $G(X)$. If a full-support mean b lay in the relative boundary of K_G , a supporting affine functional on $\text{aff } K_G$ would be nonnegative on K_G , zero at b , and positive somewhere on K_G . Its composition with G is nonnegative continuous and positive at some point of X . It has positive integral under every full-support probability, contradicting its value zero at the mean. A zero-dimensional K_G has empty relative boundary and causes no exception.

Conversely, write $b_0 = \mu_0 G$. Since b is a relative interior point, for all sufficiently small $\varepsilon > 0$ the point

$$c_\varepsilon = \frac{b - \varepsilon b_0}{1 - \varepsilon} = b + \frac{\varepsilon}{1 - \varepsilon} (b - b_0)$$

belongs to K_G . The convex-hull theorem in its affine hull represents c_ε by a probability ν with the stated number of atoms. The displayed mixture has full support and mean b . This proves equality of attainable moment sets, not uniqueness or full support of every representation of an interior moment. $\square$

9.2. A product-cone criterion and all rank defects.

Theorem 9.2 (Universal rectangular pairing). *The following three conditions are equivalent:*

- (i) $\text{rank } H_\mu = k$ for every full-support probability μ on X .
- (ii) For each $0 \neq f \in F$ there is $g \in E$ such that $fg \geq 0$ on X and fg is not identically zero.
- (iii) For each $0 \neq f \in F$, $0 \notin \text{ri conv}\{f(x)e(x) : x \in X\}$.

Proof. A function $f \in F$ is in the kernel precisely when $\mu(fe) = 0$. Lemma 9.1, applied to $G = fe$, therefore identifies the failure of (i) with the failure of (iii). If (ii) holds and a full-support μ satisfies $H_\mu f = 0$, then $\mu(fg) = 0$ for every $g \in E$. The particular product in (ii) has strictly positive integral, a contradiction. Thus (ii) implies (i).

For completeness, let K be a nonempty compact convex subset of a Euclidean space. If $0 \notin \text{ri } K$, there is a linear functional λ nonnegative on K and positive somewhere on K . When $0 \notin K$, separate the compact set K strictly from zero, for example by the nearest point of K to zero. When $0 \in K \setminus \text{ri } K$, its affine hull is a linear space; a supporting hyperplane in that space has the required property, and its functional extends to the ambient space. Apply this observation to $K = \text{conv}\{f(x)e(x)\}$. Write $\lambda(z) = \sum_a c_a z_a$ and set $g = \sum_a c_a e_a$. Then $fg \geq 0$ everywhere and is positive somewhere. This proves (iii) implies (ii). $\square$

For finite X , the kernel formulation is a positive-diagonal sign-vector alternative. This algebraic mechanism is already present in Müller–Feliu–Regensburger–Conradi–Shiu–Dickenstein [34, Lemma 2.1]. The compact-space formulation retains continuity, actual full-support measures, and the representation of those measures needed below. The finite square specialization is the strong-compatibility criterion of Banaji–Pantea [33, Definition 2.24 and Lemma 2.36]; see also Section L.1. Neither finite criterion is claimed as a new algebraic input.

For a subspace $U \subset F$ of dimension j , choose a basis $u_1, \dots, u_j$ and define

$$G_U(x) = (u_i(x)e_a(x))_{i,a} \in \mathbb{R}^{jd}, \quad K_U = \text{conv } G_U(X).$$

The condition $0 \in \text{ri } K_U$ is unchanged by any basis change. For $U = \{0\}$, use the one-point body in $\mathbb{R}^0$.

Theorem 9.3 (Rank alternatives with full-support witnesses). *There is a full-support probability for which $U \subset \ker H_\mu$ if and only if $0 \in \text{ri } K_U$. Consequently*

$$(52) \quad \min_{\text{supp } \mu = X} \text{rank } H_\mu = k - \max\{\dim U : U \subset F, 0 \in \text{ri } K_U\}.$$

Whenever the condition holds, a witnessing prior can be chosen in the form $\varepsilon\mu_0 + (1 - \varepsilon)\nu$ for any prescribed full-support μ_0 , with ν supported on at most $jd + 1$ points. A subspace of maximal permitted dimension gives a prior attaining the minimum rank in (52).

Proof. The inclusion $U \subset \ker H_\mu$ consists exactly of the jd equations $\mu G_U = 0$. The equivalence and the finite-atom mixture follow from Lemma 9.1. Every actual kernel is one of the permitted subspaces. Conversely, a prior realizing a permitted subspace has nullity at least its dimension. The possible dimensions form a nonempty subset of the finite set $\{0, \dots, k\}$, so their maximum is attained by a subspace. A witnessing prior for such a subspace cannot have larger nullity, since its whole kernel would then be a permitted subspace of larger dimension. Rank–nullity gives (52) and actual attainment of its left side. There is no assertion of a common positive mixture weight for all the witnesses. $\square$

9.3. A quantitative margin and posterior normalization. Fix the Euclidean coefficient norm on F . For $f \in F$ write $a_f = \mu_0(fe)$ and $K_f = \text{conv}\{f(x)e(x) : x \in X\}$.

Proposition 9.4 (The dominated-prior margin). *For $0 < c \leq 1$,*

$$(53) \quad \inf_{\mu \geq c\mu_0} s_{\min}(H_\mu) = \min_{\|f\|=1} \text{dist}(0, ca_f + (1 - c)K_f).$$

Under Theorem 9.2, this number is positive. Here $s_{\min}$ is the least of the k singular values. The margin also remains positive over a compact continuous family of pairs of bases satisfying that theorem, with their dimensions fixed.

Proof. For $c < 1$, every admitted measure has the unique form $\mu = c\mu_0 + (1 - c)\nu$ with $\nu \in \mathcal{P}(X)$. For a fixed unit f , the vectors $H_\mu f$ therefore fill $ca_f + (1 - c)K_f$. Taking the infimum over f and ν in either order gives (53). For $c = 1$ only μ_0 occurs and the same formula is immediate. The unit sphere and $\mathcal{P}(X)$ are compact, the latter in the weak topology, and the integrands are

continuous. Thus the joint minimum is attained. Every admitted measure has full support, so a zero minimum would contradict full column rank. The same compactness argument applies with an additional compact parameter and bases continuous in the uniform norm. It requires the criterion at every parameter. It makes no assertion of uniformity over all full-support priors without domination. $\square$

Corollary 9.5 (Rectangular positive-history attainment). *Let $P(u, x)$ be a positive history product, twice continuously differentiable in an interior command neighborhood of u_* , and let $\Phi = \text{span}\{1, \phi_1, \dots, \phi_{d_\Phi-1}\}$ be its finite future test space. Put*

$$T = \text{span}\{P(u_*, \cdot)\} + \text{im } DP(u_*), \quad k = \min(\dim T, \dim \Phi).$$

Apply Theorem 9.2 with F the smaller of T, Φ and E the other. Its equivalent conditions hold if and only if, for every full-support prior, the normalized prediction map has rank $k - 1$ at u_ . For $k > 1$, independent commands having a density bounded below near u_* then give an unconditional subprobability minorization on a $(k - 1)$ -dimensional projection of predictions.*

The minorization constants are uniform on compact continuous families satisfying these conditions, for priors $\mu \geq c\mu_0$, provided the spaces have fixed dimensions and continuous bases, the products and tests have uniform bounds through two command derivatives, and there are common positive likelihood, interior-neighborhood, and command-density bounds. The output projection may be selected locally in the family.

Proof. Let $L : T \rightarrow \Phi^*$ be $Lv = (\mu(v\phi_j))_{j=0}^{d_\Phi-1}$, where $\phi_0 = 1$. Write $Y = LP = (Z, Zp)^t$ with $Z = \mu P > 0$. The derivative of normalization at Y sends (z, w) to $(w - pz)/Z$. Its kernel is precisely $\mathbb{R}Y$. The image of L is the sum of $\mathbb{R}Y$ and $L(\text{im } DP)$. Since $Y \neq 0$ belongs to that image, normalization of the actual command derivative has rank $\text{rank } L - 1$. Transposing the pairing when necessary does not change its rank. Theorem 9.2 now proves the equivalence, without assuming that the radial direction P is a physical command derivative.

Write $r = k - 1 > 0$, choose an r -coordinate output projection z with onto differential at u_* , and let K be an orthonormal coordinate map on its differential kernel. The augmented map $u \mapsto (z(u), K(u - u_*))$ has invertible derivative. The inverse function theorem gives a product box in these coordinates. Integrating the command density over the entire kernel-coordinate box, and multiplying by the probability of the specified report word, gives a positive density on the output box. That word probability is $\mu P(u)$, bounded below by the assumed likelihood bound. Thus the minorization is a subprobability of the original experiment, not a conditional-on-word law and not mass placed on a section of zero ambient measure. This is the construction in Proposition 11.1, applied after augmentation.

In a compact dominated family, the least nonzero singular value of the normalized command derivative is positive uniformly: its rank is always r , its entries are continuous, and the parameter-measure space is compact. Equivalently one can first use Proposition 9.4 on the pairing and then normalization and the actual tangent map. A maximal output minor and an interior command box may be selected near each parameter-measure pair. A finite subcover, the uniform second-derivative bounds, and the common lower density and likelihood bounds give uniform inverse radii and density constants. The case $k = 1$ asserts rank zero only; it asserts no positive-dimensional mass. $\square$

9.4. A genuinely rectangular physical acquisition. The next construction explains why a fixed evidence-preserving square reduction would discard admissible experiments.

Proposition 9.6 (A five-point rectangular experiment). *Let $X = \{z_0, \dots, z_4\} \subset \mathbb{R}^2$ with*

$$z_0 = (0, 0), \quad z_1 = (2, 0), \quad z_2 = (3, 1), \quad z_3 = (1, 3), \quad z_4 = (-1, 1), \quad h(z_i) = i.$$

The spaces $E = \text{span}\{1, x, y\}$ and $F = \text{span}\{1, h\}$ satisfy the universal rectangular criterion. No two-dimensional subspace of E containing 1 has a universal nonsingular square pairing with F .

These spaces arise from the positive acquisition

$$L_{\pm}(u, z) = \frac{1}{2}(1 \pm u^t f(z)), \quad f(z) = (x/8, y/8)^t, \quad u \in [-1, 1]^2,$$

followed by one physical binary query of success probability $g(z) = 1/4 + h(z)/8$. At the interior command $u = 0$, every full-support prior gives normalized prediction rank one. The full one-step memory profile is the affine profile with the nonzero row covariance $\text{Cov}_{\mu}(g, f)$.

Proof. The listed evaluations show that E, F have dimensions three and two. For each $j = 0, 1, 2, 3$, the following affine functions are negative on $z_0, \dots, z_j$ and positive on $z_{j+1}, \dots, z_4$:

j	function	z_0	z_1	z_2	z_3	z_4
0	$x + 3y - 1$	-1	1	5	9	1
1	$2y - 1$	-1	-1	1	5	1
2	$-x + y - 1$	-1	-3	-3	1	1
3	$-2x - 1$	-1	-5	-7	-3	1

Every nonzero $a + bh$ either has one weak sign, in which case the constant 1 or -1 is an admissible multiplier, or changes sign once along this list. In the latter case choose the indicated separator, with its sign reversed when $b < 0$. If the zero is at a listed vertex, either adjacent separator gives a nonnegative product, positive at a nonzero value of $a + bh$. This proves condition (ii) of Theorem 9.2.

Every two-dimensional subspace of E containing 1 is $\text{span}\{1, ax + by\}$ for $(a, b) \neq (0, 0)$. By the square criterion, its evaluations must be weakly monotone or weakly antitone along the strictly increasing values of h . For weak monotonicity the consecutive differences require

$$2a \geq 0, \quad a + b \geq 0, \quad -2a + 2b \geq 0, \quad -2a - 2b \geq 0.$$

They imply $a \geq 0$, $b \geq a$, $b \leq -a$, hence $a = b = 0$. Reversing all inequalities gives the same conclusion. The finite square criterion used here is precisely the corresponding-minor criterion attributed above.

Finally $|x| + |y| \leq 4$ on X , so $\sum_i |f_i| \leq 1/2$ and $L_{\pm} \geq 1/4$. The query values lie in $[1/4, 3/4]$. At $u = 0$, the positive history product and its derivatives span $\{1, x, y\}$, while the future tests span $\{1, h\}$. Corollary 9.5 proves the normalized rank assertion. Equivalently the row covariance cannot vanish for any positive probability vector. Theorem 7.1 gives its full profile, and a fixed dominated family has uniform positive mass constants. No uniform nonzero covariance lower bound is asserted as arbitrary positive probabilities approach the boundary of the probability simplex. $\square$

10. EXACT KERNELS OF POSITIVE MOMENT PAIRINGS

The annihilation of a subspace and the realization of that subspace as an entire kernel are different questions. The distinction is governed by multiplication of functions, before a prior is chosen. We describe it by a linear quotient and a determinantal test, and then construct full-support witnesses arbitrarily close to any admissible prior.

Let X be a nonempty compact metric space, and let $E, F \subset C(X; \mathbb{R})$ have dimensions d, k . No ordering of these dimensions is required in this section. Measures μ below are Borel probabilities. For $U \subset F$ put

$$(54) \quad \begin{aligned} W_U &= \text{span}\{eu : e \in E, u \in U\}, & V &= \text{span}(\{1\} \cup EF), \\ \mathcal{P}_U &= \{\mu : \text{supp } \mu = X, \mu W_U = 0\}. \end{aligned}$$

Here EF denotes the set of pointwise products, not a tensor product. The condition $\mu W_U = 0$ means that every member of W_U has integral zero. It is equivalent to $U \subset \ker H_\mu$, where $H_\mu f = (\mu(e_a f))_{a=1}^d$ in a fixed basis of E .

Suppose for the moment that $\mathcal{P}_U$ is nonempty. Then $1 \notin W_U$. Choose $v_1, \dots, v_s \in V$ so that

$$[1], [v_1], \dots, [v_s] \text{ is a basis of } V/W_U, \quad s = \dim V - \dim W_U - 1.$$

Choose representatives $\bar{f}_1, \dots, \bar{f}_q$ of a basis of F/U , where $q = k - \dim U$. There are unique coefficients with

$$(55) \quad e_a \bar{f}_i \equiv c_{ai0} + \sum_{\nu=1}^s c_{ai\nu} v_\nu \pmod{W_U}, \quad \mathcal{H}_U(z) = C_0 + \sum_{\nu=1}^s z_\nu C_\nu.$$

Thus $\mathcal{H}_U$ is a d by q affine matrix pencil determined by the multiplication table modulo W_U . Define its algebraic rank by

$$(56) \quad r_U = \text{rank}_{\mathbb{R}(z_1, \dots, z_s)} \mathcal{H}_U.$$

For $s = 0$ this is the rank of the constant matrix. For $q = 0$ it is zero. A change of bases makes an invertible affine change of moment coordinates and invertible row or column operations, so r_U is intrinsic.

Lemma 10.1 (Local coordinates on a constrained prior slice). *If $\mu_* \in \mathcal{P}_U$, the set*

$$\mathcal{O}_U = \{(\mu v_1, \dots, \mu v_s) : \mu \in \mathcal{P}_U\}$$

is a nonempty open convex subset of $\mathbb{R}^s$. At every μ_ it admits a local parametrization by bounded continuous density perturbations that preserve all the equations $\mu W_U = 0$. More precisely, there are $h_1, \dots, h_s \in C(X)$ and a positive definite matrix B such that*

$$(57) \quad d\mu_t = (1 + \sum_\nu t_\nu h_\nu) d\mu_*, \quad \mu_t v = \mu_* v + Bt, \quad \mu_t W_U = 0,$$

and these are full-support probabilities whenever $\|\sum_\nu t_\nu h_\nu\|_\infty < 1$.

Proof. Full support makes the $L^2(\mu_*)$ inner product positive definite on any finite-dimensional space of continuous functions: a nonzero continuous function has nonzero magnitude on a nonempty open set. Since $\mu_* W_U = 0$, that space is orthogonal to the constants. Let Π_U be its $L^2(\mu_*)$ orthogonal projection and set

$$h_\nu = v_\nu - \mu_* v_\nu - \Pi_U v_\nu.$$

These are continuous, have mean zero, and are orthogonal to W_U . They are linearly independent, since a relation among them would put the same linear combination of the v_ν in $\mathbb{R}1 + W_U$. Consequently $B_{\alpha\nu} = \mu_*(h_\alpha h_\nu)$ is positive definite. Orthogonality gives $\mu_*(v_\alpha h_\nu) = B_{\alpha\nu}$, proving all identities in (57). The strict uniform bound on the tilt gives a strictly positive density, normalization follows from the zero means, and full support is retained. Since B is invertible, the image contains a neighborhood of $\mu_* v$. Repeating this argument at every point proves openness. Convex mixtures of full-support probabilities prove convexity. When $s = 0$, the moment space is the one-point space $\mathbb{R}^0$ and the assertion uses the empty family of tilts. $\square$

Theorem 10.2 (Exact-kernel criterion and full-support realization). *A subspace $U \subset F$ is the kernel of H_μ for some full-support probability if and only if the following two conditions hold:*

$$(58) \quad W_U \cap C(X; \mathbb{R})_+ = \{0\}, \quad r_U = k - \dim U.$$

The first condition is equivalent to $0 \in \text{ri } K_U$ in Theorem 9.3; it permits the definition of the pencil in (55). The second condition says that at least one maximal column minor of that pencil is a nonzero polynomial. When $q = 0$, the empty minor is one.

More generally, subject to the first condition,

$$(59) \quad \max_{\mu \in \mathcal{P}_U} \text{rank } H_\mu = r_U.$$

The priors attaining this maximum form a relatively weakly open dense subset of $\mathcal{P}_U$. For every $\mu_* \in \mathcal{P}_U$ and $0 < \varepsilon < 1$, such a prior can be chosen with continuous density $1 + h$ relative to μ_* and $\|h\|_\infty < \varepsilon$. In the chart of Lemma 10.1, a tensor grid with $r_U + 1$ points per coordinate inside any sufficiently small box contains a witness. This is a finite algebraic test of at most $(r_U + 1)^s$ points, not an assertion that unspecified real moments are computable.

When (58) holds, for any prescribed full-support probability μ_0 an exact witness can also be chosen as $\delta\mu_0 + (1 - \delta)\nu$, where $0 < \delta < 1$ and ν has at most $\dim V$ atoms. No common positive value of δ is asserted.

Proof. If $\mu \in \mathcal{P}_U$, a nonnegative nonzero continuous function in W_U would have strictly positive integral, a contradiction. Conversely, if $0 \notin \text{ri } K_U$, finite-dimensional separation, as in Theorem 9.2, gives a linear combination of the coordinates of G_U which is nonnegative on X and nonzero somewhere. This is a nonzero member of $W_U \cap C(X)_+$. Thus the first condition is equivalent to $0 \in \text{ri } K_U$. Theorem 9.3 then gives $\mathcal{P}_U \neq \emptyset$. In particular it also gives $1 \notin W_U$.

For $\mu \in \mathcal{P}_U$, the matrix induced by H_μ on F/U is exactly $\mathcal{H}_U(\mu\nu)$. All minors of order $r_U + 1$ vanish identically, and some minor of order r_U is a nonzero polynomial, unless $r_U = 0$, when every induced matrix is zero. A nonzero real polynomial cannot vanish on a nonempty open set. By Lemma 10.1 the nonzero minor therefore has a nonzero value at an admissible full-support moment. This proves (59). Rank-nullity now proves (58). Necessity follows from the same matrix identity and the fact that there are only q columns.

For the stronger local assertion fix μ_* . In its chart, the chosen minor becomes a nonzero polynomial in t , since B is invertible. Its degree is at most r_U . Choose a box so small that every density perturbation in it has uniform norm less than ε . A polynomial of degree at most r_U in each variable that vanishes on a product of $r_U + 1$ distinct values per variable is identically zero: apply the univariate root bound to the last variable and induct on the number of variables. Hence the grid contains a nonzero value of the minor. The perturbation retains full support and every annihilation equation. Arbitrarily small boxes prove density, even by uniform density perturbations. Maximal rank is relatively weakly open because the entries are integrals of continuous functions and a nonzero minor stays nonzero nearby. If $r_U = 0$, all priors are witnesses and the grid may consist of its center. The same convention handles $s = 0$.

Finally, take an exact full-support witness already constructed and record all its moments in a basis of V beginning with 1. The nonconstant moment body has affine dimension $\dim V - 1$: an affine relation would be a linear dependence in that basis. Apply Lemma 9.1 to this vector of moments and the prescribed μ_0 . Its finite-atom representation has at most $\dim V$ atoms and preserves every product in EF . The whole pairing matrix, and therefore its exact kernel, is unchanged. $\square$

The polynomial in this criterion is tested as an identity on the full affine quotient, not by an unverified search on a constrained set of priors. Positivity and the annihilation equations have already been resolved by the first condition and the local chart. The argument uses finite-dimensional separation, orthogonal projection, and elementary determinantal geometry; it is not a new generic-matrix rank theorem. Its content here is the exact compatibility with full support, prescribed annihilation, and arbitrarily small positive prior perturbations.

10.1. Forced directions and a positive block classification. A useful preliminary test is the multiplication closure

$$(60) \quad \text{cl}_E(U) = \{f \in F : ef \in W_U \text{ for every } e \in E\}.$$

It records directions forced to vanish by the annihilation equations alone, whereas the pencil in (55) retains all remaining determinantal restrictions.

Proposition 10.3 (The forced kernel). *If $\mathcal{P}_U \neq \emptyset$, then*

$$(61) \quad \bigcap_{\mu \in \mathcal{P}_U} \ker H_\mu = \text{cl}_E(U).$$

Moreover $W_{\text{cl}_E(U)} = W_U$, and this closure is idempotent. In particular, exact realization of U requires $\text{cl}_E(U) = U$. The exact sufficient condition is the maximal-minor test in Theorem 10.2.

Proof. Every member of the right side has all its products in W_U , so is annihilated by every admissible prior. Conversely, let f belong to every kernel and fix $e \in E$. Put $g = ef$ and choose $\mu_* \in \mathcal{P}_U$. Then $\mu_* g = 0$. Orthogonally project g onto W_U in $L^2(\mu_*)$ and put $h = g - \Pi_U g$. It has mean zero, is orthogonal to W_U , and is bounded and continuous. For all sufficiently small real t , the measure $(1 + th)\mu_*$ belongs to $\mathcal{P}_U$. Its integral of g is $t\mu_*(h^2)$, which must vanish. Thus $h = 0$ in $L^2(\mu_*)$, hence identically by continuity and full support. Therefore $g \in W_U$. Since this holds for every e , the claimed equality follows. The inclusions $U \subset \text{cl}_E(U)$ and $E \text{cl}_E(U) \subset W_U$ give equality of the product spaces. Applying the definition again proves idempotence. $\square$

The following family makes the closure and the exact criterion explicit. It includes the failure of an unqualified kernel-realization claim, but also classifies the kernels that are realized in the same positive experiment.

Proposition 10.4 (Exact balanced-block kernels). *Let $n \geq m + 1$, let $J = \{1, \dots, n\}$, and put*

$$X = J \times \{-1, 1\}, \quad E = \text{span}\{e_j : j \in J\}, \quad F = \mathbb{R}1 \oplus \sigma \mathbb{R}_{<m}[z].$$

Here e_j is the indicator of the j th two-point block and a polynomial p represents the function $(j, \sigma) \mapsto p(j)$. Let $0 \neq S \subset \mathbb{R}_{<m}[z]$, $U = \sigma S$, and

$$Z_S = \{j \in J : p(j) = 0 \text{ for every } p \in S\}.$$

Then $\mathcal{P}_U \neq \emptyset$,

$$(62) \quad \text{cl}_E(U) = \sigma\{p \in \mathbb{R}_{<m}[z] : p|_{Z_S} = 0\}, \quad r_U = 1 + |Z_S|.$$

Thus U is exactly realizable if and only if

$$(63) \quad S = \left(\prod_{j \in Z_S} (z - j) \right) \mathbb{R}_{<m-|Z_S|}[z].$$

All these assertions have a positive affine acquisition and a fixed finite physical query realization.

Proof. Write $s_j = \mu(j, +) + \mu(j, -) > 0$ and $d_j = \mu(j, +) - \mu(j, -)$, so $|d_j| < s_j$ and $\sum_j s_j = 1$. The equations $\mu EU = 0$ are $d_j p(j) = 0$ for every $p \in S$. They force $d_j = 0$ on $J \setminus Z_S$ and impose no condition on d_j on Z_S . The uniform probability is an admissible witness. Since a nonzero polynomial of degree less than m has at most $m - 1$ roots, $|Z_S| \leq m - 1$ and $J \setminus Z_S$ is nonempty.

In the basis $1, \sigma, \sigma z, \dots, \sigma z^{m-1}$ the rows of the pairing are

$$(s_j, d_j, d_j j, \dots, d_j j^{m-1}).$$

Any kernel element $c + \sigma p$ has $c = 0$, by a row outside Z_S . For fixed d , its remaining constraint is $p(j) = 0$ at those $j \in Z_S$ where $d_j \neq 0$. Distinct-point polynomial evaluation has rank equal to the number of such points, because that number is less than m . It follows that the maximal pairing rank is $1 + |Z_S|$. It is attained explicitly by $s_j = 1/n$, $d_j = 1/(2n)$ on Z_S , and $d_j = 0$ elsewhere. All atom masses are positive. The kernel of this witness is exactly the first space in (62); every admissible kernel contains that space. Proposition 10.3 identifies it with the closure. Rank-nullity proves the exact-realization criterion, and the factor theorem for polynomials gives (63).

For the physical realization use $n - 1$ acquisition features

$$f_i(j, \sigma) = \frac{e_i(j, \sigma)}{2(n-1)}, \quad L_y(u, j, \sigma) = \frac{1}{2}(1 + y u^t f(j, \sigma)), \quad u \in [-1, 1]^{n-1}.$$

They satisfy $L_y \geq 1/4$. At $u = 0$, the history product and its command derivatives span E , since $e_n = 1 - \sum_{i < n} e_i$. The query probabilities

$$g_\ell(j, \sigma) = \frac{1}{2} + \frac{1}{4}\sigma(j/n)^\ell, \quad 0 \leq \ell < m,$$

belong to $[1/4, 3/4]$, and their span with 1 is F . They can each be executed as a binary query with the stated success probability. The detector, acquisition cube, and query menu are fixed while the prior varies. This proves the physical assertion. $\square$

For $n = 4$, $m = 2$, and $S = \text{span}\{1\}$, the set Z_S is empty. Thus $U = \text{span}\{\sigma\}$ satisfies the annihilation criterion but its forced closure is $\text{span}\{\sigma, j\sigma\}$; the pairing rank on the entire constrained slice is one. The product-evaluation vectors are $\pm \mathbf{e}_j$, so their convex hull contains zero in its interior. There is nevertheless no exact kernel U . In contrast, for each $j_0 \in J$, the line $\text{span}\{(j - j_0)\sigma\}$ satisfies (63) and is an exact kernel of the explicit full-support witness above. The relative-interior condition and the exact-kernel condition therefore make different, verifiable predictions inside the same experiment.

10.2. Return to the acquired prediction state.

Corollary 10.5 (Exact constrained prediction ranks). *Let $P : O \rightarrow C(X; \mathbb{R})$ be a positive history-product map of class C^2 , where u_* is an interior point of the command neighborhood O , and let $\Phi = \text{span}\{1, \phi_1, \dots, \phi_b\}$ be a finite continuous future test space. Apply the preceding construction with*

$$F = \text{span}\{P(u_*, \cdot)\} + \text{im } DP(u_*), \quad E = \Phi.$$

For every $U \subset F$ with $\mathcal{P}_U \neq \emptyset$, the maximal normalized prediction rank at u_ among these priors is $r_U - 1$. It is attained by the arbitrarily small bounded-density perturbations in Theorem 10.2. If the exact-kernel condition holds, the attained rank is $\dim F - \dim U - 1$.*

For each maximizing prior with $r_U > 1$, commands with a density bounded below near u_ and the specified report word give an unconditional subprobability minorization on an $(r_U - 1)$ -dimensional projection of predictions.*

Proof. Let $L : F \rightarrow E^*$ be the moment pairing and let $Y = LP = (Z, Zp)^t$, where $Z = \mu P > 0$. Normalization has kernel $\mathbb{R}Y$ on the image of L . Moreover

$$L(F) = \mathbb{R}Y + L(\text{im } DP(u_*)).$$

Hence the actual command derivative of the normalized map has rank $\text{rank } L - 1$; the evidence column need not itself be a command derivative. The positive evidence also ensures $\text{rank } L \geq 1$. Theorem 10.2 gives the maximal rank and its local attainment, and rank-nullity gives the exact-kernel specialization.

For a maximizing prior choose $r_U - 1$ independent output rows of the command derivative and complete them by orthonormal kernel coordinates. The resulting square map has an invertible derivative. A sufficiently small product box in its image has inverse contained in the command neighborhood, and its inverse Jacobian determinant is bounded below. The joint density of the commands and the specified report word is the command density times $\mu P(u)$, which has a positive lower bound on a smaller neighborhood. Change variables and integrate over the entire kernel-coordinate box. The resulting positive lower density on the output box proves the asserted subprobability minorization. It is a statement about actual histories, not mass assigned to a section. $\square$

For the balanced-block witnesses, normalization yields the prediction rank $|Z_S|$. Their full one-step resolution law is then supplied by Theorem 7.1, using the covariance of the fixed acquisition and query functions. In the general multi-step problem, the present kernel criterion supplies local attainment information; the whole-image cover and stable causal update remain separate requirements. In the monomial experiment these are precisely the other two parts of Theorems 5.3 and 5.4.

11. POSITIVE PRODUCTS AND ATTAINABLE INFORMATION

The lower bound in an information problem requires a measure on histories, not merely a rank calculation in an ambient test space. We isolate the passage from a product tangent to such a measure. This formulation also separates the part of the argument that uses an ordered parameter space from the part that does not.

Let X be a compact space with probability μ . A command ranges over a cube $U \subset \mathbb{R}^b$. Fix one report with likelihood $0 < f(u, x) \leq 1$, twice continuously differentiable in u as a map into $C(X)$. For a tuple $\mathbf{u} = (u_1, \dots, u_n)$ write

$$P_{\mathbf{u}} = \prod_{i=1}^n f(u_i, \cdot), \quad Z_{\mathbf{u}} = \mu P_{\mathbf{u}}, \quad T_{\mathbf{u}} = \text{im } D_{\mathbf{u}} P.$$

For bounded tests $\phi = (\phi_1, \dots, \phi_p)$, define

$$z(\mathbf{u}) = \frac{\mu(P_{\mathbf{u}}\phi)}{Z_{\mathbf{u}}}, \quad LQ = (\mu Q, \mu(Q\phi)) \in \mathbb{R}^{p+1}.$$

The command law below has a density bounded below on a neighborhood of the displayed tuple. The report word is not conditioned away.

Proposition 11.1 (Normalized product criterion). *Suppose $\mathbf{u}_*$ is interior, $P_{\mathbf{u}_*} \in T_{\mathbf{u}_*}$, and $L : T_{\mathbf{u}_*} \rightarrow \mathbb{R}^{p+1}$ is onto. Then $Dz(\mathbf{u}_*)$ is onto $\mathbb{R}^p$. More generally, on any product tangent containing P , normalization decreases the rank of L by exactly one.*

For a family of such maps, suppose the tuples stay a fixed distance from the command boundary, the least row singular value of Dz is at least $\sigma > 0$, and the first two command derivatives are uniformly bounded. Suppose also $f \geq \kappa > 0$ and the joint command density is at least $a > 0$ on a fixed-radius neighborhood of those tuples. There are uniform $\rho, \beta > 0$ such that

$$(64) \quad z_*(Z_{\mathbf{u}} dQ(\mathbf{u})) \geq \beta \text{Unif}(z(\mathbf{u}_*) + [-\rho, \rho]^p).$$

Here the measure on the left is the subprobability law of the commands and the specified report word. The constants do not depend on any singular values of a subsequent observation map.

Proof. Put $v = LP/Z$ and let e_0 select its constant coordinate. Direct differentiation gives the normalized derivative on T as

$$Z^{-1}(LQ - v e_0 LQ).$$

Since $e_0 v = 1$ and $v \in L(T)$, the kernel of $w \mapsto w - v e_0 w$ on $L(T)$ is exactly $\text{span}\{v\}$. Its range has constant coordinate zero. Thus precisely one rank is lost; in particular surjectivity of L gives surjectivity onto the p nonconstant tests.

For the quantitative assertion let $d = nb$, let $J = Dz(\mathbf{u}_*)$, and choose a matrix K whose rows form an orthonormal basis of $\ker J$. Use the square map

$$F(\mathbf{u}) = (z(\mathbf{u}), K(\mathbf{u} - \mathbf{u}_*)).$$

Its central derivative has least singular value at least $\min(\sigma, 1)$: its two row spaces are orthogonal. The second derivative bound gives a ball of uniform radius r , inside the command cube, on which $\|DF - DF(\mathbf{u}_*)\| \leq \min(\sigma, 1)/2$. Writing the inverse equation as

$$u = u_* + DF(u_*)^{-1}\{y - F(u_*) - [F(u) - F(u_*) - DF(u_*)(u - u_*)]\}$$

shows, by contraction on that ball, that a uniform ball about $F(u_*)$ is in its image. Choose a product cube in this output ball. If the forward derivative norm is at most B , then $|\det DF| \leq B^d$; hence the inverse Jacobian on the cube is at least B^{-d} . The subprobability density in command coordinates is at least $a\kappa^n$. Change variables and integrate over the entire kernel-coordinate cube. This gives a positive lower density on the p -cube, and therefore (64). The integration over kernel coordinates is essential; a section by itself has zero ambient command measure. $\square$

11.1. A criterion valid for every full-support prior. For $v_0, \dots, v_p \in C(I)$ on a compact interval I , call this a strict Chebyshev system when its evaluation determinant is nonzero at every ordered $(p+1)$ -tuple of distinct interior points. Reversing one basis vector fixes its sign. Endpoint values are understood by continuity. We use only this finite determinant property, not an unproved assertion about individual derivatives selected from an incomplete Hermite block.

Theorem 11.2 (Positive-history attainment from two test systems). *Let $W \subset C(I)$ be finite dimensional and contain 1. Suppose a positive finite detector spans W , with command cube as in Section 2. At an interior tuple of its failure factors $F_1, \dots, F_n$, suppose the product tangent*

$$T = \sum_{i=1}^n W \prod_{j \neq i} F_j$$

contains a $(p+1)$ -dimensional strict Chebyshev system. Suppose $(1, \phi_1, \dots, \phi_p)$ is another strict Chebyshev system. Then, for every full-support probability μ on I , the posterior map to these p tests is a submersion at that tuple. Independent uniform commands give a positive subprobability minorization of a p -cube.

These assertions are uniform in a compact parameter family provided the chosen command tuples vary continuously and remain uniformly interior, the factors and tests are continuous in uniform norm, the command derivatives through order two are continuous and bounded, and both determinant properties hold at every parameter. One may take any finite union of such continuous families. The assertion also holds uniformly over $c_- \mu_0 \leq \mu \leq c_+ \mu_0$, where $c_- > 0$ and μ_0 has full support.

Proof. Choose a basis $v_0, \dots, v_p$ of the stated subsystem of T , and put $\phi_0 = 1$. Both evaluation determinants have constant nonzero sign on the connected set of ordered interior points. Orient their bases so the signs agree. Determinant expansion and Fubini give

$$\det[\mu(v_i \phi_j)]_{i,j=0}^p = \frac{1}{(p+1)!} \int_{I^{p+1}} \det[v_i(t_k)]_{i,k=0}^p \det[\phi_j(t_k)]_{j,k=0}^p \prod_{k=0}^p \mu(dt_k).$$

The integrand is nonnegative everywhere and strictly positive when the points are distinct and interior. Full support assigns positive mass to a product of disjoint ordered interior intervals. The determinant is therefore nonzero and $L : T \rightarrow \mathbb{R}^{p+1}$ is onto. Also $P \in T$, by varying any one factor along itself. The command-to-factor differential is onto W , so Proposition 11.1 applies to actual commands, not to fictitious factor variations.

For a fixed prior, continuity and compactness give a positive lower bound on the least row singular value. For the dominated family, argue by compactness of the measures as well. Every sequence of probabilities on the compact interval has a weakly convergent subsequence; the two measure inequalities pass to its limit by testing against nonnegative continuous functions. The limiting measure still dominates $c\text{-}\mu_0$ and therefore has full support. Uniform-norm continuity of the factors, tests and their command derivatives makes the normalized Jacobian jointly continuous in the parameter and the weakly varying measure. A sequence of least row singular values tending to zero would thus give a limiting Jacobian of deficient rank, contradicting the determinant argument just proved for every full-support prior. This establishes the uniform bound. The quantitative part of Proposition 11.1 proves the minorization. Finitely many parameter charts or test orders require only finitely many minima. $\square$

In the monomial experiment, Lemma 3.2 supplies the first system and a complete confluent future flag supplies the second. Thus Theorem 11.2 identifies precisely the positive-history statement beyond a spectral comparison. Its hypotheses are verified in Lemma 5.2; neither generic rank nor full support alone is substituted for them. A local minorization still does not prove a global cover or a causal realization. Those are separate assertions in Theorems 5.3 and 5.4.

The circle experiment of Section 13 uses the first proposition but not the ordered-space hypothesis of the theorem. There the required submersion comes from elementary symmetric polynomials, and both acquisition and observation contribute to the disappearing scales.

12. REALIZATION AND DEGENERATION OF ACQUISITION GEOMETRY

The affine classification has two consequences that are not supplied by a rank count. First, every small rectangular covariance matrix can be realized by varying the prior inside one fixed positive experiment. Second, along an analytic arc, the determinantal orders give the entire joint memory–degeneration law. The realization uses uniformly full-support priors; thus these phenomena need not be caused by loss of latent support.

The product characters used in the construction are the usual orthonormal Fourier basis on a finite sign cube [36, Section 2.2]. Their use in a positive density is an elementary specialization of Fourier inversion. The conclusions proved here concern fixed physical acquisition and query functions, and the uniform memory law supplied by Theorem 7.1.

12.1. A local realization theorem.

Theorem 12.1 (Covariance realization in a fixed experiment). *For every $b, q \geq 1$ there is one fixed positive binary affine acquisition, with b command coordinates and a fixed menu of q physical binary queries, having the following property. Every $A \in \mathbb{R}^{q \times b}$ with $\sum_{j,i} |A_{ji}| \leq 1/2$ is realized by a prior μ_A satisfying*

$$(65) \quad \frac{1}{2}\mu_0 \leq \mu_A \leq \frac{3}{2}\mu_0, \quad C_{\mu_A} = \frac{1}{8b\sqrt{q}} A,$$

where μ_0 is one fixed full-support probability and C is the weighted acquisition–query covariance of Theorem 7.1. The map $A \mapsto \mu_A$ is affine and injective.

Consequently every finite-dimensional real analytic matrix germ, after multiplication by a fixed sufficiently small positive scalar, is the covariance germ of priors in this same dominated family. All its matrix-rank strata are realized exactly, not just the maximal rank.

Proof. Take $X = \{-1, 1\}^{b+q}$ with uniform probability μ_0 . Denote its coordinate functions by $\xi_1, \dots, \xi_b, \zeta_1, \dots, \zeta_q$. Set

$$f_i = \frac{\xi_i}{2b}, \quad L_y(u, x) = \frac{1 + y \sum_i u_i f_i(x)}{2}, \quad y \in \{-1, 1\}, \quad u \in [-1, 1]^b.$$

The sum $\sum_i |f_i| = 1/2$, so both report likelihoods are at least $1/4$ on the entire command cube. Query j has success probability $g_j = 1/2 + \zeta_j/4$, and is selected with probability $1/q$ independently after the retained label is formed. These are fixed physical queries with values in $[1/4, 3/4]$. Their weighted functions are $h_j = g_j/\sqrt{q}$.

Define

$$(66) \quad \frac{d\mu_A}{d\mu_0} = 1 + \sum_{j=1}^q \sum_{i=1}^b A_{ji} \xi_i \zeta_j.$$

Each nonconstant coordinate product has mean zero under μ_0 . The density integrates to one, and the entrywise bound gives the two inequalities in (65). In particular every μ_A has full support on the same finite space. Independence and symmetry of the coordinate signs give, exactly,

$$\mu_A \xi_i = \mu_A \zeta_j = 0, \quad \mu_A(\xi_i \zeta_j) = A_{ji}.$$

For the last identity, the average of $\xi_i \zeta_j \xi_{i'} \zeta_{j'}$ is one when both indices coincide and zero otherwise. Thus

$$\text{Cov}_{\mu_A}(h_j, f_i) = \mu_A \left(\frac{\zeta_j}{4\sqrt{q}} \frac{\xi_i}{2b} \right) = \frac{A_{ji}}{8b\sqrt{q}}.$$

This also proves injectivity. An analytic matrix germ has bounded entries on some sufficiently small closed parameter neighborhood. One scalar multiplication places it in the entrywise ball used above. Formula (66) then produces analytic prior weights, and the nonzero scalar in (65) preserves every determinantal rank locus. The assertion concerns matrix germs and their rank loci; it does not posit simultaneous Smith diagonalization in several parameters. $\square$

In this construction the acquisition means are zero for every prior. Thus the projective coordinate of Lemma 7.2 is simply $\theta = yu$. The same command distribution and physical query menu serve all the matrices. The changing geometry in (65) is entirely prior geometry.

12.2. Determinantal orders on an analytic arc. We use the local Smith form over the ring of convergent power series. The connection between its invariant orders and singular values is classical; in particular Kaveh–Makhnatch [35, Theorem 1.1 and Section 6] establish the singular-value logarithmic limit. The elementary version below includes rectangular matrices and identically zero singular directions, and records the two-sided estimates needed for memory budgets varying with the parameter.

Lemma 12.2 (Analytic determinantal orders). *Let $C(t)$ be a real analytic $q \times b$ matrix near $t = 0$, with rank $r > 0$ over the field of convergent Laurent series. There are unique integers $0 \leq \alpha_1 \leq \dots \leq \alpha_r$ and analytic invertible matrices $U(t), V(t)$ near zero such that*

$$(67) \quad U(t)C(t)V(t) = \text{diag}(t^{\alpha_1}, \dots, t^{\alpha_r}, 0),$$

with rectangular zero padding. Both matrices and their inverses are bounded on a sufficiently small closed interval. Put $A_0 = 0$ and $A_\ell = \sum_{j=1}^\ell \alpha_j$. Then

$$(68) \quad A_\ell = \min_{|I|=|J|=\ell} \text{ord}_0 \det C_{I,J}(t), \quad 1 \leq \ell \leq r,$$

where the order of an identically zero minor is $+\infty$. The decreasing singular values satisfy, uniformly for $0 < t \leq t_0$,

$$(69) \quad s_j(C(t)) \asymp t^{\alpha_j} \quad (j \leq r), \quad s_j(C(t)) = 0 \quad (j > r).$$

Proof. A nonzero convergent power series is t^a times an analytic unit. Choose a nonzero entry of smallest order, move it to the upper left corner, and multiply its row by a unit so that this entry becomes t^a . Every other entry is divisible by t^a in the convergent power-series ring. Analytic row operations kill the rest of its column, and analytic column operations then kill the rest of its row. The remaining block still has every entry divisible by t^a : its entries have the form $c_{ij} - c_{i1}c_{1j}/t^a$ after the unit normalization. Repeat on that block. Each operation is invertible analytically at zero. The process stops after r pivots, with a zero remaining block, and the pivot orders are nondecreasing. This proves (67). Nonvanishing determinants of the multipliers at zero give the asserted bounds for them and their inverses.

By Cauchy–Binet, analytic invertible row and column operations preserve the ideal generated by all ℓ -minors; apply the same argument to the inverse operations to get equality of ideals. In a one-variable power-series ring the least order of a finite set of generators is the order of its ideal. For the diagonal form this order is A_ℓ . This proves (68), and its successive differences prove uniqueness of the α_j .

For any matrices B, Q with compatible sizes and Q invertible, the variational characterization of singular values gives $s_j(BQ) \leq \|Q\|s_j(B)$ and $s_j(BQ) \geq s_j(B)/\|Q^{-1}\|$; the analogous inequalities hold for left multiplication. Apply these four inequalities to (67) and its inverse. For $0 < t \leq \min(t_0, 1)$ the diagonal entries are already in decreasing order. The multiplier bounds give (69), including exact zeros. No inverse power of a vanishing diagonal entry occurs in the comparison constants. $\square$

Theorem 12.3 (The joint memory–degeneration law). *Fix a positive affine acquisition and a physical query menu as in Theorem 7.1. Let μ_t be priors whose weighted covariance matrix $C(t)$ is real analytic near zero. If its generic rank is $r > 0$, use the integers A_ℓ from Lemma 12.2. For every integer $M \geq 1$ and $0 < t \leq t_0$, both checkpoint criteria satisfy*

$$(70) \quad R_{M,1}^{\text{av}}(t) \asymp R_{M,1}^{\text{max}}(t) \asymp \max_{1 \leq \ell \leq r} t^{2A_\ell/\ell} M^{-2/\ell}.$$

The constants are independent of M, t ; they may depend on the fixed analytic germ and the physical bounds. At $t = 0$, the same expression holds by continuity with $0^0 = 1$: only the indices for which $A_\ell = 0$ remain. If all $\alpha_j > 0$, the endpoint risk is zero. If C vanishes identically near zero, all these risks are zero there.

Proof. For the stated physical experiment, the affine theorem gives uniform two-sided comparisons with

$$\max_{1 \leq \ell \leq \min(b,q)} \left(\frac{\prod_{j=1}^\ell s_j(C(t))}{M} \right)^{2/\ell}.$$

Its constants do not depend on the prior or on a nonzero singular value. Products with $\ell > r$ vanish; the other products are comparable to t^{A_ℓ} by Lemma 12.2. There are only finitely many indices, so the constants can be chosen jointly for all of them and all integer budgets. This proves (70). At zero, the analytic invertible multipliers in (67) show that the rank is exactly the number of zero orders. Apply the affine theorem at that prior itself. If there are no zero

orders, all prediction probabilities are constant after acquisition and one label is exact. The same argument applies to an identically zero covariance. The endpoint claim does not use a limiting interchange of a minimization over encoders with $t \rightarrow 0$. $\square$

Corollary 12.4 (Phase exponents, their inverse, and realization). *Under Theorem 12.3, put $M_\gamma(t) = \lceil t^{-\gamma} \rceil$ for $\gamma \geq 0$ and $0 < t < 1$. For either checkpoint criterion,*

$$(71) \quad \lim_{t \downarrow 0} \frac{\log R_{M_\gamma(t),1}(t)}{\log t} = \min_{1 \leq \ell \leq r} \frac{2(A_\ell + \gamma)}{\ell}.$$

The successive transition exponents are

$$(72) \quad \beta_\ell = \ell \alpha_{\ell+1} - A_\ell, \quad 1 \leq \ell < r.$$

They are nonnegative and nondecreasing. On $\beta_{\ell-1} \leq \gamma \leq \beta_\ell$ the ℓ th branch is minimal, with $\beta_0 = 0$ and $\beta_r = +\infty$; repeated transition values give intervals of length zero.

For the full checkpoint curve $R_t(M)$ and $1 \leq \ell \leq r$,

$$(73) \quad \lim_{t \downarrow 0} \frac{\log \mathcal{I}_\ell(R_t)}{\log t} = A_\ell, \quad \mathcal{I}_\ell(R) = \inf_{M \in \mathbb{N}, M \geq 1} MR(M)^{\ell/2}.$$

For $\ell > r$ the transform is zero. Thus its orders determine the α_j by successive differences. Every nondecreasing list of nonnegative integers of length $r \leq \min(b, q)$ occurs in (71) inside the single fixed experiment of Theorem 12.1, with the same two-sided density envelope.

Proof. The integer ceiling satisfies $t^{-\gamma} \leq M_\gamma(t) \leq 2t^{-\gamma}$. Therefore (70) is comparable to the maximum of the finitely many powers $t^{2(A_\ell + \gamma)/\ell}$. The smallest exponent gives this maximum up to a factor at most r . Dividing its logarithm by $\log t$, whose magnitude tends to infinity, proves (71); all fixed comparison constants disappear. Adjacent affine functions of γ intersect at (72). These values are nonnegative since $A_\ell \leq \ell \alpha_{\ell+1}$, and

$$\beta_{\ell+1} - \beta_\ell = (\ell + 1)(\alpha_{\ell+2} - \alpha_{\ell+1}) \geq 0.$$

Between two successive transitions, comparisons with all earlier branches follow by chaining adjacent comparisons on their right, and comparisons with later branches by chaining them on their left. This proves the phase assertion, including ties.

Apply Corollary 7.4, equivalently the integer-envelope argument in Theorem 14.2, to each full curve. The uniform affine constants give $\mathcal{I}_\ell(R_t) \asymp \prod_{j \leq \ell} s_j(C(t)) \asymp t^{A_\ell}$ for $\ell \leq r$, and zero for larger ℓ . This proves (73). For realization choose $A(t)$ to be a rectangular diagonal matrix with entries $\varepsilon t^{\alpha_1}, \dots, \varepsilon t^{\alpha_r}$, and all other entries zero. Taking $0 < \varepsilon \leq 1/(2r)$ on $|t| \leq 1$ meets the entrywise bound in Theorem 12.1. Its nonzero constant covariance factor does not change any minor order. That theorem supplies fixed physical likelihoods, fixed query weights, and priors between $\mu_0/2$ and $3\mu_0/2$, completing the realization. $\square$

12.3. An interior two-scale degeneration. For instance, in the construction with $b = q = 2$, take

$$A(t) = \frac{1}{16} \begin{pmatrix} t & t^2 \\ t^2 & t^3 + t^5 \end{pmatrix}, \quad 0 \leq t \leq 1.$$

Its entrywise norm is at most $5/16$, its first minor order is one, and its determinant is $t^6/256$. Thus $\alpha = (1, 5)$ and

$$R_{M,1}(t) \asymp \max\{t^2 M^{-2}, t^6 M^{-1}\}.$$

The changeover occurs at $M \asymp t^{-4}$ although every prior remains uniformly equivalent to the same full-support probability. This is an actual physical family, not merely a matrix spectrum postulated after the risk law. More generally the realization theorem and (71) classify all the permitted one-parameter orders in arbitrary finite covariance dimensions. The algebraic

normal form is classical; the conclusion here couples it to uniform attainable mass, whole-image approximation, and the physical memory resource. It is a one-acquisition-step conclusion. The distinct multi-step collision and circular theorems, with their causal recurrences, retain their full force and their separate hypotheses.

The inverse in (73) uses the entire unbounded integer-budget curve at each $t > 0$. It does not infer orders from finitely many noisy errors, from a maximum over checkpoints, or from a risk with a positive uncertainty floor. Nor does the analytic-arc theorem assert a simultaneous monomial normal form in several parameters.

13. A CIRCULAR EXPERIMENT WITH VANISHING CONTRAST

We now give a second application in which the one-step detector itself loses contrast. The hidden parameter lies on a circle, the posterior pairing is Fourier rather than ordered Chebyshev, and acquisition and future observation each contribute a small factor. Consequently its resolution scales are not obtained by attaching a new observation matrix to a fixed, nondegenerate attainable posterior cube.

13.1. The physical experiment. Write $w = e^{i\theta}$, $0 \leq \theta < 2\pi$, and take normalized Haar measure $d\theta/(2\pi)$ as prior. Fix $0 < \eta < 1/2$. The known contrast is $0 \leq \tau \leq 1/2$. Four detector cells are

$$(74) \quad k_{1,\pm}(\theta) = \frac{1}{4}(1 \pm \tau \cos \theta), \quad k_{2,\pm}(\theta) = \frac{1}{4}(1 \pm \tau \sin \theta).$$

The controller specifies four real acceptance probabilities varying continuously over $[\eta, 1 - \eta]^4$. Thus the command alphabet is continuous, although the detector and report alphabets are finite. Reports and failures are as in (1). All five report likelihoods are bounded below by $\eta/8$. The prior and calibration in this section are known exactly; no prior-uncertainty assertion is included in the result.

Every report factor has a unique representation, for $\tau > 0$,

$$(75) \quad f(\theta) = a\{1 + \tau(zw + \bar{z}w^{-1})\}, \quad a > 0, \quad |z| \leq \frac{1}{2}.$$

The coefficients a, z extend continuously to $\tau = 0$ using the specified command and report. For accepted cells, z is respectively $1/2, -1/2, -i/2, i/2$. For failures, with rejection probabilities u ,

$$(76) \quad a = \frac{1}{4} \sum_j u_j, \quad z = \frac{u_{1,+} - u_{1,-} - i(u_{2,+} - u_{2,-})}{8a}.$$

The bound on $|z|$ follows by the triangle inequality and $u_j \geq 0$. Near the central command, z ranges over a fixed complex neighborhood of zero, independently of τ . More explicitly, the rejection vector

$$(77) \quad (u_{1,+}, u_{1,-}, u_{2,+}, u_{2,-}) = \left(\frac{1}{2} + 2\Re z, \frac{1}{2} - 2\Re z, \frac{1}{2} - 2\Im z, \frac{1}{2} + 2\Im z\right)$$

realizes $a = 1/2$ whenever all entries are in the command cube. Thus the construction uses a fixed finite interface, not a gate that reads the hidden angle.

Fix $0 < \rho < (1/2 - \eta)/8$. For an m -trial query put $L_m = 2m + 1$ and $\varphi_l = 2\pi l/L_m$, $0 \leq l < L_m$. Query l repeats the attainable failure factor with $a = 1/2$, $z = \rho e^{-i\varphi_l}$ exactly m times, and asks for the probability of all failures. Only the selected query is executed; it is selected uniformly and independently after the memory index has been formed. Independent uniform acquisition commands define the unconditional experiment. Worst-history and causal losses have the same meaning as in Definition 2.4, with this query menu.

Theorem 13.1 (Harmonic resolution from acquisition and observation). *Fix $N \geq 2$ and the interface above. There is $\tau_* > 0$, depending only on N, η, ρ , such that the following holds uniformly for $0 \leq \tau \leq \tau_*$ and all integers $M \geq 1$. At a checkpoint $n + m \leq N$, $n, m \geq 1$, put $k = \min(n, m)$ and*

$$(78) \quad \mathcal{H}_{n,m}(M, \tau) = \max_{1 \leq j \leq k} \tau^{2(j+1)} M^{-1/j}.$$

Then

$$(79) \quad c\mathcal{H}_{n,m} \leq \overline{R}_{M;n,m}^\tau \leq R_{M;n,m}^{\tau, \max} \leq C\mathcal{H}_{n,m}.$$

For horizon N , one deterministic, stage-compatible M -state transducer has worst-history loss at every checkpoint bounded by

$$(80) \quad C \max_{1 \leq j \leq \lfloor N/2 \rfloor} \tau^{2(j+1)} M^{-1/j}.$$

Every such transducer has maximum unconditional checkpoint loss at least a fixed positive multiple of the same expression. The lower laws permit independent coding randomization. For $\tau > 0$ in this interval the minimal continuous sufficient checkpoint dimension is $2 \min(n, m)$; for $\tau = 0$ it is zero.

Information convention and comparison. The Fourier representation used below is the established Bayesian representation discussed in Section L.3. The resource restriction is instead a persistent index in an alphabet of size at most M . The prior, contrast, horizon, and interface may enter a read-only program; a transition receives only its preceding index and the current command and report. In particular, an exact Fourier vector is not charged as one label. The causal assertion of Theorem 13.1 uses this convention, at fixed N and known contrast; it is not an effective numerical compiler.

For orientation, consider one past trial and one future trial. If the past factor is $a\{1 + \tau(zw + \bar{z}w^{-1})\}$, Haar normalization gives $c_1 = \tau z$. With the query parameter ρ and phases φ_l defined above, direct multiplication gives

$$q_l(z) = \frac{1}{2} + \rho\tau^2 \Re(z e^{i\varphi_l}), \quad \frac{1}{3} \sum_{l=0}^2 |q_l(z) - q_l(z')|^2 = \frac{1}{2} \rho^2 \tau^4 |z - z'|^2.$$

The two contrast factors are already visible in this elementary case. The higher-dimensional result requires more: the normalized flags must be excited by a set of acquisition histories with uniformly positive subprobability, the global image must admit a matching cover, and the successive covers must be compatible with the weighted Bayesian update. The next two lemmas and the proof of the theorem establish these properties with the actual evidence retained. This computation is an illustration of the theorem, not a separate improvement of its law.

13.2. Acquisition: a symmetric-polynomial submersion. Remove the positive scalar factors a_i , which cancel from the posterior, and expand the Laurent polynomial

$$(81) \quad P(w) = \prod_{i=1}^n \{1 + \tau(z_i w + \bar{z}_i w^{-1})\} = \sum_{j=-n}^n p_j w^j, \quad p_{-j} = \bar{p}_j.$$

Haar integration extracts p_0 , and $p_0 \geq (1 - \tau)^n$. For $j \geq 0$, each term contributing to p_j chooses $j + h$ positive and h negative powers from distinct factors. Hence

$$(82) \quad p_j = \tau^j P_{n,j}(z, \bar{z}; \tau^2), \quad P_{n,j}(z, \bar{z}; 0) = e_j(z_1, \dots, z_n),$$

where e_j is the elementary symmetric polynomial and $e_0 = 1$. This is an exact polynomial identity. Define

$$S_j = \frac{P_{n,j}(z, \bar{z}; \tau^2)}{P_{n,0}(z, \bar{z}; \tau^2)}, \quad c_j = \frac{p_j}{p_0} = \tau^j S_j, \quad Y_j = \tau^j c_j = \tau^{2j} S_j.$$

Set these coordinates to zero for $j > n$. The c_j are the posterior Fourier coefficients, or equivalently the posterior expectations of w^{-j} . The S_j and their first two command derivatives are uniformly bounded on the command cube and $0 \leq \tau \leq 1/2$, for $n \leq N$. Indeed the numerator polynomials have finitely many bounded coefficients, while their common denominator is at least 2^{-N} ; also $a \geq \eta$ for failures. These observations justify the assertion even at zero contrast without division by τ .

Lemma 13.2 (Attainment of the normalized harmonic flag). *There are $\tau_* > 0$ and, for every $n \leq N$, one interior all-failure command tuple independent of τ , at which $(S_1, \dots, S_k) : \mathbb{R}^{4n} \rightarrow \mathbb{C}^k$ has real rank $2k$ for all $k \leq n$ and $0 \leq \tau \leq \tau_*$. The least row singular values are uniformly positive. Under independent uniform commands the all-failure subprobability law of these $2k$ real coordinates minorizes uniform probability on a translated cube of fixed side and fixed mass.*

Proof. Choose distinct small positive real numbers $\zeta_1, \dots, \zeta_N$ within the interior neighborhood in (77). Use the first n as the z_i at length n . At $\tau = 0$ the map $(S_1, \dots, S_n)$ is $(e_1, \dots, e_n)$. Its complex differential is invertible at a tuple of distinct numbers. To verify this directly, if all differential coefficients of $\prod_i (\lambda + z_i)$ vanish, then

$$\sum_i \dot{z}_i \prod_{h \neq i} (\lambda + \zeta_h) = 0.$$

Evaluating at $\lambda = -\zeta_i$ gives $\dot{z}_i = 0$ for every i . The corresponding real differential on $\mathbb{C}^n$ is therefore invertible. Every initial collection of k complex rows has real rank $2k$. The command map onto z is a submersion at the implementing tuples by (76)–(77).

Continuity in $s = \tau^2$ now preserves all these lower singular-value bounds on one interval $[0, \tau_*]$: there are only finitely many $n, k \leq N$. For precision about this choice, let σ_0 be the minimum of the finitely many positive central row singular values at zero. The rational formulas give a finite C_* with $\|DS(\tau) - DS(0)\| \leq C_* \tau^2$ at the chosen tuples. One can take the minimum of $1/2$ and $\sqrt{\sigma_0 / (2 \max\{C_*, 1\})}$. No assertion of a large-contrast uniform interval is needed for this small-contrast theorem. The first two derivative bounds established above, the fixed interior margin, and $f \geq \eta/8$ verify the quantitative hypotheses of Proposition 11.1. Its change-of-variables argument applies directly to the maps S , including their continuous extension at zero, and proves the stated minorization. The actual failure evidence is $p_0 \prod_i a_i$; it has not been replaced by one. $\square$

13.3. Observation: a finite Fourier menu.

Lemma 13.3 (The physical query metric). *For histories h, h' at the same checkpoint, let $q_l(h)$ denote the probability of query l . Then*

$$(83) \quad \frac{1}{2m+1} \sum_{l=0}^{2m} |q_l(h) - q_l(h')|^2 = 2^{1-2m} \sum_{j=1}^m B_{m,j}(\tau^2)^2 |Y_j(h) - Y_j(h')|^2,$$

where

$$(84) \quad B_{m,j}(s) = \sum_{h=0}^{\lfloor (m-j)/2 \rfloor} \frac{m! \rho^{j+2h} s^h}{(j+h)! h! (m-j-2h)!}, \quad 0 \leq j \leq m.$$

For $j \geq 1$ these factors are bounded above and away from zero uniformly on $0 \leq \tau \leq 1/2$. Thus the physical loss metric is uniformly equivalent to Euclidean distance in $(Y_1, \dots, Y_m)$.

Proof. Expanding the repeated query factor, its coefficient of w^j is $2^{-m}\tau^j B_{m,j}(\tau^2)e^{-ij\varphi_l}$. Multiplication by the posterior density and Haar integration therefore give

$$q_l(h) = 2^{-m}B_{m,0}(\tau^2) + 2^{1-m}\Re \sum_{j=1}^m B_{m,j}(\tau^2)Y_j(h)e^{ij\varphi_l}.$$

The discrete identity $(2m+1)^{-1}\sum_l e^{ia\varphi_l} = 0$ for $0 < |a| \leq 2m$ makes the distinct sine and cosine columns orthogonal, with squared norm $1/2$. This proves (83). All coefficients in (84) are nonnegative, and its constant term is $\binom{m}{j}\rho^j > 0$. Compactness gives the other bound. Orthogonal projection of an arbitrary prediction vector onto this affine Fourier span cannot increase loss, so the same metric comparison is available for lower bounds with unrestricted decoders. $\square$

13.4. Proof of the resolution and causal laws.

Proof of Theorem 13.1. At a checkpoint only the first $k = \min(n, m)$ complex coordinates Y_j can be nonzero. Their real and imaginary parts lie in a rectangle with paired half-widths

$$C_N\tau^2, C_N\tau^2, C_N\tau^4, C_N\tau^4, \dots, C_N\tau^{2k}, C_N\tau^{2k}.$$

For ordered half-widths $a_1 \geq \dots \geq a_d$, a rectangular grid has covering number at most $C_d \sum_{l=0}^d r^{-l} \prod_{i \leq l} a_i$ at radius r . Indeed the product of the d one-dimensional counts expands into subset products, each bounded by the corresponding initial product. Put $r = A \max_{l \geq 1} (M^{-1} \prod_{i \leq l} a_i)^{1/l}$ with a fixed large A . For $M \geq 2C_d$, choose A so the nonconstant terms are at most $M/2$; the constant term is also at most $M/2$. For the finitely many smaller budgets one point and a larger fixed constant suffice, since $r \geq a_1/M$. Replace every grid point whose ball meets the reachable set by a point of that set, doubling the radius. This enforces at most M reachable representatives for every integer budget, not only asymptotically.

For the lower bound, Lemma 13.2 minorizes a $2k$ -cube in the normalized real coordinates. Its first l -coordinate projection, scaled by the first l half-widths without the harmless constant C_N , is a rectangle with volume comparable to $\prod_{i \leq l} a_i$. The union of M radius- b balls covers at most $C_l M b^l / \prod_{i \leq l} a_i$ of this minorized probability. Taking b a small multiple of $(\prod_{i \leq l} a_i / M)^{1/l}$ leaves a fixed positive mass and gives a squared-error lower bound of order b^2 . Lemma 13.3 transfers it to the actual query loss. Conditional averaging removes decoder randomization; randomized label assignment cannot beat the nearest of its M centers. Independent public coding randomness may be fixed and then averaged. The minorization retains the acquisition probability throughout.

For $l = 2j$ the squared radius just obtained is $\tau^{2(j+1)}M^{-1/j}$. No odd index increases their maximum. For $l = 1$ its term is at most the $l = 2$ term since $M \geq 1$. For $l = 2j - 1$, $j \geq 2$, its logarithm is a convex combination of the logarithms for $l = 2j - 2$ and $l = 2j$; this follows by inserting the equal last two half-widths into $2l^{-1}(\sum_{i \leq l} \log a_i - \log M)$. This proves (79) for $\tau > 0$. At $\tau = 0$ every report is independent of the hidden angle and every query probability is known; both sides of the asserted law are zero.

It remains to construct one causal filter. Use the raw weighted Fourier moments Y_j , not S_j . Extend by $Y_0 = 1$ and $Y_{-j} = \bar{Y}_j$. Multiplication by a new factor (75) gives, for $j \geq 1$,

$$(85) \quad Y'_j = \frac{Y_j + \tau^2 z Y_{j-1} + \bar{z} Y_{j+1}}{1 + z \bar{Y}_1 + \bar{z} Y_1}.$$

Only $j \leq m - 1$ is needed after the report. The denominator is the posterior expectation of f/a and is at least $1 - \tau \geq 1/2$; this also holds on the segment between any two reachable posterior states. The numerator and denominator have uniformly bounded coefficients and states have bounded norm. Direct subtraction of the two fractions therefore gives a Lipschitz constant independent of τ . In particular no reciprocal contrast occurs.

At each stage take the reachable representative codebook constructed above. Update its current representative by (85) with the current command and report, then select a nearest representative in the next codebook with a fixed tie rule. An updated representative is reachable. If e_n is its error from the true state, then $e_{n+1} \leq L e_n + r_{n+1}$, $e_0 = 0$, where r_{n+1} is the next covering radius. Finite iteration gives $e_n \leq C_N \max_s r_s$ and the asserted maximum-checkpoint upper law by (83). Missing harmonics above degree n are exactly zero before the changeover; after it, the shrinking future state contains every index used in (85). The stage is public, while the persistent state is one of M labels. The lower bounds use prefixes of the same independent uniform acquisition law and so apply to this same causal resource model. The maximum of $\min(n, N - n)$ is $\lfloor N/2 \rfloor$, proving (80) with matching lower order.

Finally, for positive contrast the local submersion and the invertible physical Fourier menu give an open $2k$ -dimensional prediction patch. A continuous sufficient encoding restricted to a local section is injective, so invariance of domain forbids fewer than $2k$ real coordinates. Every m -trial continuation likelihood is a trigonometric polynomial of order at most m , so these Fourier moments determine all such future laws, not merely the displayed menu. The Fourier state with its identically zero high harmonics uses exactly $2k$, and (85) realizes it causally. This proves the exact dimension assertion. $\square$

Corollary 13.4 (Resolved harmonic phases). *For $0 < \tau < \min(\tau_*, 1)$, adjacent terms $j, j + 1$ in the circular causal profile exchange dominance at*

$$M = \tau^{-2j(j+1)}.$$

Every harmonic pair occurs in this ordered sequence. For $N = 4$ and $N = 6$ the respective laws are

$$\max\{\tau^4 M^{-1}, \tau^6 M^{-1/2}\}, \quad \max\{\tau^4 M^{-1}, \tau^6 M^{-1/2}, \tau^8 M^{-1/3}\}.$$

Proof. Equating the two displayed adjacent powers yields the asserted value of M . These values increase strictly with j for $\tau < 1$. Moreover the logarithms of the terms are lines in $\log M$ with strictly increasing slopes $-1/j$. Each line is maximal between its crossings with its two neighbors; successive comparison also excludes every nonadjacent line on that interval. The two examples substitute $\lfloor N/2 \rfloor = 2, 3$. $\square$

The exact dimension agrees with that of a saturated three-dimensional polynomial detector, but the resolved law does not follow from that agreement. Here each j th harmonic is $O(\tau^j)$ in the acquired posterior and another $O(\tau^j)$ in the future query. The physical width is therefore τ^{2j} . Keeping only the future attenuation would already give the wrong τ power at $n = m = 1$. Haar orthogonality and the symmetric-polynomial submersion are explicit hypotheses and mechanisms of this application; an arbitrary prior on the circle is not silently substituted. The original full-support-prior monomial theorem retains its separate scope.

14. RECONSTRUCTION OF THE ATTAINABLE VOLUME PROFILE

The preceding theorems determine a prediction law from an attainable flag. There is a converse: the entire integer-budget regret curve recovers every product in that flag, up to uniform constants. Thus the truncated volume profile is also necessary for uniform comparison of prediction laws. We first prove the elementary envelope identity on which this converse rests.

Its role is to extract the invariant from the two experiment-specific classifications; it is not a replacement for their attainment, global covering, or causal arguments.

14.1. An envelope identity with integer budgets. For an ordered list $s_1 \geq \dots \geq s_p \geq 0$, put

$$V_0(s) = 1, \quad V_\ell(s) = \prod_{i=1}^{\ell} s_i, \quad e_s(b) = \max_{1 \leq j \leq p} \left(\frac{V_j(s)}{b} \right)^{1/j} \quad (b \geq 1).$$

Zero entries and their products are retained. The extension to real b is an analytic device; an actual state alphabet always has an integer cardinality.

Lemma 14.1 (Recovery of all initial products). *For every $1 \leq \ell \leq p$,*

$$(86) \quad V_\ell(s) = \inf_{b \geq 1} b e_s(b)^\ell.$$

For integer budgets one has

$$(87) \quad V_\ell(s) \leq \inf_{M \in \mathbb{N}, M \geq 1} M e_s(M)^\ell \leq 2V_\ell(s).$$

Both assertions include zero products. If $s_\ell > 0$, equality in (86) holds at $b_\ell = V_\ell(s)/s_\ell^\ell \geq 1$.

Proof. The ℓ th term in the maximum gives $b e_s(b)^\ell \geq V_\ell(s)$. Suppose first that $s_\ell > 0$. For $j < \ell$, monotonicity of the list gives

$$\frac{V_\ell}{V_j} = \prod_{i=j+1}^{\ell} s_i \geq s_\ell^{\ell-j},$$

and hence $V_j/b_\ell \leq s_\ell^j$. For $j > \ell$, the reverse product estimate $V_j/V_\ell \leq s_\ell^{j-\ell}$ gives the same inequality. The ℓ th term is equal to s_ℓ , so $e_s(b_\ell) = s_\ell$. Taking $M_\ell = \lceil b_\ell \rceil$ and using that e_s is nonincreasing, we obtain

$$M_\ell e_s(M_\ell)^\ell \leq \lceil b_\ell \rceil s_\ell^\ell \leq 2b_\ell s_\ell^\ell = 2V_\ell.$$

This proves both assertions for positive products without rounding a real budget down to an inadmissible alphabet size.

Let r be the number of positive entries. If $r = 0$, every quantity in the assertions is zero. If $0 < r < \ell$, then for $b \geq b_r = V_r/s_r^r$ the last positive term is maximal. Indeed, with $t = (V_r/b)^{1/r} \leq s_r$, the inequality $V_r/V_j \geq s_r^{r-j} \geq t^{r-j}$ gives $V_j/b \leq t^j$ for $j < r$. Consequently

$$b e_s(b)^\ell = V_r^{\ell/r} b^{1-\ell/r} \rightarrow 0.$$

The same limit holds along the integers. Since $V_\ell = 0$, this settles every remaining case. $\square$

The equality uses the ordered-product structure. It would be incorrect to apply it to an arbitrary list of coefficients in a maximum of power laws: some such coefficients can lie strictly below the envelope. Here $\log V_\ell$ is concave on the positive indices, and each product has the supporting budget exhibited in the proof. Repeated scales may share that budget; their products are still recovered.

14.2. A complete invariant for uniform regret comparison. For any nonnegative curve $R : \mathbb{N} \rightarrow [0, \infty)$ define

$$(88) \quad \mathcal{I}_0(R) = 1, \quad \mathcal{I}_\ell(R) = \inf_{M \in \mathbb{N}, M \geq 1} M R(M)^{\ell/2} \quad (\ell \geq 1).$$

The infimum refers to the whole optimal-regret curve, not to finitely many observations of an implemented filter. No numerical inversion procedure or finite-budget estimator is implicit in this definition.

Theorem 14.2 (Operational reconstruction and comparison). *Suppose a family of checkpoint regret curves satisfies*

$$(89) \quad ce_s(M)^2 \leq R_s(M) \leq Ce_s(M)^2 \quad (M \in \mathbb{N}, M \geq 1)$$

with $c, C > 0$ uniform in the family and ordered lists of a fixed maximum length p , padded with zeros when necessary. Then

$$(90) \quad c^{\ell/2} V_\ell(s) \leq \mathcal{I}_\ell(R_s) \leq 2C^{\ell/2} V_\ell(s), \quad 1 \leq \ell \leq p.$$

In particular the positive products, the scale orders of their successive ratios, and their uniform orders of degeneration are invariants of the regret curve up to comparison constants.

For two such families, their curves are uniformly comparable for all integer $M \geq 1$ if and only if all their corresponding products V_ℓ are uniformly comparable, including their zero sets. The comparison concerns finite-resolution prediction laws, not an isomorphism of experiments or a reconstruction of their hidden parameters.

Proof. Raise (89) to the power $\ell/2$, multiply by M , and take the infimum. Lemma 14.1 gives (90). For a positive product the ratio $\mathcal{I}_\ell/\mathcal{I}_{\ell-1}$ is comparable to $V_\ell/V_{\ell-1} = s_\ell$, with $\mathcal{I}_0 = V_0 = 1$. The first zero product and all its successors are detected as exact zeros of the transform; no quotient $0/0$ is formed.

Suppose $A^{-1}R_s(M) \leq \tilde{R}_t(M) \leq AR_s(M)$ uniformly. The transform preserves these inequalities with constants $A^{\ell/2}$. Applying (90) to both curves proves uniform comparison of their products. Conversely, if $B^{-1}V_\ell(s) \leq V_\ell(t) \leq BV_\ell(s)$ for all ℓ , with $B \geq 1$, then

$$B^{-2}e_s(M)^2 \leq e_t(M)^2 \leq B^2e_s(M)^2.$$

Indeed each branch has comparison factor $B^{2/\ell} \leq B^2$; taking maxima preserves the bounds. The two instances of (89) give the asserted comparison of curves. Every constant depends only on the stated uniform constants and the fixed maximum length, not on a positive lower bound for any individual scale. $\square$

For positive rank r , the proof of the lemma also gives

$$\lim_{M \rightarrow \infty} \frac{\log R_s(M)}{\log M} = -\frac{2}{r}.$$

Thus the usual dimension slope is one consequence of the curve; the products recovered in (90) give its additional finite-resolution information. Uniformly equivalent norms on the same prediction vector change every loss, and hence every optimal curve, by uniform constants. The recovered product orders are therefore independent of a uniformly conditioned choice of prediction coordinates. Parameter-dependent changes with unbounded condition number are not covered by this observation.

14.3. The collision and contrast invariants. For a monomial checkpoint define the past-truncated products

$$\hat{\mathcal{V}}_{n,m;\ell}(a) = \begin{cases} \mathcal{V}_{m,\ell}(a), & 1 \leq \ell \leq p_{n,m}, \\ 0, & \ell > p_{n,m}. \end{cases}$$

The truncation is part of the invariant, not a convention for suppressing unattained directions.

Corollary 14.3 (Recovery at every additive collision). *Under the hypotheses of Theorem 5.3, let $R^a(M)$ be either $\bar{R}_{M;n,m}^a$ or $R_{M;n,m}^{a,\max}$. For each fixed finite maximum index,*

$$(91) \quad \mathcal{I}_\ell(R^a) \asymp \hat{\mathcal{V}}_{n,m;\ell}(a), \quad \ell \geq 1,$$

uniformly on K , including all additive collision strata. Consequently these truncated volumes are a complete invariant for uniform comparison of the corresponding checkpoint regret curves.

Proof. Take $s_j = d_j$ for $1 \leq j \leq p_{n,m}$ from Lemma 5.1 and pad by zeros. The same lemma gives $V_\ell(s) \leq \mathcal{V}_{m,\ell} \leq \ell! V_\ell(s)$ through the truncation. These factors are uniform at fixed horizon, including when both sides vanish. Theorem 5.3 therefore verifies (89) for either regret curve. Theorem 14.2 proves the assertion, including zero products beyond the truncation. $\square$

The conclusion recovers the observable volume orders, not the complete arrangement of the future exponents. In particular it does not recover directions inaccessible to this past. The latter may still occur in the prior-ambiguity body of Section 15, whose information convention is different.

Corollary 14.4 (Recovery of paired acquisition–observation scales). *For either checkpoint regret curve in Theorem 13.1, with $k = \min(n, m)$, uniformly on its fixed-horizon contrast interval,*

$$(92) \quad \begin{aligned} \mathcal{I}_{2j-1}(R^\tau) &\asymp \tau^{2j^2}, \\ \mathcal{I}_{2j}(R^\tau) &\asymp \tau^{2j(j+1)}, \quad 1 \leq j \leq k. \end{aligned}$$

The transforms vanish at every higher index. For $\tau > 0$ the successive nonzero ratios recover both copies of each scale τ^{2j} ; at $\tau = 0$ all positive-order transforms vanish.

Proof. Use the real list $(\tau^2, \tau^2, \tau^4, \tau^4, \dots, \tau^{2k}, \tau^{2k})$. Its products of lengths $2j - 1$ and $2j$ have respectively the exponents $2j^2$ and $2j(j + 1)$. The proof of Theorem 13.1 identifies its full envelope with $\mathcal{H}_{n,m}$, including the odd products even though they add no larger branch. Theorem 14.2 now applies. Ratios of consecutive products have exponent $2j$ on both members of the j th pair. Zero contrast and indices above $2k$ are the zero-product cases of Lemma 14.1. $\square$

These reconstructions concern checkpoint curves with the experiment known. Taking the maximum over checkpoints produces the causal laws already proved, but need not retain the individual checkpoint curves. Nor is a curve with a positive common-name input-error floor an instance of (89). The distinct uncertainty law in Theorem 16.1 and all of its hypotheses remain in force.

15. THE GEOMETRY OF UNRESOLVED PRIOR INFORMATION

The acquired-history image and uncertainty about the prior are different geometric objects. The first has its dimension bounded by $n(r - 1)$. The second can occupy every future direction, even after a fixed history. We describe the second object here. This also identifies the directions which are suppressed when an uncertainty problem is summarized by a single squared-error floor.

15.1. The bounded-test principle. The finite-radius mechanism in this section is independent of monomials. We state it before its confluent specialization. This formulation and the normalization below were isolated in the preceding mathematical audit [27]; we retain that attribution. The result is a bounded right-inverse principle, not a second positive-history attainment theorem.

Proposition 15.1 (Relative prior perturbations under an observation map). *Let μ be a probability, $0 < c \leq \ell \leq C < \infty$, and $\nu = \ell\mu/(\mu\ell)$. Let $\phi = (\phi_1, \dots, \phi_q)$ be bounded real tests with $\|\phi_j\|_\infty \leq B$ and $\text{Cov}_\nu(\phi, \phi) \geq \lambda I_q$, $\lambda > 0$. For $0 < \varepsilon \leq 1/2$, vary μ' over*

$$\{(1 + u)\mu : \mu u = 0, \|u\|_\infty \leq \varepsilon\}, \quad \nu' = \ell\mu'/(\mu'\ell).$$

For every linear map $A : \mathbb{R}^q \rightarrow \mathbb{R}^d$ the displacement set satisfies

$$c_0 \varepsilon A[-1, 1]^q \subset \{A(\nu' \phi - \nu \phi)\} \subset C_0 \varepsilon A[-1, 1]^q,$$

where $c_0, C_0 > 0$ depend only on q, B, λ , not on the singular values or rank of A .

Proof. For $\mu' = (1 + u)\mu$, the exact identity

$$\nu' \phi - \nu \phi = \frac{\nu \{u(\phi - \nu \phi)\}}{1 + \nu u}$$

bounds each coordinate by $4B\varepsilon$. For the other inclusion put $\Gamma = \text{Cov}_\nu(\phi, \phi)$ and, for $v \in [-1, 1]^q$, set

$$f_v = v^\top \Gamma^{-1}(\phi - \nu \phi), \quad K = \max\{1, 2qB/\lambda\}, \quad s = \varepsilon/(4K).$$

Then $\nu f_v = 0$, $\nu(\phi f_v) = v$, and $\|f_v\|_\infty \leq K$. Define a prior, rather than only a posterior perturbation, by

$$\frac{d\mu'}{d\mu} = \frac{1 + s f_v}{1 + s \mu f_v}.$$

It has total mass one, is positive, and obeys

$$\left\| \frac{d\mu'}{d\mu} - 1 \right\|_\infty \leq \frac{2sK}{1 - sK} = \frac{\varepsilon/2}{1 - \varepsilon/4} \leq \varepsilon.$$

The evidence is $\mu \ell / (1 + s \mu f_v)$, so Bayes normalization gives exactly $\nu' = (1 + s f_v)\nu$ and hence displacement sv . This realizes one common cube, not merely separate directional alternatives. Applying A to both inclusions completes the proof, also when A has a kernel. $\square$

15.2. Complete confluent flags and all width orders. Throughout this section the calibration is known. Write $q = q_m$ and use one of the Leja orders in Lemma 5.1. Put

$$\phi_a = (g_1^a, \dots, g_q^a), \quad D_a = \text{diag}(d_1, \dots, d_q), \quad B_\infty^q = [-1, 1]^q,$$

and let L_a be the matrix in (17). All repeated formal labels are retained. The convention $d_{q+1} = 0$ will be useful. For a probability ν , let

$$X_a(\nu) = (\nu(t^{Hx_i}))_{i=1}^q = L_a D_a \nu \phi_a.$$

The norm of a difference of physical query vectors is equivalent to the Euclidean norm of the corresponding difference of X_a , with the fixed constants already established for G_m .

Fix a full-support prior μ and a history h of length n , where $n + m \leq N$. Its likelihood is denoted by ℓ_h , and its posterior by $\nu_h = \ell_h \mu / \mu \ell_h$. For $0 < \varepsilon \leq 1/2$ set

$$(93) \quad \mathcal{B}_\varepsilon(\mu) = \{(1 + u)\mu : \mu u = 0, \|u\|_{L^\infty(\mu)} \leq \varepsilon\}.$$

Every member is a full-support probability. Let ν'_h denote the posterior obtained from $\mu' \in \mathcal{B}_\varepsilon(\mu)$ using *the same* likelihood, and define the centered ambiguity set

$$(94) \quad \mathcal{A}_h(a, \varepsilon) = \{X_a(\nu'_h) - X_a(\nu_h) : \mu' \in \mathcal{B}_\varepsilon(\mu)\}.$$

This is a set at a specified history; choosing a point in it does not select a different prior after each realized history in a statistical experiment. A separate common-prior decision statement is given below.

Lemma 15.2 (A uniformly bounded dual for the complete test flag). *The covariance matrix*

$$\Gamma_{a,h} = \text{Cov}_{\nu_h}(\phi_a, \phi_a)$$

has least eigenvalue bounded below by a positive constant, uniformly in $a \in K$, histories of length at most $N - m$, and all orders of the formal nodes, including their repeated-node limits. Moreover there is $C_0 \geq 1$ such that, for every $v \in B_\infty^q$, the function

$$(95) \quad f_v(t) = v^\top \Gamma_{a,h}^{-1}(\phi_a(t) - \nu_h \phi_a)$$

satisfies $\nu_h f_v = 0$, $\nu_h(\phi_a f_v) = v$, and $\|f_v\|_\infty \leq C_0$. The constants can be chosen uniformly for centers in any fixed dominated prior family $c_- \mu_0 \leq \mu \leq c_+ \mu_0$ with $c_- > 0$ and full-support μ_0 .

Proof. For every ordered multiset of q positive nodes, the functions $1, g_1^a, \dots, g_q^a$ are linearly independent. Indeed, Lemma 4.1 identifies their span with the constant and complete positive-exponent Hermite blocks. Linear independence of these exponential-polynomial functions on $(0, 1)$ follows from the confluent Chebyshev argument in Lemma 4.3. A nonzero linear combination of the g_j^a cannot therefore be constant on the support of a full-support measure. Consequently its variance is positive, and the covariance matrix under μ_0 is positive definite.

The uniform bounds and continuity in the proof of Lemma 5.2 apply to the full prefix of length q , regardless of the number of distinct node values. They show that this covariance matrix is continuous on K for each fixed permutation. Compactness, and the finite number of permutations, give $\text{Cov}_{\mu_0}(\phi_a, \phi_a) \geq \lambda_0 I_q$ for some $\lambda_0 > 0$. This statement concerns the normalized functions, not physically observable coordinates divided by a zero pivot.

Since $\kappa^n \leq \ell_h \leq 1$, one has $\nu_h \geq \kappa^n c_- \mu_0$. For any $w \in \mathbb{R}^q$, the variational formula for variance gives

$$\inf_b \nu_h((w^\top \phi_a - b)^2) \geq \kappa^n c_- \inf_b \mu_0((w^\top \phi_a - b)^2) \geq \kappa^n c_- \lambda_0 \|w\|^2.$$

For a single fixed prior use $\mu_0 = \mu$ and $c_- = c_+ = 1$. This proves the eigenvalue assertion without any compactness assumption on the family of priors. Formula (95) now has zero mean, and multiplication by the centered test vector gives v . The uniform bound on ϕ_a and the just obtained bound on $\Gamma_{a,h}^{-1}$ give C_0 , after increasing it to at least one. $\square$

Proposition 15.3 (All collision scales of the prior ambiguity body). *There are constants $c, C > 0$, with the uniformity of Lemma 15.2, such that*

$$(96) \quad c\varepsilon L_a D_a B_\infty^q \subset \mathcal{A}_h(a, \varepsilon) \subset C\varepsilon L_a D_a B_\infty^q.$$

If

$$w_k(\mathcal{A}) = \inf_{\dim V \leq k} \sup_{z \in \mathcal{A}} \text{dist}(z, V), \quad 0 \leq k \leq q,$$

then

$$(97) \quad c\varepsilon d_{k+1} \leq w_k(\mathcal{A}_h(a, \varepsilon)) \leq C\varepsilon d_{k+1}.$$

In particular all widths past the number of distinct positive future exponents are zero. No past-dimension truncation occurs in this assertion.

Proof. Write $\nu = \nu_h$. For $\mu' = (1 + u)\mu$ in (93), direct normalization yields

$$\nu' \phi_a - \nu \phi_a = \frac{\nu(u(\phi_a - \nu \phi_a))}{1 + \nu u}.$$

If every g_j^a is bounded by B , the right side has coordinatewise absolute value at most $2B\varepsilon/(1 - \varepsilon) \leq 4B\varepsilon$. Multiplication by $L_a D_a$ proves the upper inclusion.

For the lower inclusion take any $v \in B_\infty^q$, let $f = f_v$ be (95), and put $s = \varepsilon/(4C_0)$. Define

$$(98) \quad d\mu' = \frac{1 + sf}{1 + s\mu f} d\mu.$$

This is a positive probability, and

$$\left\| \frac{d\mu'}{d\mu} - 1 \right\|_{\infty} \leq \frac{2sC_0}{1-sC_0} = \frac{\varepsilon/2}{1-\varepsilon/4} \leq \varepsilon.$$

Because $\nu f = 0$, its posterior at the same history is exactly $d\nu' = (1 + sf) d\nu$. Hence $\nu' \phi_a - \nu \phi_a = sv$, proving the lower inclusion with $c = 1/(4C_0)$. Formula (98) enforces prior normalization; simply using $(1 + sf)\mu$ here would not do so.

Let s_a be the number of distinct positive future exponents. The active q by s_a block of L_a is bounded above and below as an operator, by its leading triangular block in Lemma 5.1. Thus the nonzero singular values of $L_a \text{diag}(d_1, \dots, d_{s_a})$ are comparable, in order, to $d_1, \dots, d_{s_a}$. The comparison follows directly by applying the min-max formula to the two quadratic forms, bounded above and below by fixed multiples of $\sum_j d_j^2 v_j^2$. The Euclidean unit ball lies in the cube, which lies in the ball of radius $\sqrt{q}$. An ellipsoid with ordered semiaxes b_j has k th linear width b_{k+1} : projection on its first k principal axes gives the upper bound, while any k -dimensional space has a nonzero orthogonal direction in the span of its first $k + 1$ axes, giving the lower bound. Apply this fact to the two inclusions in (96). If $k \geq s_a$, all vectors lie in a space of dimension s_a , and both sides of (97) are zero. $\square$

15.3. Exact lower moments and a vanishing uncertainty direction. The preceding proposition resolves every direction, rather than only the longest one. An equivalent formulation specifies which moments have already been supplied. Let s_a again denote the number of distinct positive nodes. For $0 \leq k \leq q$, restrict (93) by the finite conditions

$$(99) \quad \nu'_h(t^{Hx_i}) = \nu_h(t^{Hx_i}), \quad 1 \leq i \leq \min\{k, s_a\},$$

and call the restricted class $\mathcal{B}_{h,k}$. Denote by $Q_m^a(\nu)$ the vector of probabilities of the ordered physical m -trial queries, with its normalized Euclidean norm $\|\cdot\|_{\text{av}}$. Define its squared minimax radius by

$$\mathbf{a}_{h,k}(a, \varepsilon) = \inf_z \sup_{\mu' \in \mathcal{B}_{h,k}} \|z - Q_m^a(\nu'_h)\|_{\text{av}}^2.$$

The center μ , the relative envelope, the calibration and the values in (99) specify this local problem. These are exact conditional-moment constraints at h , not the pre-acquisition, absolute-error moment name of Section 16.

Corollary 15.4 (Sharp residual uncertainty after a moment prefix). *With uniform constants as above,*

$$(100) \quad c\varepsilon^2 d_{k+1}^2 \leq \mathbf{a}_{h,k}(a, \varepsilon) \leq C\varepsilon^2 d_{k+1}^2.$$

For a constant-failure history, the same order is realized as an unconditional, event-weighted physical prediction problem with exact prior-moment advice. In particular, for $A_\theta = \{0, 1, 2 + \theta\}$, $N = 3$, $n = 1$, $m = 2$ and sufficiently small $|\theta|$, one has

$$(101) \quad \Xi_3(M, A_\theta) \asymp M^{-1}, \quad \mathbf{a}_{h,4}(A_\theta, \varepsilon) \asymp \varepsilon^2 \theta^2.$$

The latter is zero at $\theta = 0$.

Proof. For $k < s_a$, the first k raw constraints in (99) are equivalent to equality of the first k normalized moments. This follows by triangular back substitution in the leading block of $L_a D_a$, whose first k pivots are nonzero. This equivalence is an exact identity, not a stability assertion for approximate constraints. The upper-inclusion proof in Proposition 15.3 therefore has its first k coordinates zero. Its remaining scaled coordinates are bounded by $C\varepsilon d_{k+1}$. The central prediction gives the upper bound. For the lower bound use the construction (98) with $v = e_{k+1}$ and $v = -e_{k+1}$. Both priors satisfy the constraints and their query vectors differ by at

least $c\varepsilon d_{k+1}$. For any proposed output, one of its two distances is at least half this separation. Query-norm equivalence proves the lower bound. If $k \geq s_a$, all distinct raw future moments are fixed, and Proposition 2.3 gives radius zero.

For the physical assertion issue n constant commands whose failure factor is $1/2$, and consider the event E of n failures. Its likelihood is the constant 2^{-n} , and its probability is 2^{-n} under every prior in the class. On E the posterior is the prior, so (99) is now exact *prior*-moment advice fixed before acquisition. Score the squared prediction loss for the selected m -trial query on E , and assign zero loss off E . Every trial and the probability of E remain counted. A single rule has the same prediction kernel on E for all compatible priors. The preceding two-point argument, multiplied by 2^{-n} , gives the unconditional lower bound; the central prediction gives the upper bound. This is a specific event-weighted decision problem, not a claim about the risk on all other report histories under a different payoff.

For the displayed family, the five nonconstant formal two-trial nodes, before the fixed normalization by H , are

$$1, \quad 2, \quad 2 + \theta, \quad 3 + \theta, \quad 4 + 2\theta.$$

They form four uniformly separated groups, with the only merging pair $2, 2 + \theta$. Thus $\mathcal{V}_{2,j} \asymp 1$ for $1 \leq j \leq 4$ and $\mathcal{V}_{2,5} \asymp |\theta|$. Lemma 5.1 gives $d_1, \dots, d_4 \asymp 1$ and $d_5 \asymp |\theta|$, including $d_5 = 0$ at the collision. Formula (100) proves the second assertion. At checkpoint $(1, 2)$ the attainable truncation is $p_{1,2} = 2$, so its regret profile is $\asymp M^{-1}$. At checkpoint $(2, 1)$ there are two uniformly separated positive one-trial nodes, and the same profile results. Taking the maximum proves the first assertion in (101). $\square$

The example separates a collision which is invisible to the acquired memory exponent at this horizon from a vanishing direction of unresolved prior information. More generally, (97) uses all future scales, whereas (16) uses only those which the past can excite. The isotropic common-name law in the next section allows uncertainty in a uniformly visible direction; its δ^2 term is therefore not, by itself, a classification of the weaker directions. No calibration-only converse is inferred from the prior perturbations constructed here.

16. SHARP MEMORY UNDER A COMMON MOMENT NAME

We now determine the joint effect of persistent memory and uncertainty in the finite moment data. The upper construction in Corollary J.8 serves every experiment consistent with the same name. A converse requires the consistency class to contain actual alternatives; an arbitrary boundary name may determine a singleton. The following formulation makes the necessary interior condition explicit. It applies at every additive collision stratum, without changing the fixed-prior classification.

16.1. The consistency class. Fix the detector, $K, N \geq 2$, and $r \geq 2$ as in the introduction. Choose the attainable failure basis to be

$$(102) \quad F_0 = \tfrac{1}{2}, \quad F_j = \tfrac{1}{2} + \beta t^{a_j}, \quad 1 \leq j < r,$$

where $\beta > 0$ is fixed and sufficiently small. The right inverse in Lemma 2.1 supplies one command for each formal coefficient vector, independent of the calibration. Thus this is one of the fixed physical bases already allowed in the intrinsic theorem, not an extra observation channel. At future length m , use its ordered m -fold products, selected uniformly after the checkpoint index is formed.

Let μ_0 be a fixed full-support probability on $[0, 1]$, and take $0 < c_- < 1 < c_+ < \infty$. For a fixed $0 < \lambda < \min\{1 - c_-, c_+ - 1\}$, set

$$\mathcal{D} = \{\nu : c_- \mu_0 \leq \nu \leq c_+ \mu_0\}, \quad \mathcal{D}_\lambda = \{\nu : (c_- + \lambda) \mu_0 \leq \nu \leq (c_+ - \lambda) \mu_0\}.$$

All members are probability measures. These inequalities are inequalities of measures; no Lebesgue density is assumed. Let $\mathbf{m}_N(a, \mu)$ denote the vector of all formal moments of degrees $0, \dots, N$, with repeated exponent labels retained. For a rational vector y of the same format and $\delta > 0$, write

$$(103) \quad \mathcal{U}(y, \delta) = \{(a', \nu) \in K \times \mathcal{D} : \|\mathbf{m}_N(a', \nu) - y\|_\infty \leq \delta\}.$$

The detector is fixed; certified approximations to its coefficients may be supplied separately. The vector y need not itself be a moment vector.

A common-name program is chosen from y, δ and the fixed task data, without consulting the unknown member of $\mathcal{U}(y, \delta)$. It retains at most M labels at every stage. Let $\mathcal{R}_{M,N}^{\text{av}}(y, \delta)$ be the infimum over such programs of the supremum, over this class, of maximum unconditional checkpoint regret under the common independent command exploration. Define $\mathcal{R}_{M,N}^{\text{max}}(y, \delta)$ by using worst-history checkpoint regret instead. At each checkpoint the query is the fresh random choice specified above. The admissible programs include the integer/dyadic programs with memoryless command coding. The lower bounds below hold even for randomized programs with unrestricted offline advice that is the same in all members of the consistency class.

Theorem 16.1 (Joint memory and moment-uncertainty law). *There are constants $c, C, \delta_0 > 0$, depending only on the fixed data and λ , such that the following holds. Suppose $(a, \mu) \in K \times \mathcal{D}_\lambda$, $M \geq 1$ is an integer, $0 < \delta \leq \delta_0$, and*

$$(104) \quad \|y - \mathbf{m}_N(a, \mu)\|_\infty \leq \delta/2.$$

Then

$$(105) \quad c(\Xi_N(M, a) + \delta^2) \leq \mathcal{R}_{M,N}^{\text{av}}(y, \delta) \leq \mathcal{R}_{M,N}^{\text{max}}(y, \delta) \leq C(\Xi_N(M, a) + \delta^2).$$

The upper bound is attained in order by one finite integer/dyadic program constructed from y , not from (a, μ) . Its selected numerical scale is comparable to $\min\{1, \sqrt{\Xi_N(M, a)} + \delta\}$. The constants are uniform on every additive collision stratum of K and across their intersections.

The interior point in this statement is used to describe the order of the risk and to prove a lower bound. It is not supplied to the algorithm. In particular the theorem does not replace the consistency class by alternatives selected after the true model has been revealed. The hypothesis (104) leaves room for a two-sided physical perturbation. Omitting this hypothesis, or replacing $\mathcal{D}_\lambda$ by an arbitrary boundary point of $\mathcal{D}$, would not be justified by the argument.

16.2. Uniform ambiguity after positive histories. The converse rests on the following elementary identity. It records why acquisition does not resolve the local ambiguity at a fixed horizon.

Lemma 16.2 (Posterior separation with overlapping history laws). *Let μ be a probability, let $\mu f = 0$ and $|f| \leq 1$, and put $\mu_\pm = (1 \pm \varepsilon f)\mu$ for $0 < \varepsilon \leq 1/2$. For a positive likelihood ℓ_w , let $\nu_w = \ell_w \mu / \mu \ell_w$. For every bounded query q ,*

$$(106) \quad \nu_{w,+} q - \nu_{w,-} q = \frac{2\varepsilon \text{Cov}_{\nu_w}(q, f)}{1 - \varepsilon^2(\nu_w f)^2}.$$

Under the same parameter-independent acquisition rule, the history laws satisfy $P_\pm \geq (1 - \varepsilon)P_0$. If one prediction kernel Y is used in both experiments and $q_\pm(w) = \nu_{w,\pm} q$, then

$$(107) \quad \frac{1}{2} \sum_{\sigma \in \{-, +\}} \mathbb{E}_{P_\sigma}(Y - q_\sigma)^2 \geq \frac{1 - \varepsilon}{4} \int (q_+(w) - q_-(w))^2 P_0(dw).$$

The expectation includes any common independent coding randomization.

Proof. Both perturbations have mass one and are positive. Direct normalization gives

$$\nu_{w,\pm}q = \frac{\nu_w q \pm \varepsilon \nu_w(qf)}{1 \pm \varepsilon \nu_w f}.$$

Subtract these fractions to obtain (106). The density of $P_{\pm}$ relative to P_0 is $1 \pm \varepsilon \nu_w f$, by integration of the same likelihood against the two priors. It lies between $1 - \varepsilon$ and $1 + \varepsilon$. Conditional on a given history, the prediction kernel is identical because the data supplied to the program are identical. For every real Y and real u, v ,

$$\frac{1}{2}\{(Y - u)^2 + (Y - v)^2\} = (Y - (u + v)/2)^2 + (u - v)^2/4.$$

Dominates both history measures by $(1 - \varepsilon)P_0$, integrate this identity, and discard its nonnegative first term. This proves (107). The same calculation integrates randomized predictions. No equality of the two history laws is required. $\square$

Lemma 16.3 (A uniform physical moment ambiguity). *Under the hypotheses of Theorem 16.1, the class $\mathcal{U}(y, \delta)$ contains two experiments with the same calibration a such that, at every checkpoint $1 \leq n < N$, every common prediction rule has maximum unconditional regret at least $c_0 \delta^2$. The rule may retain the entire acquired history. The constant $c_0 > 0$ is uniform in $(a, \mu), y, \delta$ in the stated ranges. Both alternatives are positive experiments in $\mathcal{D}$.*

Proof. Put $z_a(t) = t^{a_1}$ and

$$v_0 = \min_{a \in K} \text{Var}_{\mu_0}(z_a) > 0.$$

The variance is positive for each a : full support gives positive mass to two separated intervals on which the strictly increasing function z_a has disjoint ranges. Its dependence on a is continuous in the uniform norm, including at $t = 0$, because a_1 has a positive lower bound on K . Compactness therefore gives the displayed positive minimum.

Set $f = z_a - \mu z_a$ and $\varepsilon = \delta/2$. Then $\mu f = 0$ and $|f| \leq 1$. If $\varepsilon \leq \lambda/c_+$, the measure bounds for $\mu \in \mathcal{D}_{\lambda}$ imply

$$c_- \mu_0 \leq (1 \pm \varepsilon f) \mu \leq c_+ \mu_0.$$

Indeed, the absolute change of the measure is bounded by $\varepsilon \mu \leq \varepsilon c_+ \mu_0 \leq \lambda \mu_0$. For every formal label through degree N , its monomial v lies in $[0, 1]$, and hence

$$|\mu_{\pm} v - \mu v| = \varepsilon |\mu(vf)| \leq \varepsilon.$$

Together with (104) this puts both $(a, \mu_{\pm})$ in (103). This uses the same y in both experiments; none of its entries is changed.

Let w be any length- n history and ν_w its reference posterior. Its likelihood satisfies $\kappa^n \leq \ell_w \leq 1$, so $\nu_w \geq \kappa^n \mu$. The variational formula for variance gives

$$(108) \quad \text{Var}_{\nu_w}(z_a) = \inf_{b \in \mathbb{R}} \nu_w((z_a - b)^2) \geq \kappa^n \text{Var}_{\mu}(z_a) \geq \kappa^n c_- v_0.$$

This bound is uniform over commands and report words, including failures. Let $m = N - n$ and take the actually executable ordered product probe $F_1 F_0^{m-1}$. Its event function is

$$q(t) = 2^{-m} + \beta 2^{-(m-1)} z_a(t).$$

It is selected with probability r^{-m} by the prescribed menu and uses exactly its m trials. Since subtracting the constant μz_a from z_a does not change covariance under ν_w ,

$$\text{Cov}_{\nu_w}(q, f) = \beta 2^{-(m-1)} \text{Var}_{\nu_w}(z_a).$$

Equation (106), whose denominator is in $(0, 1]$, therefore bounds the two conditional query means apart by

$$2\varepsilon \beta 2^{-(m-1)} \kappa^n c_- v_0.$$

Apply (107) and multiply by r^{-m} . Taking the minimum of the resulting positive constants over the finitely many n gives $c_0\delta^2$. For example, the coefficient for a given n may be taken as $r^{-m}\beta^22^{-2(m-1)}\kappa^{2n}c_-^2v_0^2/8$. Here $\varepsilon = \delta/2$ and $1 - \varepsilon \geq 1/2$ were used. The argument applies to every rule on the complete history, and hence also to every finite-label rule. It neither resolves nor divides by any additive exponent difference. $\square$

Proof of Theorem 16.1. For every $(a', \nu) \in \mathcal{U}(y, \delta)$, the same finite table is δ -accurate. Corollary J.8 supplies one program with worst-history regret at most $C(\Xi_N(M, a') + \delta^2)$ in each such experiment. Its finite input and prior-family hypotheses hold uniformly by Lemma J.5. The moment vectors of (a', ν) and (a, μ) differ by at most $3\delta/2$. Consequently Corollary J.6, applied with this bound, gives

$$\Xi_N(M, a') \leq C'(\Xi_N(M, a) + \delta^2).$$

Increasing the fixed constant proves the upper inequality in (105); no point of the consistency class has been selected in the construction. The stated numerical scale follows by applying the same corollary to the center, which belongs to the class. An unconditional risk is bounded by its worst-history counterpart.

For the memory lower bound, the center itself is an admissible member of $\mathcal{U}(y, \delta)$. Every common-name M -label program is an admissible M -label program for that fixed experiment. The intrinsic causal lower theorem and Lemma J.5 imply a uniform lower bound $c_1\Xi_N(M, a)$, including independent coding randomization. Lemma 16.3 supplies the independent lower bound $c_0\delta^2$ for every such program. Use $\max\{u, v\} \geq (u + v)/2$ and take the infimum over programs. Finally choose δ_0 to satisfy the positive-evidence restrictions in the upper construction, $\delta_0 \leq 1$, and $\delta_0/2 \leq \lambda/c_+$. All restrictions depend only on the fixed data. This proves the theorem. $\square$

The two-point method in these lemmas is elementary. The collision-sensitive content of (105) is the full attained profile Ξ_N , in both its memory converse and its common-program realization. The uncertainty converse is now valid uniformly at colliding calibrations rather than only in a noncolliding example. It uses prior ambiguity; it is not a calibration-only converse for a known exact prior, nor a lower bound for arbitrary degenerate consistency classes.

16.3. The anisotropic saturation threshold. The full profile gives a more precise resource threshold than its separated-chain floor. Define

$$(109) \quad \mathcal{M}_\delta(a) = \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n, N-n}}} \mathcal{V}_{N-n, \ell}(a)\delta^{-\ell}.$$

Terms with zero volume are zero, not omitted by a numerical rank decision.

Corollary 16.4 (Collision-sensitive saturation). *For every $a \in K$, $0 < \delta \leq 1$, and integer $M \geq 1$,*

$$(110) \quad \Xi_N(M, a) \leq \delta^2 \iff M \geq \mathcal{M}_\delta(a).$$

Thus the least integer satisfying the profile inequality is $\lceil \mathcal{M}_\delta(a) \rceil$. Under the hypotheses of Theorem 16.1, the minimax order is δ^2 above this threshold and is $\Xi_N(M, a)$ below it, with the same fixed comparison constants. This is an order statement about risk, not an exact numerical value of the minimax risk.

Suppose along a positive one-parameter path $a(\theta)$ each nonzero volume satisfies $\mathcal{V}_{m, \ell}(a(\theta)) \asymp \theta^{\beta_{m, \ell}}$, with constants uniform for sufficiently small $\theta > 0$. Then, uniformly also in $0 < \delta \leq 1$,

$$(111) \quad \mathcal{M}_\delta(a(\theta)) \asymp \max_{n, \ell} \theta^{\beta_{N-n, \ell}} \delta^{-\ell}.$$

Identically zero volumes contribute no term. In particular all contact orders in the collision-tree law enter the advice saturation threshold.

Proof. Each term $(\mathcal{V}_{m,\ell}/M)^{2/\ell}$ is at most δ^2 if and only if $M \geq \mathcal{V}_{m,\ell}\delta^{-\ell}$. Take the maximum over all checkpoints and active formal indices. This proves (110), including exact zero volumes. The term $\ell = 1$ gives $\mathcal{M}_\delta \geq \delta^{-1}$, so the integer statement has no small-budget exception. On each side of the threshold the sum $\Xi_N + \delta^2$ is between its larger term and twice that term; apply (105). Finally there are only finitely many volume terms. Substitute their uniform two-sided estimates into (109) and take the smallest and largest of their positive constants. This proves (111). The contact exponents are those already determined by the collision-tree argument; the corollary introduces no new equality test. $\square$

Corollary 16.5 (Uncertain moments at intersecting collision strata). *Consider the fixed four-cell detector and square of calibrations in Corollary 5.7. Put $\rho = \max\{|u|, |v|\}$ and $\tau = \min\{|u|, |v - u|, |v - 2u|\}$, as in that corollary. Under the prior-interior and common-name hypotheses of Theorem 16.1, both common-name minimax criteria at $N = 5$ have order*

$$(112) \quad \max \left\{ M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau)^{2/9} M^{-2/9}, \delta^2 \right\}.$$

The profile saturation threshold satisfies, uniformly on the entire square,

$$(113) \quad \mathcal{M}_\delta(u, v) \asymp \max \{ \delta^{-6}, \rho^2 \delta^{-8}, \rho^2 \tau \delta^{-9} \}.$$

In particular, on $u = \theta$, $v = \theta + \theta^k$, $k \geq 2$ fixed, and for sufficiently small $\theta > 0$,

$$(114) \quad \mathcal{M}_\delta \asymp \begin{cases} \delta^{-6}, & \delta \geq \theta, \\ \theta^2 \delta^{-8}, & \theta^k \leq \delta \leq \theta, \\ \theta^{k+2} \delta^{-9}, & 0 < \delta \leq \theta^k. \end{cases}$$

These are uniform comparison orders, including their boundary scales. Thus a fixed moment accuracy can obscure one collision scale while resolving another; the same common-name program attains the appropriate risk order without being told the stratum or the contact order.

Proof. Theorem 16.1 and Corollary 5.7 give (112), since the sum of a nonnegative profile and δ^2 is comparable to their maximum. For the threshold, the determinant calculation in that corollary gives, at $(n, m) = (3, 2)$, volumes comparable to one through order six, to ρ at order seven, to ρ^2 at order eight, and to $\rho^2 \tau$ at order nine. Other checkpoints contribute at most $C\delta^{-6}$ to (109), since their attainable dimensions are at most six and their normalized volumes are bounded. The seven-dimensional term introduces no further order:

$$\rho \delta^{-7} = \sqrt{\delta^{-6} \rho^2 \delta^{-8}} \leq \max \{ \delta^{-6}, \rho^2 \delta^{-8} \}.$$

This proves (113), also when a volume is exactly zero. Along the stated path, $\rho \asymp \theta$ and $\tau \asymp \theta^k$. The ratios of the middle threshold term to the first, and of the last to the middle, have orders θ^2/δ^2 and θ^k/δ , respectively. The ordering $\theta^k \leq \theta$ gives the three cases in (114). The implementation assertion is the common-program conclusion of Theorem 16.1, not a separate procedure that decides which case holds. $\square$

APPENDIX A. THE OBSERVATION ALGEBRA AND ASYMMETRIC PROFILES

For integral exponents define the standard graded algebra

$$\mathcal{R}_A = \mathbb{R}[zt^a : a \in A] \subset \mathbb{R}[z, t], \quad \deg(zt^a) = 1.$$

Its degree- m component has basis $\{z^m t^s : s \in mA\}$, so its Hilbert function is exactly $h_A(m)$. The general sumset–Hilbert–function connection is classical [2, 3]. Its role here is operational: $h_A(m) - 1$ is the number of independent future observable tests, and Theorem 3.3 proves when the attainable past can excite all of them. The algebra alone would not prove that transversality.

Proposition A.1 (Three-cell classification). *Let $A = \{0, a, b\}$ with integers $0 < a < b$, put $g = \gcd(a, b)$, $a' = a/g$ and $B = b/g$. Then*

$$(115) \quad h_A(m) = \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2}.$$

The homogeneous relation is generated by $x_1^B - x_0^{B-a'} x_2^{a'}$ under the map $(x_0, x_1, x_2) \mapsto (z, zt^a, zt^b)$. In particular,

$$d_A(n, m) = \min \left\{ 2n, \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2} - 1 \right\}.$$

For nonnegative real exponents $0 < a < b$ with a/b irrational, $h_A(m) = \binom{m+2}{2}$ for every m .

Proof. A monomial $x_0^k x_1^i x_2^j$ of degree m maps to $z^m t^{ai+bj}$. Two exponent triples of the same degree have the same image exactly when their difference is an integer multiple of $-(B - a'), B, -a'$: this follows by dividing $a(i - i') + b(j - j') = 0$ by g and using coprimality of a', B . After cancelling a common monomial, each corresponding binomial is a multiple of $x_1^B - x_0^{B-a'} x_2^{a'}$, by the difference-of-powers identity. The kernel is spanned by these binomials, because monomials with distinct images are linearly independent and each fiber's coefficient sum must vanish. Thus this one relation generates the kernel.

The degree- m polynomials in three variables have dimension $\binom{m+2}{2}$. Multiplication by the displayed nonzero degree- B relation is injective in the polynomial ring, so the degree- m part of the ideal has dimension zero for $m < B$ and $\binom{m-B+2}{2}$ otherwise. Subtraction proves (115). If a/b is irrational, $a(i - i') + b(j - j') = 0$ forces both differences to vanish; all $\binom{m+2}{2}$ possible pairs are distinct. The operational formula now follows from Theorem 3.3. $\square$

Corollary A.2 (The sparse space that separates rank from future dimension). *For $A = \{0, 2, 5\}$, the first values of $h_A(m)$, starting at $m = 0$, are 1, 3, 6, 10, 15, 20, 25. At $n = 3, m = 2$ the exact information dimension is five, for every full-support prior. For a total budget of six the minimal causal dimensions, starting at $n = 0$, are*

$$0, 2, 4, 6, 5, 2, 0.$$

Thus the peak need not be the peak of a rank-based symmetric tent.

Proof. Apply (115) with $B = 5$ and then (5). In particular $2A = \{0, 2, 4, 5, 7, 10\}$, while the binomial witness at $n = 3$ has $B_{A,3} = \{0, 2, 5, 7, 10, 12, 15\}$. The mixed matrix has rank six and normalization leaves five. This proves the example uniformly over full-support priors without relying on a single large rational determinant. The uniform-prior three-cell example and its numerical witness first appeared in the source-pinned report [10]; the present corollary is an application of the general theorem, not a new attribution of that example. $\square$

There is also an exact finite computation for any integral A : start with $S_0 = \{0\}$, form $S_{m+1} = \{s + a : s \in S_m, a \in A\}$, and count distinct exponents. This handles all additive collisions, not just a nominal degree range. For real exponents the formula is equally exact as mathematics, but an effective implementation requires exact equality information about the supplied exponents. Neither fixed rank nor arbitrarily accurate floating-point exponents substitute for that information.

APPENDIX B. CONFLUENT OBSERVATION ALGEBRAS AND RESOLVED DIRECTIONS

The cardinality of a sumset is discontinuous when two of its elements coalesce. The corresponding experiment need not be discontinuous. We now identify the scales of the directions lost at such a collision. The parameter θ below indexes a *known calibration*; the unknown parameter of each experiment is still t .

B.1. Affine exponent families. Fix $0 = a_0 < a_1 < \dots < a_{r-1}$ and real numbers b_i , with $b_0 = 0$. On a sufficiently small compact interval $0 \leq \theta \leq \theta_0 \leq 1$ put

$$A_\theta = \{a_i + \theta b_i : 0 \leq i < r\}.$$

The order of these exponents is assumed unchanged. Take $I = [0, 1]$ and one fixed full-support prior μ . A fixed coefficient matrix specifies positive detector cells

$$(116) \quad k_j^\theta(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i + \theta b_i}, \quad \sum_j k_j^\theta = 1.$$

It has column rank r , and the cells have a common positive lower bound. The comparator cube is fixed. These hypotheses include any sufficiently small affine perturbation of a strictly positive calibration of full rank. No continuity of an unknown prior density is assumed.

For a fixed positive future length $m \geq 1$, let

$$\mathcal{B}_m = \left\{ \left(\sum_{\nu=1}^m a_{i_\nu}, \sum_{\nu=1}^m b_{i_\nu} \right) : 0 \leq i_\nu < r \right\}.$$

Repeated *pairs* are identified. Distinct pairs give distinct exponents $\lambda + \theta\gamma$ for $0 < \theta \leq \theta_0$; shrink θ_0 to exclude crossings if necessary. For each first coordinate λ , order its second coordinates as $\gamma_{\lambda,0} < \dots < \gamma_{\lambda,s_\lambda-1}$. Set $K_m = \sum_\lambda s_\lambda$. The cluster at $\lambda = 0$ consists only of $(0, 0)$.

Define the divided-difference functions

$$(117) \quad \psi_{\lambda,k}^\theta(t) = [\lambda + \theta\gamma_{\lambda,0}, \dots, \lambda + \theta\gamma_{\lambda,k}] (x \mapsto t^x), \quad 0 \leq k < s_\lambda.$$

At $\theta = 0$ their continuous values are

$$(118) \quad \psi_{\lambda,k}^0(t) = t^\lambda (\log t)^k / k!.$$

For $\lambda > 0$ these functions have value zero at $t = 0$; the sole zero-exponent function is the constant one. The bracket in (117) is a divided difference in the exponent, not a new apparatus output. For a history h let $z_\theta(h)$ be its posterior expectations of these functions, omitting the constant coordinate.

List the orders k of the nonconstant functions in nondecreasing order:

$$0 = \nu_1 \leq \dots \leq \nu_d, \quad d = K_m - 1.$$

There are $|mA_0| - 1$ zero orders. The associated resolution function is

$$(119) \quad \Phi_\theta(M) = \max_{1 \leq \ell \leq d} \theta^{2(\nu_1 + \dots + \nu_\ell)/\ell} M^{-2/\ell}, \quad M \geq 1.$$

We set $0^0 = 1$ and $0^a = 0$ for $a > 0$ in this formula. In particular $\Phi_0(M) = M^{-2/(|mA_0|-1)}$.

Choose r attainable failure factors which form a basis in the formal monomials $t^{a_i + \theta b_i}$, with coefficients independent of θ . For example use $1/2$ and $1/2 + \delta t^{a_i + \theta b_i}$, $i > 0$, with δ sufficiently small. The fixed right inverse of the calibration matrix realizes these by fixed command labels. All ordered m -fold products form a common finite probe menu. Any other such fixed basis, and its full product menu, gives equivalent bounds below.

Theorem B.1 (Uniform resolved checkpoint law). *Suppose*

$$(120) \quad K_m - 1 \leq n(r - 1).$$

Supply independent uniform commands for the first n trials, then reveal an independent uniform query from the preceding product menu. Let $\bar{R}_{M;n,m}^\theta$ be the optimal unconditional squared-prediction regret of an arbitrary M -message checkpoint encoder. Let $R_{M;n,m}^{\theta,\max}$ be its minimax counterpart over admitted histories. There are constants $0 < c \leq C < \infty$, independent of M and θ , such that

$$(121) \quad c\Phi_\theta(M) \leq \bar{R}_{M;n,m}^\theta \leq R_{M;n,m}^{\theta,\max} \leq C\Phi_\theta(M), \quad 0 \leq \theta \leq \theta_0.$$

The encoders may be Borel and randomized as in Definition 2.4. At the common interior binomial command tuple the normalized map $h \mapsto z_\theta(h)$ is a submersion of rank d , including at $\theta = 0$. Its local inverse bounds and an unconditional minorization can be chosen uniformly in θ .

The upper bound in this theorem is a checkpoint bound. It does not assume that access to an exact prefix is free during repeated compression. Every streaming filter is a checkpoint encoder, so the lower bound applies to such filters as well. Section E proves a matching genuinely streaming bound across an entire singular family. Theorem D.1 below removes the full-future restriction by truncating the resolution flag, and Theorem D.2 proves the fixed-horizon causal version for the affine class. When $m = 0$ there is no nonconstant future query and the prediction regret is zero; formula (119) is not used in that case.

B.2. Confluent positivity and the physical prediction metric. We give the details of the analytic and geometric steps in Theorem B.1.

The proof of Lemma 4.3 appears in the principal analytic inputs.

Lemma B.2 (Desingularization and observable scales). *The functions in (117), and the needed derivatives in θ , are uniformly bounded on $[0, 1] \times [0, \theta_0]$. The posterior vectors $z_\theta(h)$ therefore lie in a fixed box. For the common product probe menu let $p_\theta(h)$ be its probability vector. There is a fixed matrix G of full column rank such that*

$$(122) \quad p_\theta(h) - p_\theta(\tilde{h}) = G \operatorname{diag}(\theta^{\nu_i})(z_\theta(h) - z_\theta(\tilde{h})).$$

In particular the squared norm on the left is bounded above and below by positive constants times $\sum_i \theta^{2\nu_i} |z_i(h) - z_i(\tilde{h})|^2$, with constants independent of θ .

Proof. The divided difference of order k is the integral of $f^{(k)}(\sum_j w_j x_j)$ over the standard k -simplex, whose volume is $1/k!$. This identity follows by iterating the fundamental theorem of calculus, starting with $(f(y) - f(x))/(y - x) = \int_0^1 f'(x + s(y - x)) ds$. It also gives the repeated-node value. Here $f(x) = t^x$, so every integrand is $t^x(\log t)^k$. At a positive cluster the exponent is bounded below by some $a > 0$ uniformly on the fixed parameter interval. For every fixed j , $\sup_{0 < t \leq 1} t^a |\log t|^j < \infty$. The same domination applies after finitely many θ derivatives. There are no higher divided differences at zero. Posterior expectations of bounded functions are bounded independently of the history.

Newton's interpolation identity, proved by successive subtraction of the interpolating polynomials, is exact at each node:

$$(123) \quad t^{\lambda + \theta \gamma_{\lambda,j}} = \sum_{k=0}^j \theta^k \left(\prod_{v < k} (\gamma_{\lambda,j} - \gamma_{\lambda,v}) \right) \psi_{\lambda,k}^\theta(t).$$

The matrix multiplying $\text{diag}(\theta^k)$ is fixed, triangular within each cluster, and has nonzero diagonal. The standard identities used here are classical divided-difference calculus [12]; the integral proof makes the uniformity at coalescence explicit.

The expansion matrix from the formal pair monomials in $\mathcal{B}_m$ to the probe products has full column rank. To see this, invert the fixed one-step basis change and multiply it m times. Every formal monomial product is a linear combination of the probe products. Identifying equal pairs is a surjective linear quotient, so the spanning statement survives that identification. Both this expansion matrix and the triangular matrix in (123) are independent of θ . Their product has full column rank. Subtracting two normalized posterior vectors removes its constant column and gives (122). The remaining columns are still independent. Their extreme singular values give the two constants. The equality extends to $\theta = 0$ by continuity; vanishing weights then remove precisely the unresolved jets. $\square$

Lemma B.3 (Uniform attainable patch with its evidence). *Under (120), there are $\rho, \beta > 0$, independent of θ , such that the subprobability distribution of z_θ on all-failure prefixes dominates β times uniform probability on a translate of $[-\rho, \rho]^d$.*

Proof. Use the binomial factors $(1 + c_i t^{D_\theta})/2$ of Lemma 3.2, with distinct small c_i and $D_\theta = a_{r-1} + \theta b_{r-1}$. Since the coefficient calibration is fixed, one common interior command tuple implements this family for every θ . Its unnormalized tangent has $n(r-1) + 1$ distinct monomials, also at zero. At positive θ , pairing with the K_m distinct future exponents has rank K_m by Lemma 3.1. The divided-difference basis change is invertible there. At zero, Lemma 4.3 gives the same rank directly, without passing a rank assertion through a singular limit. The product itself lies in the tangent. The normalization map $v \mapsto v - p e_0 v$ has kernel exactly $\text{span}\{p\}$ on this full image, as in (10). Thus the normalized jet map has rank d at every θ , including zero.

Here are uniform local details. Let q be the total number of command coordinates and g_* the common interior tuple. The matrices $J_\theta = D_g z_\theta(g_*)$ are continuous in θ and surjective. Their least row singular value has a positive minimum σ on the compact parameter interval. Choose an orthonormal basis E_θ of the row space and a complementary orthonormal basis F_θ of its kernel, and use coordinates $g = g_* + E_\theta u + F_\theta w$. The derivative at zero of

$$(u, w) \longmapsto (z_\theta(g_* + E_\theta u + F_\theta w) - z_\theta(g_*), w)$$

is block diagonal with invertible upper block, least singular value at least σ , and lower block the identity. Its derivative norm and its second derivative have common finite bounds on a fixed neighborhood. This follows from Lemma B.2, finite product rules, and the prefix evidence lower bound κ^n .

Choose a ball inside the command cube on which the derivative differs from its central value by at most half its least singular value. This radius can be chosen uniformly. For a sufficiently small target ball, the map obtained by subtracting the nonlinear remainder and applying the inverse central derivative is a contraction of that command ball into itself. The contraction theorem gives a differentiable inverse. Hence a product of a d -cube and a $(q-d)$ -cube of common positive side length lies in its image, uniformly in θ . All inverse Jacobian determinants are bounded below by a positive constant: the forward derivative is uniformly bounded, so its determinant is uniformly bounded above. The orthogonal command-coordinate change has determinant of absolute value one. No continuity of the chosen orthonormal frames in θ is needed for these uniform bounds.

Uniform commands have density $(1 - 2\eta)^{-q}$. Their joint law with the all-failure word has density

$$(1 - 2\eta)^{-q} \mu \left(\prod_{i=1}^n F_{g_i} \right) \geq (1 - 2\eta)^{-q} \eta^n.$$

Change variables using the inverse just constructed and integrate over the complementary cube. This gives the asserted minorization, with $\beta > 0$ the product of this density bound, the inverse-Jacobian bound, and the two cube volumes. The convention for a zero-dimensional complement is volume one. This is a subprobability bound retaining the failure evidence, not conditioning away a rare prefix. $\square$

B.3. A scale-sensitive finite-state calculation.

Lemma B.4 (Rectangular quantization at every budget). *For $a_1 \geq \dots \geq a_d > 0$ set*

$$e_M(a) = \max_{1 \leq \ell \leq d} \left(\frac{a_1 \cdots a_\ell}{M} \right)^{1/\ell}.$$

A box with these side lengths has an at-most- M -cell rectangular partition of diameter at most $2\sqrt{d}e_M(a)$ in every cell. The optimal mean squared distortion of its uniform distribution using M arbitrary centers is at least $c_d e_M(a)^2$, with $c_d > 0$ depending only on d . Zero side lengths may be omitted.

Proof. Put $e = e_M(a)$ and take $N_i = \max\{1, \lfloor a_i/e \rfloor\}$ equal intervals in coordinate i . If ℓ coordinates satisfy $a_i \geq e$, then $\prod_i N_i \leq \prod_{i \leq \ell} a_i/e^\ell \leq M$. Every side length in a cell is at most $2e$: for $a_i/e \geq 1$ use $\lfloor a_i/e \rfloor \geq a_i/(2e)$, and otherwise use $a_i < e$. Choose an actual reachable representative in each nonempty cell when covering a reachable subset of the box. The diameter bound remains valid.

For the lower bound project onto the first ℓ coordinates. The projected uniform density is $(a_1 \cdots a_\ell)^{-1}$ on its box. The union of M balls of radius r has probability at most $Mv_\ell r^\ell / (a_1 \cdots a_\ell)$, where v_ℓ is the volume of the unit ball. Taking $r = (a_1 \cdots a_\ell / (2Mv_\ell))^{1/\ell}$ shows that mean squared distance is at least $r^2/2$. Projection cannot increase distance. Apply this for every ℓ and take the smallest of the finitely many positive dimensional constants. Centers need not lie in the box. $\square$

Proof of Theorem B.1. Let $D_\theta = \text{diag}(\theta^{\nu_i})$; this temporary matrix notation is unrelated to the peak dimension $D_A(N)$. By Lemma B.2, $D_\theta z_\theta(h)$ lies in a translated box of side lengths at most $2B\theta^{\nu_i}$ for a fixed B . Lemma B.4 partitions this box into at most M cells with squared diameter at most $C\Phi_\theta(M)$. Choose a reachable representative in each nonempty cell. At the checkpoint, encode the cell of the exact history and decode its representative's probe probabilities. These are probabilities in $[0, 1]$. Equation (122) gives the uniform upper regret bound. Finite cells with deterministic boundary conventions give Borel encoders.

For the lower bound first average private decoder coins conditional on the message and query. Squared loss cannot increase, leaving at most M centers in probe space. Orthogonally project these centers onto the affine space $p_* + \text{range}(G)$; projection cannot increase distance to a true prediction. Invert G on its range. Its lower singular bound reduces the problem to distance from at most M arbitrary centers in the scaled z coordinates. The law there dominates β times the image of the cube from Lemma B.3; its side lengths are $2\rho\theta^{\nu_i}$. The lower half of Lemma B.4 gives $c\Phi_\theta(M)$ after division by the fixed number of queries. Encoder randomization cannot beat distance to the closest center at a fixed history. Fresh public randomness can be fixed and then averaged because it is independent of the entire prefix. At zero, omit zero-weight coordinates in this argument. The same constants, reduced if necessary over the finitely many possible dimensions, work throughout the compact parameter interval. This proves both criteria and the claimed submersion statement. $\square$

Thus the relevant finite-resolution data are the multiplicities of additive collisions and their confluent orders, not only the cardinality of the nearby sumset. The result is a uniform

comparison up to constants. It neither asserts a new general quantization principle nor identifies a universal leading distortion constant.

APPENDIX C. FINITE-STATE FILTERING AFTER EVERY REPORT

Fix $N \geq 2$ and abbreviate $d_n = d_A(n, N - n)$ and $D_* = D_A(N) > 0$. The exact state from Theorem 3.4 will be denoted s_n . A clock n , calibration, prior moments and a precomputed finite program are read-only data. The charged resource is the persistent state between successive input symbols.

The resource model and loss are defined in Section 2.

Lemma C.1 (Uniform finite-horizon transition bounds). *On the compact exact state sets there are constants $L_n < \infty$ such that the update $T_n(s, g, x)$ satisfies*

$$\|T_n(s, g, x) - T_n(\tilde{s}, g, x)\| \leq L_n \|s - \tilde{s}\|$$

for every admitted g, x and $s, \tilde{s} \in \mathcal{S}_n$. Every fixed finite probe prediction vector is likewise Lipschitz in the state, with a finite constant independent of the history.

Proof. In factor coordinates an update before changeover appends the new factor $f/\mu f$. The old factor coordinates are unchanged. All unnormalized report probabilities lie in $[\kappa, 1]$; consequently the normalized factors lie pointwise in $[\kappa, 1/\kappa]$. Their coordinates are bounded, since they are continuous images of a compact cube and finite alphabet. At changeover each needed moment is a ratio of polynomial expressions in these coordinates and the fixed prior moments, with denominator the integral of a product of normalized factors, bounded below by κ^{n+1} . The same lower bound holds on line segments between factor tuples, because each individual interpolated normalized factor is still at least κ . Finite differentiation and the mean value theorem give a uniform derivative bound there.

In moment coordinates the update is (11). Its denominator is at least κ at every state; on a line segment between two such moment vectors the corresponding posterior mixture has the same lower bound. Numerators, coefficients and moment coordinates range over compact sets. The quotient rule therefore gives a uniform bound. Probe predictions are linear in the relevant moment coordinates, or normalized polynomial expressions in the factor coordinates with the same positive denominator control. The argument gives their claimed Lipschitz constants. All constants use only finite algebra and the supplied moments; they may grow with N or deteriorate with κ . $\square$

Theorem C.2 (Finite-state streaming upper bound). *For every $M \geq 1$ there is a deterministic M -state filter, updated after every report, whose prediction regret at every checkpoint and every admitted command-report history is at most*

$$(124) \quad C_N M^{-2/D_*},$$

where $C_N < \infty$ depends on the fixed experiment, prior, probe lists and total horizon, but not on M or the input history. The representatives and read-only transition functions are supplied existentially from the exact model; this is not an effective synthesis theorem from finite-precision data.

Proof. Enclose $\mathcal{S}_n \subset \mathbb{R}^{d_n}$ in $[-B_n, B_n]^{d_n}$. For $d_n > 0$ partition each coordinate interval into $k_n = \lfloor M^{1/d_n} \rfloor$ intervals. In each nonempty cell choose one representative in $\mathcal{S}_n$. There are at most $k_n^{d_n} \leq M$ representatives, and assigning a state to its cell representative has error at most

$$\delta_n \leq 2B_n \sqrt{d_n}/k_n \leq 4B_n \sqrt{d_n} M^{-1/d_n} \leq CM^{-1/D_*}.$$

Here C is the maximum of the finite prefactors. Empty cells need no representative. For $d_n = 0$ use the unique state. Deterministic boundary and tie conventions give Borel quantizers Q_n .

The filter stores only the index of a representative $\hat{s}_n$. Its transition is exactly

$$(125) \quad \hat{s}_{n+1} = Q_{n+1}(T_n(\hat{s}_n, g_{n+1}, x_{n+1})).$$

Since each representative is reachable and every report has positive probability, its exact image lies in $\mathcal{S}_{n+1}$. Thus the quantizer is always applied inside its domain and no infeasible moment update is used. The finite representative choices and the functions in (125) are read-only program data. They may be obtained by selecting points in nonempty cells of the compact state sets; no efficient global synthesis or computability of unsupplied real prior moments is asserted.

Let $e_n = \|s_n - \hat{s}_n\|$. On the actual command-report stream, Lemma C.1 gives

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C M^{-1/D_*}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C$. Induction proves $e_n \leq b_n M^{-1/D_*}$. This explicitly accumulates the errors from *all* previous lossy updates. At a query use the probe predictions of $\hat{s}_n$, which are actual probabilities. If the vector prediction map has Lipschitz constant K_n , (4) bounds regret by $(K_n^2/s_n^q)b_n^2 M^{-2/D_*}$. The maximum of these finitely many constants proves (124). No exact prefix was kept for the changeover: the factor-to-moment computation uses the representative factors already encoded by i_n . $\square$

Theorem C.3 (Matching exponent in one fixed streaming experiment). *Choose n_* attaining $D_* = d_A(n_*, N - n_*)$. For the first n_* trials supply independent uniform commands from $\mathcal{G}$, independent of the parameter, cartridges and filtering coins. The realized command is read once by the filter. At n_* reveal the independent probe query and execute its remaining trials. This is one fixed experiment, known before the filter is chosen. Its optimal unconditional regret over M -state filters satisfies*

$$(126) \quad 0 < c_N M^{-2/D_*} \leq \overline{\mathcal{R}}_{M,N}^{\text{stream}} \leq C_N M^{-2/D_*}.$$

The same power is sharp for the minimax streaming criterion taking the maximum conditional regret over checkpoints and input histories. Fresh private update and decoder randomization does not defeat the lower bound; a fresh public coding seed independent of the complete prefix may also be allowed. Thus for fixed N , the required persistent bits for this prediction task satisfy $B_(\varepsilon) = \frac{D_*}{2} \log_2(1/\varepsilon) + O_N(1)$.*

Proof. Write $n = n_*$, $d = D_*$, and $s = s_n^q$. Theorem 3.3 gives an interior command tuple g_0 where the probe map $p(g)$ on all-failure prefixes has rank d . Select d independent probe coordinates, and let E denote their coordinate projection, of operator norm one. Choose complementary command coordinates w so that $g \mapsto (Ep(g), w(g))$ has an invertible derivative at g_0 . Let Φ be its local inverse. There is a compact product $B_\rho^d \times W$ inside its domain, centered at the projected prediction point, such that Φ stays in the command interior and $|\det D\Phi| \geq j_0 > 0$. The complementary box W has positive volume, with unit-volume convention in zero dimensions.

If $k = Kn$ is the number of real command entries, their density is $(1 - 2\eta)^{-k}$. The joint density of commands and an all-failure prefix is this density times $Z(g) = \mu \prod_i F_{g_i} \geq \eta^n$. Change variables with Φ and integrate over W . The subprobability law of the projected predictions on this prefix dominates β times uniform probability on B_ρ^d , with

$$\beta = \frac{\eta^n j_0 |W| |B_\rho^d|}{(1 - 2\eta)^k} > 0.$$

In particular the lower bound retains the probability of the failure prefix. It does not assign an atom to an exact real command or assume a flat ball in the entire curved prediction image.

Every streaming filter is in particular an M -message encoder at this checkpoint. Given any message, average decoder coins conditional on each query; squared loss can only decrease. This gives at most M prediction vectors. Randomizing the encoder cannot beat the closest of these vectors at a fixed history. Projecting their centers by E cannot increase squared distance. For a uniform point X in a d -ball of radius ρ and any M centers, the union-of-balls estimate gives

$$\Pr\{\min_i \|X - z_i\| \leq r\} \leq \min\{1, M(r/\rho)^d\}.$$

Integrating its complementary tail yields

$$\mathbb{E} \min_i \|X - z_i\|^2 \geq \int_0^{\rho^2 M^{-2/d}} (1 - Mt^{d/2}/\rho^d) dt = \frac{d}{d+2} \rho^2 M^{-2/d}.$$

Multiply by β/s to obtain $c_N > 0$. Fresh public randomness can be fixed and then averaged because it is independent of the prefix. Private online coins may equivalently be drawn on an independent tape for this proof; fixing them makes a deterministic transducer and the same bound holds uniformly. That proof device supplies no tape to the implemented filter. The prescribed exploration is independent of that tape, a condition that is essential to this particular argument. An old command-generation seed is not fresh coding randomness and remains charged.

Theorem C.2 gives the upper bound in the same experiment. The minimax conditional criterion is at least its unconditional regret, while the universal upper filter works at all checkpoints. Finally put $M = 2^B$ and solve the two inequalities. The Euclidean covering mechanism is classical [4]; the identified sparse dimension and the actual repeated-update construction are the experiment-specific inputs. $\square$

Theorem C.4 (A separate finite-state control bound). *For the same finite detector and horizon, suppose stage rewards $r_n(g, x) \in [0, 1]$ are continuous in g for each report, and the objective is their expected sum. There is an M -state feedback controller whose loss relative to the full-history optimum is at most*

$$(127) \quad C'_N M^{-1/D*}.$$

No matching lower power for arbitrary control tasks is asserted. An additional finite payoff automaton must be counted as task state, multiplying the number of states if it is included.

Proof. Exact dynamic programming on $\mathcal{S}_n$ is valid by Proposition 2.3. With terminal value zero, its recursion is

$$V_n(s) = \max_{g \in \mathcal{G}} \sum_{x \in \mathcal{X}} p_x(s, g) \{r_{n+1}(g, x) + V_{n+1}(T_n(s, g, x))\}.$$

Here p_x is a linear moment functional or a normalized factor expression. The positive denominator bounds, compact action set and continuous rewards make the displayed objective continuous, and the maximum is attained. By backward induction it is uniformly Lipschitz in s , since finite sums, products of bounded Lipschitz functions, and the updates in Lemma C.1 preserve that property. The maximum of functions with a common Lipschitz constant is again Lipschitz. Denote by H_n a uniform Lipschitz constant of the action objective, before its maximization.

At each representative choose an exact maximizing action and store that action in the finite output table. At the true state s_n , the loss from using the action optimal at $\hat{s}_n$ is at most $2H_n e_n$. Update the representative by (125). The true exact state is computed only for analysis along the controller's actual observations; the implemented controller still stores only its index. Bellman telescoping and conditional expectation give total loss at most $2 \sum_{n < N} H_n e_n \leq 2 \sum_{n < N} H_n b_n M^{-1/D*}$. This proves (127). It uses no continuity of

the optimizing action as a function of state, and it does not differentiate an optimizer. Its approximation principle is related to [8]; it is not claimed as a new general dynamic-programming principle. $\square$

C.1. Horizon and calibration remain part of the constants. If every factor in a probe menu satisfies $0 < F \leq v < 1$, then $H_{n,j} \leq v^{N-n}$ and every posterior prediction satisfies the same bound. The one-state decoder that always announces zero therefore has

$$(128) \quad \mathcal{R} \leq v^{2(N-n)}.$$

For the shared physical menu $F = c + \delta k_j$ one may take $v = c + \delta$. Thus a checkpoint implementation may choose, in advance, between the streaming filter and the one-state decoder and obtain

$$\mathcal{R} \leq \min\{C_N M^{-2/D^*}, (c + \delta)^{2(N-n)}\}.$$

At $c = 1/2$, $\delta = 1/8$, $N - n = 20$, the second bound is $(5/8)^{40} < 10^{-8}$. This does not contradict a fixed- N high-resolution exponent. Neither the new sparse rank theorem nor the streaming construction turns it into a horizon-uniform useful-memory law. We have not rescaled the loss or discarded rare histories to avoid this bound.

The upper constants involve state bounds, transition Lipschitz constants and repeated error propagation. The lower constants involve an attainable Jacobian, a chart radius, evidence and the prior pairing. They are not uniform over vanishing detector amplitudes, all full-support priors or increasing horizons. With a finite fixed command alphabet and finite N there are finitely many histories and an exact finite code eventually suffices; the matching asymptotic lower bound above expressly uses the fixed continuous-command exploration protocol.

APPENDIX D. ATTAINABLE RESOLUTION FLAGS AND UNIFORM CAUSAL MEMORY

The full-future hypothesis in Section B makes every nonconstant future jet attainable. It is not needed for a resolution law, but it cannot simply be omitted from the formula there. When the past is shorter, the relevant list is an initial part of the future resolution flag. The lower bound requires transversality to this flag; the upper bound requires an estimate on the whole, generally curved, attainable image. We establish both and then prove their compatibility with every successive report update.

D.1. Statement of the attainable law. Retain the affine calibration and positivity assumptions of Section B, but do not impose (120). Fix $n, m \geq 1$. There are K_m formal sum pairs in $\mathcal{B}_m$; the zero pair is the constant, and the other $q_m = K_m - 1$ divided-difference coordinates have orders

$$\nu_{m,1} \leq \cdots \leq \nu_{m,q_m}.$$

Order equal orders by increasing limiting exponent, once and for all. Thus every initial part of the list includes, within each cluster, all jets preceding any jet that it contains. Set

$$(129) \quad p_{n,m} = \min\{n(r-1), q_m\}, \quad S_{m,\ell} = \sum_{i=1}^{\ell} \nu_{m,i}, \quad \Phi_{n,m}^{\text{att}}(M, \theta) = \max_{1 \leq \ell \leq p_{n,m}} \theta^{2S_{m,\ell}/\ell} M^{-2/\ell}.$$

Here $0^0 = 1$ and $0^s = 0$ for $s > 0$. If $n = 0$ or $m = 0$, define $\Phi_{n,m}^{\text{att}} = 0$ separately. No division by a zero dimension is then involved. Throughout, constants may depend on the fixed experiment, prior, horizon and calibration interval, but not on M or θ .

Theorem D.1 (Attainable confluent checkpoint law). *For every integer $M \geq 1$ and every calibration in the compact interval of Section B, the unconditional optimal M -message checkpoint regret under independent uniform commands and the minimax checkpoint regret both have order $\Phi_{n,m}^{\text{att}}(M, \theta)$, with positive uniform comparison constants. The upper code has at most M reachable representatives and works on every admitted history. The lower bound permits the independent encoder and decoder randomization specified in Definition 2.4.*

In particular, past limitation truncates the ordered ambient list at $p_{n,m}$; it does not introduce the remaining ambient orders into the rate. At zero the exponent is $2/\min\{n(r-1), |mA_0| - 1\}$, and at each fixed positive calibration it is $2/p_{n,m}$.

For the sequential assertion fix $N \geq 2$ and choose the compact calibration interval so that distinct formal pairs in $\mathcal{B}_m$, $1 \leq m < N$, do not meet at a positive calibration. Such an interval always exists after decreasing its positive endpoint: there are finitely many pairs and their exponent differences are affine. Put

$$(130) \quad \Omega_N(M, \theta) = \max_{1 \leq n < N} \Phi_{n, N-n}^{\text{att}}(M, \theta).$$

Use the same independent uniform command law at every acquisition time. For each selected checkpoint reveal its independent product query only after the index has been formed. As in Section C, only one checkpoint and its remaining query trials are executed in a run. Let $\mathfrak{R}_{M,N}^{\text{av}}$ be the infimum, over one streaming filter, of the maximum of its unconditional checkpoint regrets. Let $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding infimum of the maximum conditional regret over checkpoints and histories.

Theorem D.2 (Uniform causal realization of the resolution flag). *In this fixed-horizon experiment there are $0 < c \leq C < \infty$ such that*

$$(131) \quad c\Omega_N(M, \theta) \leq \mathfrak{R}_{M,N}^{\text{av}}(\theta) \leq \mathfrak{R}_{M,N}^{\text{max}}(\theta) \leq C\Omega_N(M, \theta)$$

for every $M \geq 1$ and every admitted θ , including zero. For each known θ and each M , there exists a single deterministic M -state transducer attaining the upper bound after every report. Its codebooks and transitions may depend on θ and M ; the comparison constants do not. No joint-in-calibration measurable or computable selection of codebooks is asserted. It stores only a representative index; its transitions use physical moments, never division by a collision parameter. Read-only prior moments at the formal exponents in $\mathcal{B}_N$ suffice. The lower bound holds in the stated common exploration experiment, including for the independent randomized filters of Definition 2.4.

These are statements about finite persistent memory in the exact-real program model already defined, not about finite-precision synthesis. In particular the number of real coordinates used to describe a representative is not the persistent state count: the latter is the number of representative labels, at most M .

D.2. A global covering estimate. We first record the real-geometric input with its anisotropic consequence. A bounded semialgebraic format means a bound on the number of variables, polynomial relations and their degrees in a defining formula; arbitrary real coefficients are allowed. Quantified formulas of fixed format can also be used, since real quantifier elimination gives a bound depending only on this format. Neither a coefficient norm nor a lower bound on a nonzero coefficient is part of the definition.

The proof of Lemma 4.4 appears in the principal analytic inputs.

The bounded-format hypothesis is essential to a uniform statement of this kind. An unrestricted smooth image could develop arbitrarily many sheets. In the present experiment this

hypothesis is a consequence of the finite product likelihood, not an extra regularity assumption on the prior or on an unknown attainable variety.

Lemma D.3 (The whole attainable image has bounded format). *Let $z_\theta(h) \in \mathbb{R}^{q_m}$ be the nonconstant divided-difference moment vector, in the ordered coordinates of (129), and put*

$$D_{m,\theta} = \text{diag}(\theta^{\nu_{m,i}}), \quad S_{n,m}^\theta = \{D_{m,\theta} z_\theta(h) : |h| = n\}.$$

The set $S_{n,m}^\theta$ is compact, has semialgebraic format bounded independently of θ , and has dimension at most $p_{n,m}$. It is contained in a rectangle with sides $[-B\theta^{\nu_{m,i}}, B\theta^{\nu_{m,i}}]$, with B uniform. At zero all positive-order coordinates are zero.

Proof. Fix the finite report word $x_1, \dots, x_n$ and write u for its command entries. Every report factor is affine in its current command. Thus the evidence $Z(u) = \mu \prod_{i=1}^n f_{x_i, g_i}^\theta$ and each numerator

$$N_j(u) = \mu \left(\psi_j^\theta \prod_{i=1}^n f_{x_i, g_i}^\theta \right)$$

are polynomials of degree at most n in u . Their coefficients are real prior integrals, well defined also at zero. The moment image is specified by the existence of u in its fixed cube with $Z(u)y_j = \theta^{\nu_{m,j}} N_j(u)$ for every j . Moreover $Z(u) \geq \kappa^n > 0$. This is a semialgebraic formula of fixed format, uniformly in its real coefficients. Finite union over the report words preserves a format bound, and continuity on compact command cubes proves compactness. There is no requirement that these real coefficients, as functions of θ , be semialgebraic.

For the dimension bound, divide each report factor by its prior integral. Its coefficient vector lies in the affine hyperplane $\mu f = 1$ of the r -dimensional one-step space. The constant monomial allows its constant coefficient to be eliminated, leaving at most $r - 1$ coordinates. The product posterior and all displayed moments are rational functions of these $n(r - 1)$ coordinates. A rational map cannot increase semialgebraic dimension. Multiplication by $D_{m,\theta}$ cannot increase it either; the ambient dimension is q_m . Finally the uniform bounds for ψ_j^θ in Lemma B.2 give $|z_j| \leq B$ before any command integration: they are expectations under probability measures. This proves the rectangle assertion without an inverse power of the prefix evidence. $\square$

D.3. Transversality to every initial resolution flag.

Lemma D.4 (Attainment of the truncated flag). *At one common interior binomial command tuple, projection of $z_\theta(h)$ onto its first $p_{n,m}$ coordinates has full row rank for every calibration, including zero. Under independent uniform commands, the all-failure subprobability distribution of this projection dominates β times uniform probability on a translated cube of side 2ρ , where $\beta, \rho > 0$ are independent of the calibration.*

Proof. Choose distinct sufficiently small $c_1, \dots, c_n$ and realize the factors $(1 + c_i t^{D_\theta})/2$ by the fixed right inverse of the calibration matrix, as in Lemma 2.1. Their commands stay in a common interior neighborhood. Lemma 3.2 identifies their unnormalized tangent with a monomial space of dimension $n(r - 1) + 1$, for every calibration, including zero.

Adjoin the constant to the selected $p = p_{n,m}$ future tests. Within each limiting exponent cluster, the selected tests are exactly an initial block of its divided differences. At a positive calibration this block is an invertible linear transformation of the monomials at its first nodes. Their union consists of distinct monomials. The mixed pairing with the binomial tangent therefore has column rank $p + 1$, by Lemma 3.1. At zero these same blocks become complete initial logarithmic-jet blocks. Lemma 4.3 gives column rank $p + 1$ directly at zero. This is not an inference of rank at zero from rank on a punctured interval.

Let L be this pairing, let F be the product at the witness, and write $v = LF/\mu F$, with constant coordinate $e_0 v = 1$. The normalized differential has image

$$(\mu F)^{-1}\{LQ - v e_0 LQ : Q \text{ in the product tangent}\}.$$

Since L is onto $\mathbb{R}^{p+1}$, the bracket is precisely the p -dimensional hyperplane $e_0 = 0$. Deleting the constant proves the asserted row rank. The remaining, unselected future coordinates need not be independent and are not included in this submersion assertion.

For clarity we give the uniformity and probability argument. On the compact calibration interval, the least row singular value of the selected derivative has a positive minimum, by the rank just proved and continuity of all divided-difference derivatives. The second command derivatives have a common bound, since the factors stay positive and the test functions are uniformly bounded. Complete the row coordinates with orthonormal kernel coordinates. On a fixed small command ball the resulting square derivative differs from its value at the center by at most half its uniformly positive least singular value. The contraction proof of the inverse theorem gives a common product box in the selected output and complementary coordinates. Shrink it so its inverse stays in the command interior. Uniform forward derivative bounds give a positive lower bound for the absolute inverse Jacobian. These estimates use only orthogonality of the chosen frames; a continuous choice of frames is unnecessary.

If k is the number of command entries, their joint density is $(1 - 2\eta)^{-k}$. The density of commands together with the all-failure word is this density multiplied by its evidence, at least η^n . Change variables using the local inverse and integrate over the common complementary box. The resulting measure dominates a fixed positive multiple of Lebesgue measure on the selected p -cube. Absorb the cube volume into β . This proves the stated subprobability minorization; it does not condition away the probability of the failure word. A full-support prior without a density is permitted throughout, since the continuous density used here is that of the commands. $\square$

Proof of Theorem D.1. The physical prediction identity of Lemma B.2 reads, after the fixed coordinate permutation,

$$(132) \quad p_\theta(h) - p_\theta(\tilde{h}) = G_m D_{m,\theta}(z_\theta(h) - z_\theta(\tilde{h})),$$

where G_m has full column rank and is independent of θ . Apply Lemmas 4.4 and D.3 to the complete attainable image. Since $0 \leq \theta \leq 1$, its ordered side lengths are $B\theta^{\nu_{m,i}}$. Equation (13) supplies at most M reachable representatives with squared covering radius at most $C\Phi_{n,m}^{\text{att}}$. The fixed norm of G_m and the finite query average convert this into the required uniform prediction upper bound. This is a cover of all report strata, not just the local patch used in the lower bound.

For that lower bound, first average decoder randomization for each message and query. The outputs form at most M centers. Orthogonally project them to the affine span $p_* + \text{im } G_m$ and apply the fixed left inverse of G_m . Distances can decrease only by a fixed factor. Project further onto the first $p_{n,m}$ coordinates. Lemma D.4 then gives a subprobability multiple of uniform measure on a translated rectangle of side lengths $2\rho\theta^{\nu_{m,i}}$, $i \leq p_{n,m}$. The lower half of Lemma B.4, projecting onto each first ℓ axes, gives $c\Phi_{n,m}^{\text{att}}$. Centers may be outside the image; randomized assignment cannot improve on the nearest center. Fixing and averaging an independent public coding seed gives the same bound. At zero, omit the zero-width axes. There are $\min\{n(r-1), |mA_0| - 1\}$ surviving axes, which also proves the stated limiting exponent. For positive calibration the last power $M^{-2/p_{n,m}}$ dominates as $M \rightarrow \infty$. The unconditional lower bound is also a minimax lower bound, completing the proof. $\square$

D.4. Physical moments and successive updates. For $b = (\lambda, \gamma) \in \mathcal{B}_m$, write

$$v_b(h) = \mu(P_h t^{\lambda+\theta\gamma}), \quad v_{(0,0)} = 1.$$

We keep the formal labels even if the functions coincide at zero. Let $v^{(m)}$ denote the nonconstant vector. Newton interpolation within each cluster, as in Lemma B.2, gives a fixed invertible matrix T_m with

$$(133) \quad v^{(m)} = T_m D_{m,\theta} z_\theta.$$

The constant cluster has no other member, so no deleted constant mixes with this equation. Both T_m and its inverse have fixed finite norms. At zero the equation still holds with the appropriate repeated raw moments. It is the analysis of this equation, not inversion of its vanishing diagonal, that identifies the metric scales.

Lemma D.5 (Uniform updates in the physical metric). *The map from reachable raw moments at remaining length m to those at length $m-1$, after a command and a report, is uniformly Lipschitz in the raw Euclidean metric, independently of calibration. Every reachable representative maps to a reachable next vector.*

Proof. Write $e_i = (a_i, b_i)$ and the current report factor as $f = \sum_i f_i(g, x) t^{a_i+\theta b_i}$. Its coefficients are uniformly bounded, and $f \geq \kappa > 0$. Addition of formal pairs gives $b + e_i \in \mathcal{B}_m$ whenever $b \in \mathcal{B}_{m-1}$. Consequently the exact update is

$$(134) \quad v'_b = \frac{\sum_i f_i(g, x) v_{b+e_i}}{\sum_i f_i(g, x) v_{e_i}}, \quad b \in \mathcal{B}_{m-1}.$$

Every denominator is the posterior expectation of f , hence at least κ . A line segment between reachable raw vectors represents a mixture of their posterior probability measures, so the same bound holds along the segment. All numerators, coefficients and coordinates have uniform bounds. The quotient rule and the mean value theorem prove the Lipschitz assertion. No exponent separation occurs in a denominator. The formula is valid also when formal labels have equal values. Finally a history realizing the representative, followed by the specified input, realizes its updated vector; positivity admits every report. $\square$

Proof of Theorem D.2. For each $1 \leq n < N$, Lemmas 4.4 and D.3, followed by the fixed matrix T_{N-n} , give a raw-moment quantizer Q_n with at most M reachable representatives and radius

$$\delta_n \leq C_n \sqrt{\Phi_{n,N-n}^{\text{att}}(M, \theta)} \leq C_0 \sqrt{\Omega_N(M, \theta)}.$$

The maximum C_0 is over finitely many stages. At time zero the prior moment vector is one fixed state; at time N the remaining state is constant. Initialize at the prior vector and implement

$$\hat{v}_{n+1} = Q_{n+1}(U_n(\hat{v}_n, g_{n+1}, x_{n+1})),$$

where U_n is (134). The transition reads the old representative index and the current input only. Reachability makes its argument an element of the next quantizer's domain. The finite codebooks and these real-valued functions are read-only program data.

Let $e_n = \|v_n - \hat{v}_n\|$. With $e_0 = 0$, Lemma D.5 yields

$$e_{n+1} \leq L_n e_n + C_0 \sqrt{\Omega_N(M, \theta)}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C_0$. Induction gives $e_n \leq b_n \sqrt{\Omega_N(M, \theta)}$ after every update. Each query prediction is a linear form with fixed coefficients in the raw formal moments. Its Lipschitz constant is therefore uniform, and (4) proves the last inequality in (131). There is no exact-prefix recomputation: all intermediate vectors in this implementation are functions of a finite representative label and the current command-report pair.

For every streaming filter, its index at checkpoint n is an M -message encoder of the prefix. Theorem D.1 therefore bounds that checkpoint's unconditional regret below by $c_n \Phi_{n, N-n}^{\text{att}}$. The commands in all these assertions are prefixes of the same independent uniform acquisition rule. Taking the maximum over n , and then the infimum over filters, gives the first inequality with $c = \min_n c_n > 0$. Independent filtering coins can be fixed for this argument because they do not generate the acquisition commands; averaging restores the randomized statement. Finally a maximum conditional risk dominates its unconditional average, which proves the middle inequality. $\square$

Corollary D.6 (A uniform bit profile). *Let $B_N(\varepsilon, \theta)$ be the least number of persistent bits needed to make the minimax streaming regret at most ε . For $0 < \varepsilon < 1$, uniformly in the admitted calibration,*

$$(135) \quad B_N(\varepsilon, \theta) = \max \left\{ 0, \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n, N-n}}} \left(\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} - S_{N-n, \ell} \log_2 \frac{1}{\theta} \right) \right\} + O(1).$$

At zero retain only terms with $S_{N-n, \ell} = 0$. The same formula holds for the maximum unconditional criterion.

Proof. For positive θ , the condition $\Omega_N(M, \theta) \leq \varepsilon$ is equivalent to all inequalities $M \geq \theta^{S_{N-n, \ell}} \varepsilon^{-\ell/2}$. Taking the largest right-hand side, adjoining $M \geq 1$, and then taking base-two logarithms gives the expression. The two fixed comparison constants in (131) change it by a bounded amount, since the possible ℓ are bounded by the fixed horizon. Rounding to a power of two changes it by at most one further bit. The zero-calibration statement follows by omitting vanishing terms. $\square$

D.5. A past-limited peak with two weak directions. The next consequence answers the concrete seven-trial boundary in the source-pinned v7 technical note [19]. It is a consequence of the attainable-image argument, rather than an extrapolation of the full-future formula.

Corollary D.7 (Seven-trial resolved memory). *For the cells (137), their fixed command cube and the product probes (138), take total horizon seven. For the two streaming criteria above, uniformly for $0 \leq \theta \leq 1/2$ and $M \geq 1$,*

$$(136) \quad \mathfrak{R}_{M,7}(\theta) \asymp \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The lower bound already holds at checkpoint four under the same uniform exploration. The crossover budget has order θ^{-6} and the crossover regret has order θ^2 .

Proof. Put $D = 2 + \theta$. For positive $\theta \leq 1/2$, the ten distinct three-fold exponents are

$$0, 1, 2, D, 3, D+1, D+2, 2D, 2D+1, 3D.$$

At zero their limiting clusters have multiplicities 1, 1, 2, 2, 2, 1, 1 at exponents 0, 1, 2, 3, 4, 5, 6. Thus the nonconstant order list has six zeros and three ones. At $n = 4, m = 3$, however, $p_{4,3} = 8$, and the selected list has six zeros and only two ones. Theorem D.1 gives

$$\Phi_{4,3}^{\text{att}} = \max\{M^{-1/3}, \theta^{2/7} M^{-2/7}, \theta^{1/2} M^{-1/4}\} = \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The middle term is the product of the first term to power 3/7 and the last to power 4/7, so cannot exceed their maximum. The minorized eight-dimensional flag uses, at zero, the tests

$$1, t, t^2, t^2 \log t, t^3, t^3 \log t, t^4, t^5, t^6$$

paired with the binomial tangent $\mathcal{P}_8$. Strict confluent positivity proves its rank for every full-support prior. The third ambient weak jet is not counted as an additional attainable dimension.

For completeness the extension of the upper bound to the whole stated calibration interval does not require excluding positive collisions in longer, unused sumsets. At stages $n = 1, 2, 3$, the raw reachable images have dimension at most $2n \leq 6$ and bounded format, and all raw coordinates lie in $[0, 1]$. Lemma 4.4 gives squared covering errors at most $CM^{-1/3}$ uniformly, whether or not some formal raw coordinates coincide. At stage four use the preceding anisotropic cover, valid throughout $[0, 1/2]$ because its ten exponents are distinct at every positive calibration in that interval. At stage five, $m = 2$ and the full-future cover has squared radius at most $C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\} \leq CM^{-1/3}$. Stage six has two coordinates and squared covering radius at most CM^{-1} . Formula (134) remains valid with all formal labels at every stage, even when values coincide. Its uniform Lipschitz estimate and the repeated-quantization argument prove the upper bound in (136) on the entire interval. The stage-four checkpoint lower bound proves the other direction.

Equating the two terms in (136) gives $M = \theta^{-6}$ and common order θ^2 . These are comparison orders, not a numerical integer threshold. At fixed positive calibration the rate is $M^{-1/4}$, consistently with eight attainable dimensions; the ambient nine-dimensional rate $M^{-2/9}$ does not occur. $\square$

APPENDIX E. A UNIFORM STREAMING LAW THROUGH AN ADDITIVE COLLISION

We apply the confluent law to a family in which exact dimension changes at a fixed horizon and at a fixed positivity margin. In this family all repeated quantization errors can be controlled in the physical prediction metric, uniformly at the collision.

E.1. One fixed five-trial protocol. Let $0 \leq \theta \leq 1/2$, $A_\theta = \{0, 1, 2 + \theta\}$, and let μ be any fixed full-support probability on $[0, 1]$. Define

$$(137) \quad \begin{aligned} k_1^\theta &= \frac{1}{3} + \frac{1}{12}t, & k_2^\theta &= \frac{1}{3} + \frac{1}{12}t^{2+\theta}, \\ k_0^\theta &= \frac{1}{3} - \frac{1}{12}(t + t^{2+\theta}). \end{aligned}$$

These cells are at least $1/6$, sum to one, and span W_{A_θ} . Use the fixed comparator cube $[1/4, 3/4]^3$; every report factor is at least $\kappa = 1/24$. The three fixed probe commands have failure factors

$$(138) \quad F_j^\theta = \frac{1}{2} + \frac{1}{8}k_j^\theta, \quad j = 0, 1, 2.$$

The command accepts cell j with probability $3/8$ and the other cells with probability $1/2$. Their sum is $13/8$, so these factors span W_{A_θ} . At a checkpoint n use the 3^{5-n} ordered product queries of length $5 - n$. Exactly one checkpoint and one query are used in a run; all five trials are counted.

For the unconditional experiment the first three commands are independent uniform vectors in the comparator cube. After their reports have been compressed, reveal the independent uniform two-step query. The controller knows θ and μ as read-only calibration, but has access to past commands and reports only through its persistent index. The input law and all query labels are independent of θ .

For a history write $M_s = \mu(P_h t^s)$ and use the five coordinates

$$(139) \quad w = (M_1, M_2, M_{3+\theta}, M_{4+2\theta}, M_{2+\theta} - M_2).$$

They determine every length-two future law. They are ordinary posterior expectations of physical tests and their differences. For analysis put

$$(140) \quad z = (w_1, w_2, w_3, w_4, w_5/\theta) \quad (\theta > 0), \quad z_5|_{\theta=0} = \mu(P_h t^2 \log t).$$

This fifth coordinate at zero is an analytic continuation, not an extra observable supplied to the filter. The zero-calibration future law uses only the first four coordinates.

Theorem E.1 (Sharp two-parameter streaming order). *Set*

$$(141) \quad \Psi_\theta(M) = \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}, \quad M \geq 1,$$

where the second term is zero at $\theta = 0$. There are constants $0 < c \leq C < \infty$, depending on the fixed prior and protocol but not on M or $\theta \in [0, 1/2]$, with the following properties.

There is a deterministic M -state filter, updated after every report, whose conditional prediction regret at every checkpoint and every admitted history is at most $C\Psi_\theta(M)$. In the prescribed three-trial exploration experiment, every M -state filter has unconditional regret at least $c\Psi_\theta(M)$. The same lower bound therefore holds for the minimax criterion over histories and checkpoints. The lower assertion permits the independent randomization of Definition 2.4.

In particular the two regimes are

$$(142) \quad \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}} \asymp \begin{cases} M^{-1/2}, & 1 \leq M \leq \theta^{-4}, \\ \theta^{2/5} M^{-2/5}, & M \geq \theta^{-4}, \end{cases}$$

with uniform comparison constants. This is an order-sharp crossover, not a claim of an exact transition budget or a leading distortion constant. The family and its base-model perturbation estimate were introduced in the source-pinned technical note [14]. The conclusion here is the matching, uniform two-parameter law and its online implementation.

E.2. Attainment of the fifth desingularized direction.

Lemma E.2 (A nondegenerate limiting patch). *At a common interior three-command tuple the map to z in (140) has rank five for every $\theta \in [0, 1/2]$. Its all-failure subprobability law dominates a fixed positive multiple of uniform probability on a translated five-dimensional cube of fixed side length, uniformly in θ .*

Proof. The formal two-fold sum pairs give exponents

$$0, 1, 2, 2 + \theta, 3 + \theta, 4 + 2\theta.$$

There is just one double cluster, at exponent two. Its divided difference is $(t^{2+\theta} - t^2)/\theta$. The nonconstant orders are $(0, 0, 0, 0, 1)$. The condition in (120) is $5 \leq 3(3 - 1) = 6$.

For an explicit common attainable tuple take $c_1 = 1/64$, $c_2 = 2/64$, $c_3 = 3/64$ and failure factors $(1 + c_i t^{2+\theta})/2$. They are realized by rejection probabilities

$$(1 - g_0, 1 - g_1, 1 - g_2) = (1/2 - 2c_i, 1/2 - 2c_i, 1/2 + 4c_i),$$

strictly inside $[1/4, 3/4]^3$. At zero their product tangent is the entire polynomial space of degree at most six. The future confluent space has basis

$$1, \quad t, \quad t^2, \quad t^2 \log t, \quad t^3, \quad t^4.$$

Lemma 4.3 makes its mixed pairing with that tangent surjective. Normalization removes one dimension, not two. For example, for the uniform prior, the minor with rows t^i , $0 \leq i \leq 5$, is exactly

$$(143) \quad \det \left[\frac{1}{i+1}, \frac{1}{i+2}, \frac{1}{i+3}, -\frac{1}{(i+3)^2}, \frac{1}{i+4}, \frac{1}{i+5} \right]_{i=0}^5 = \frac{1}{338751673344000000} > 0.$$

This rational identity is only a diagnostic of the preceding all-prior argument.

For every positive $\theta \leq 1/2$ the six displayed exponents are distinct and the binomial tangent has seven distinct exponents. Strict mixed-moment positivity gives surjectivity throughout this interval. All derivatives in the desingularized coordinates extend continuously to zero, as proved in Lemma B.2. Compactness and the uniform inverse construction in Lemma B.3 therefore apply on the *whole* interval $[0, 1/2]$, not only on an unspecified punctured neighborhood. The exact change of order in (140) is a fixed permutation of the general jet coordinates and changes no rank or minorization conclusion. $\square$

E.3. Uniform causal quantization.

Proof of Theorem E.1. At times one and two retain, exactly for the moment, the chronological normalized factors $f_i/\mu f_i$, with two coefficient coordinates per factor. At time three change to w in (139). At time four retain $(M_1, M_{2+\theta})$, and at time five use the unique terminal state. These states are sufficient for the remaining observable future law. All their reachable sets are compact.

The first two factor sets lie in fixed boxes of dimensions two and four, uniformly in θ . Indeed report coefficients are uniformly bounded, $\mu f \geq \kappa$, and the coefficient hyperplane $\mu f = 1$ can be coordinatized by the two nonconstant coefficients, its constant coefficient then being $1 - \sum_{a>0} f_a \mu(t^a)$. The required prior moments are fixed read-only data for each known calibration. Uniform boundedness of these affine formulas on $[0, 1/2]$ is immediate.

At time three the first four coordinates of w belong to $[0, 1]$. For $0 < t \leq 1$, the fundamental theorem of calculus in the exponent gives

$$|t^{2+\theta} - t^2| \leq \theta t^2 |\log t| \leq \theta.$$

This extends to zero. Consequently $-\theta \leq w_5 \leq 0$ for every posterior, with no evidence denominator in this bound. At time four both retained moments lie in $[0, 1]$. Lemma B.4, with reachable representatives in nonempty cells, now gives at most M representatives at each time and squared covering errors

$$(144) \quad \delta_1^2 \leq C_1 M^{-1}, \quad \delta_2^2 \leq C_2 M^{-1/2}, \quad \delta_3^2 \leq C_3 \Psi_\theta(M), \quad \delta_4^2 \leq C_4 M^{-1}.$$

The C_i are independent of θ . At zero simply omit the last, constant coordinate of the third state. Every bound in (144) is at most $C_0 \Psi_\theta(M)$.

It remains to verify that repeated updates do not divide by θ . The factor-to-factor update appends the current normalized factor. The factor-to- w update uses the product of the two retained factors and the new report factor. It is a ratio of polynomial coefficient expressions and bounded prior integrals. Along line segments between factor tuples every interpolated normalized factor is at least κ , so the evidence denominators and derivatives have positive uniform bounds. Each physical test in w is uniformly bounded; differentiating in factor coordinates therefore gives a uniform Lipschitz constant for this changeover.

For the only nontrivial shrinking update, write the new report factor as $f = a + bt + ct^{2+\theta}$. In terms of w set

$$Z_f = a + bw_1 + c(w_2 + w_5).$$

The next two moments are exactly

$$(145) \quad M'_1 = \frac{aw_1 + bw_2 + cw_3}{Z_f}, \quad M'_{2+\theta} = \frac{a(w_2 + w_5) + bw_3 + cw_4}{Z_f}.$$

Coefficients are uniformly bounded and $Z_f \geq \kappa$. A line segment between reachable w vectors represents a posterior mixture, so the same lower bound holds along that segment. The quotient

rule proves a uniform Lipschitz estimate in the *physical* w coordinates. There is no divided difference and no factor $1/\theta$ in (145). The next, terminal update is constant.

At each time the filter stores only a representative index and applies the exact transition to that representative and the current input, then requantizes. A reachable representative followed by any admitted report has a reachable image, so every quantizer is used inside its domain. If e_n is the state error in the appropriate Euclidean physical coordinates, the preceding bounds give

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_0^{1/2} \Psi_\theta(M)^{1/2},$$

with L_n independent of θ . Iterate this recurrence through the five times. At times one and two each query prediction is a normalized polynomial expression in retained factors; at times three and four it is a fixed linear expression in the retained physical moments. These prediction maps have uniform Lipschitz constants, again by the positive evidence bounds. Squaring the resulting prediction error proves the conditional upper bound at every checkpoint. The representatives and transition functions are supplied existentially from the exact model, not synthesized from finite-precision exponent data. Only their index is persistent mutable memory.

For the lower bound at time three apply Lemma E.2 and the physical-metric argument in the proof of Theorem B.1, now to the nine common queries in (138). Their coefficient change from any other fixed spanning basis is invertible before taking products, so its full product expansion has constant full column rank. Four physical directions have scale one and the last has scale θ . Lemma B.4 gives the maximum of the four-dimensional lower bound $cM^{-1/2}$ and the five-dimensional lower bound $c\theta^{2/5}M^{-2/5}$. The constants are uniform on $[0, 1/2]$. Every streaming filter is an M -message encoder at this time, and the retained all-failure evidence is at least $(1/4)^3$. The randomization argument of Theorem B.1 applies unchanged. Finally, minimax regret is at least the unconditional regret in this one prescribed experiment. This completes the proof. $\square$

Corollary E.3 (Uniform bit law and the lower-constant scale). *For sufficiently small $\varepsilon > 0$, the least persistent bits for unconditional regret at most ε in the five-trial protocol, and likewise for its checkpoint-uniform minimax criterion, satisfy*

$$(146) \quad B_*(\varepsilon, \theta) = \log_2 \max\{\varepsilon^{-2}, \theta\varepsilon^{-5/2}\} + O(1),$$

where the $O(1)$ is uniform in $\theta \in [0, 1/2]$. If

$$c_{\text{opt}}(\theta) = \inf_{M \in \mathbb{N}, M \geq 1} M^{2/5} \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}},$$

then $c_{\text{opt}}(\theta) \asymp \theta^{2/5}$ for $0 < \theta \leq 1/2$, with constants independent of θ .

Proof. Solve the two necessary and sufficient inequalities from Theorem E.1 for M , obtaining, up to constant factors, $M \geq \varepsilon^{-2}$ and $M \geq \theta\varepsilon^{-5/2}$. Rounding to $M = 2^B$ changes B by a bounded amount. The same bounds apply to the minimax criterion. The lower bound for c_{opt} follows from the five-dimensional term. For its upper bound let M tend to infinity in the matching upper estimate, so $M^{-1/10}$ disappears. No limiting leading quantization constant is needed. $\square$

The exact profiles remain $(0, 2, 4, 4, 2, 0)$ at zero and $(0, 2, 4, 5, 2, 0)$ at positive θ . The new assertion is that the extra direction is resolved at distortion scale θ^2 and state-budget scale θ^{-4} , with matching estimates on both sides. It sharpens the base-model bound $C(M^{-1/2} + \theta^2)$ without contradicting it.

APPENDIX F. THE VALUE OF A RESOLVED DIRECTION IN ONE SEQUENTIAL DECISION

We next use the same exploration histories, the same two-step query menu and the same eventual Bernoulli event as in Section E. Only the terminal decision is changed. This gives a common-payoff comparison between keeping and erasing the newly resolved direction, rather than comparing two separately normalized prediction tasks.

F.1. An independently priced observable ticket. After the time-three state has been formed, reveal the query J and a price U , independent and uniform on $[0, 1]$. The controller chooses a binary action $A \in \{0, 1\}$. Execute the two query trials and let Y be the same all-failure indicator as before. The payoff is

$$(147) \quad A(Y - U).$$

The price and the query are new inputs, not historical storage. The payoff belongs to $[-1, 1]$ and depends only on observable apparatus outcomes and the chosen action. Accepting the ticket changes the payoff, not the detector law or the five-trial budget. The exploration command program is prescribed; it is not optimized by the controller.

Let $p_j(h) = \mathbb{P}(Y = 1 \mid h, J = j)$. Write V_*^θ for the full-history optimal expected payoff and V_M^θ for the supremum over all M -state streaming policies, including randomized transitions and terminal actions. At the terminal decision the policy reads only its index, the new query and the new price.

Proposition F.1 (Exact decision–prediction identity). *For every fixed streaming encoder with index I , its optimal terminal rule accepts exactly when $U \leq \mathbb{E}[p_J(h) \mid I, J]$. Its value gap is*

$$(148) \quad V_*^\theta - V^\theta(I) = \frac{1}{2s} \sum_{j=1}^s \mathbb{E} \left[(p_j(h) - \mathbb{E}[p_j(h) \mid I])^2 \right], \quad s = 9.$$

Consequently

$$(149) \quad c\Psi_\theta(M) \leq V_*^\theta - V_M^\theta \leq C\Psi_\theta(M)$$

with positive constants independent of M and $\theta \in [0, 1/2]$.

Proof. Conditional on the full history and $J = j, U = u$, action one has expected payoff $p_j - u$ and action zero has payoff zero. The optimum is the threshold rule and its average over price is $\int_0^{p_j} (p_j - u) du = p_j^2/2$. Given only an index, replace p_j by $\bar{p}_{I,j} = \mathbb{E}[p_j \mid I]$, since price and query are independent of the prefix and encoder. A randomized action is a convex combination of the two payoffs and cannot improve the threshold optimum. Thus

$$V_*^\theta = \frac{1}{2s} \sum_j \mathbb{E} p_j^2, \quad V^\theta(I) = \frac{1}{2s} \sum_j \mathbb{E} \bar{p}_{I,j}^2.$$

Conditional-variance decomposition gives (148). Conditional expectation is also the optimal squared-loss decoder for the same encoder. Taking the infimum over the same streaming encoders therefore makes the optimal value gap exactly one half the optimal prediction regret. Theorem E.1 proves (149). The upper implementation may also threshold the representative's prediction directly: conditional on a history, integrating the mistaken decisions over the interval between that prediction and p_j gives half their squared difference. $\square$

This proposition is a decision identity, not a new general comparison principle. Its use here is to retain the entire resolved streaming law in a bounded, binary-action task and then compare a specific erased observation in that *same* task.

F.2. Erasing the weak coordinate. Let

$$(150) \quad T_\theta(h) = (M_1, M_2, M_{3+\theta}, M_{4+2\theta})$$

be the first four coordinates of (139). This is a deterministic, calibration-dependent postprocessing of the history. It is not a coarsening of a raw one-step detector symbol, and it is not claimed to be the best possible four-dimensional encoding. Let V_T^θ be the optimal payoff of a terminal policy given the *entire exact* statistic T_θ , with no memory restriction, and then the same J, U . This comparison gives the erased controller more computational and storage power than any M -message compression of T_θ .

Theorem F.2 (Uniform value of the weak direction). *There are constants $c_e, C_e, K > 0$, depending on the fixed prior and protocol but independent of $0 < \theta \leq 1/2$, such that*

$$(151) \quad c_e \theta^2 \leq V_*^\theta - V_T^\theta \leq C_e \theta^2.$$

Moreover, whenever $M \geq K\theta^{-4}$,

$$(152) \quad V_M^\theta - V_T^\theta \geq \frac{1}{2} c_e \theta^2 > 0.$$

In particular an M -state streaming policy beats every policy whose only historical input is an M -message compression of T_θ , even though V_T^θ allows that comparator uncompressed access to the four coordinates. Both values concern the same actual calibration, exploration law, query, price, target, payoff and five-trial budget. At $\theta = 0$ the statistic T_0 is sufficient and $V_^0 = V_T^0$.*

Proof. Put $Z_\theta = w_5/\theta$ for positive θ , with its continuous value at zero as in (140). The common product query expansion has the form

$$(153) \quad p_j(h) = a_j + \ell_j(T_\theta(h)) + \theta b_j Z_\theta(h),$$

where a_j, b_j and the linear forms ℓ_j are independent of θ . They are obtained by expanding the two factors in (138) into the six formal monomials and replacing $M_{2+\theta}$ by $M_2 + \theta Z_\theta$. The full coefficient matrix has rank six, as proved in Lemma B.2. Hence $\sum_j b_j^2 > 0$; otherwise its weak column would be zero.

Apply the conditional-variance identity from Proposition F.1 to the sigma-algebra generated by T_θ . With $b_* = (2s)^{-1} \sum_j b_j^2 > 0$ one obtains the exact formula

$$(154) \quad V_*^\theta - V_T^\theta = b_* \theta^2 \mathbb{E} \text{Var}(Z_\theta | T_\theta).$$

The random variable Z_θ is bounded uniformly, since $|(t^{2+\theta} - t^2)/\theta| \leq t^2 |\log t| \leq 1$ before taking any posterior expectation. Thus the conditional variance is uniformly bounded above.

For the lower bound, Lemma E.2 gives a fixed positive subprobability multiple β of uniform measure on a translated cube of side 2ρ in the coordinates (T_θ, Z_θ) . For every Borel function v of the first four coordinates,

$$\mathbb{E}|Z_\theta - v(T_\theta)|^2 \geq \beta \int |z - v(t)|^2 d \text{Unif}_{\text{cube}}(t, z) \geq \beta \rho^2 / 3.$$

The last inequality integrates the variance of the independent uniform last coordinate; it is unaffected by the translation. Taking the infimum over v proves $\mathbb{E} \text{Var}(Z_\theta | T_\theta) \geq \beta \rho^2 / 3$. This argument uses domination of an actual distribution, not an assumed uniform posterior on a formal tangent space. Equation (154) proves (151) with uniform positive constants.

By (149),

$$V_M^\theta - V_T^\theta \geq c_e \theta^2 - C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}.$$

For $M \geq K\theta^{-4}$ the maximum on the right is at most $\theta^2 \max\{K^{-1/2}, K^{-2/5}\}$. Choose $K \geq 1$ large enough that $C \max\{K^{-1/2}, K^{-2/5}\} \leq c_e/2$. This proves (152). A policy observing only a compression of T_θ cannot outperform the one observing all of T_θ : conditional on that statistic

its randomization is independent of the latent history, and averaging produces an admissible randomized T_θ -policy. Finally at zero the weak column in (153) vanishes, proving sufficiency and equality of the full and erased values. $\square$

The absolute advantage vanishes quadratically as the exponents collide. What remains nondegenerate is the advantage after normalization by the actual weak-direction scale θ^2 , once the state budget resolves that scale. The theorem is not a lower bound for arbitrary exploration objectives. It is an optimal-policy comparison in one prescribed sequential acquisition experiment, with no change of Bayes baseline and no substitution of the separate mechanical sign experiment.

Corollary F.3 (Necessary and sufficient budget order). *Define*

$$M_{\text{beat}}(\theta) = \min\{M \in \mathbb{N} : V_M^\theta > V_T^\theta\}.$$

For $0 < \theta \leq 1/2$ this minimum exists and $M_{\text{beat}}(\theta) \asymp \theta^{-4}$, with uniform constants. More explicitly, for some $0 < k < K < \infty$,

$$\begin{aligned} 1 \leq M \leq k\theta^{-4} &\implies V_M^\theta - V_T^\theta \leq -C_e\theta^2 < 0, \\ M \geq K\theta^{-4} &\implies V_M^\theta - V_T^\theta \geq \tfrac{1}{2}c_e\theta^2 > 0. \end{aligned}$$

The first assertion is vacuous if its integer range is empty.

Proof. Write $E_\theta = V_*^\theta - V_T^\theta$ and $L_M^\theta = V_*^\theta - V_M^\theta$. Theorems already proved give $E_\theta \leq C_e\theta^2$ and $L_M^\theta \geq c_s M^{-1/2}$. Their exact difference uses the same experiment and baseline, hence

$$V_M^\theta - V_T^\theta = E_\theta - L_M^\theta \leq C_e\theta^2 - c_s M^{-1/2}.$$

Choose $k = (c_s/(2C_e))^2$, decreasing it if necessary so that $k < K$. The first inequality follows. The second is Theorem F.2, with K increased if needed. The admissible filter classes are nested as M increases because extra states can be unused. Thus the first-crossing budget exists; the two bounds and integer rounding give its stated order. $\square$

This consequence was extracted in the source-pinned v7 referee technical note [19]. It concerns beating the exact, uncompressed T_θ comparator. It does not locate an exact integer transition, and it does not assert equality of the full and erased experiments at smaller *equal compressed* budgets. At zero T_0 is sufficient, so no finite-state policy has value exceeding $V_T^0 = V_*^0$.

APPENDIX G. A COLLISION BIT AT A COMMON FINITE-MEMORY BUDGET

This section compares total risks for one fixed target, not regrets against two different Bayes forecasters. It concerns an uncensored raw detector, so that erasure is literal postprocessing. It does not identify this operation with erasure *before* a commanded comparator.

G.1. The counted mechanical realization. Let the triangular lattice have fundamental area $\mathfrak{a} = \sqrt{3}/2$, let $I_R = [9/20, 47/100]$, and put a reflecting disk of radius R at each lattice point. In each cartridge prepare a uniform position in the ambient torus, a uniform velocity direction, and independent uniform apparatus coordinates. A guarded insertion fails when the position is inside a disk. Otherwise a unit-speed particle is released. Every attempted insertion consumes one cartridge.

Fix $0 < T < 3/50$ and $\epsilon > 0$ with $\epsilon(47/100)^2 < 1$. Distinct lifted disks have separation at least $3/50$; an outgoing reflected ray cannot reenter the same convex disk before another collision. Thus each released particle has at most one collision by time T . At a collision let z be its impact coordinate relative to the disk center. The fixed detector reports $+$ when its prepared uniform coordinate is at most $1/2 + \epsilon|z|^2/\pi$, and otherwise reports $-$. The controller

never receives z or the uniform seed. This is an internally exact-coordinate apparatus, not a lower bound on the complexity of physical measurement.

For completeness, incoming one-collision states are parameterized by

$$q = Rn_\alpha - sv^-, \quad v^- = -\cos \phi n_\alpha + \sin \phi t_\alpha, \quad 0 < s < T, \quad |\phi| < \pi/2.$$

The backward segment remains outside its own disk by convexity and cannot reach a distinct disk within T by separation. The forward reflected segment has the same exclusion. Every regular colliding initial state has its unique impact triple. At fixed incoming laboratory angle the position Jacobian is $R \cos \phi$, and the angular coordinate change has determinant of absolute value one. Integration against ambient preparation $dq d\vartheta/(2\pi\mathfrak{a})$ gives collision mass

$$\frac{R}{2\pi\mathfrak{a}} \int_0^T \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos \phi d\phi d\alpha ds = \frac{2TR}{\mathfrak{a}}.$$

Tangencies are null, not an excluded positive-mass neighborhood. Since $|z| = R$ at impact, the four uncensored raw probabilities are

$$(155) \quad \begin{aligned} k_\partial &= \pi R^2/\mathfrak{a}, & k_0 &= 1 - \pi R^2/\mathfrak{a} - 2TR/\mathfrak{a}, \\ k_\pm &= TR/\mathfrak{a} \pm 2T\epsilon R^3/(\pi\mathfrak{a}). \end{aligned}$$

They are positive on I_R and sum to one. For k_0 , positivity follows already at $R = 47/100$, $T = 3/50$, using $\mathfrak{a} > 0.86$ and $\pi < 22/7$; the other inequalities follow from the stated bounds. Their coefficient matrix in degrees zero through three is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -2T/\mathfrak{a} & T/\mathfrak{a} & T/\mathfrak{a} \\ \pi/\mathfrak{a} & -\pi/\mathfrak{a} & 0 & 0 \\ 0 & 0 & 2T\epsilon/(\pi\mathfrak{a}) & -2T\epsilon/(\pi\mathfrak{a}) \end{pmatrix},$$

with determinant of absolute value $4T^2\epsilon/\mathfrak{a}^3$. A bounded lookup comparator therefore realizes $A = \{0, 1, 2, 3\}$. Merging the signs gives the three probabilities $k_\partial, k_0, k_+ + k_-$, which span $A = \{0, 1, 2\}$. This supplies the dense special cases of Theorem 2.5 from one counted apparatus. The stronger complete-record response, positive filter, boundary-control and surrogate derivations are preserved in the source companion [9]; the collision calculation above uses their whole-preparation normalization rather than conditioning away insertion failures.

G.2. One target and all state budgets. Consume two cartridges without censoring. Let $X \in \{\partial, 0, +, -\}$ be the first raw report, and let Y be the placement-failure indicator on the second cartridge. After the first report the controller stores one of M states, then announces its forecast $a \in [0, 1]$ and incurs $(a - Y)^2$. The erased experiment observes $e(X) \in \{\partial, 0, H\}$ with $e(+)=e(-)=H$. Both experiments use the same prior, both ambient attempts, and the same second-cartridge variable Y . There is no acceptance charge or saved cartridge. This is a separate uncensored experiment; it is not subject to the lower acceptance margin of the streaming theorem.

Put $h(R) = \pi R^2/\mathfrak{a}$ and define

$$p_x = \mu k_x > 0, \quad z_x = \mu(k_x h), \quad b_x = z_x/p_x, \quad p_B = \sum_{x \in B} p_x, \quad z_B = \sum_{x \in B} z_x.$$

For a nonempty subset B set

$$(156) \quad \delta(B) = \sum_{x \in B} \frac{z_x^2}{p_x} - \frac{z_B^2}{p_B} = \frac{1}{p_B} \sum_{\{x,y\} \subset B} p_x p_y (b_x - b_y)^2.$$

All these quantities are finite combinations of prior moments through order five and the exact physical constants.

Theorem G.1 (Complete common-risk finite-state comparison). *For either alphabet $\mathcal{X}$ (four raw symbols or three erased tags), its optimal M -state risk, allowing independent randomization, is*

$$(157) \quad R_M(\mathcal{X}) = \mu h - \max_{\substack{\mathcal{Q} \text{ partition of } \mathcal{X} \\ |\mathcal{Q}| \leq M}} \sum_{B \in \mathcal{Q}} \frac{z_B^2}{p_B}.$$

A maximizing partition, its block index and block forecasts z_B/p_B give an attained finite-state implementation.

Let R_M^+ and R_M^0 denote the raw and erased risks and $\Gamma_M = R_M^0 - R_M^+$. Then $\Gamma_M \geq 0$ for every M , $\Gamma_1 = 0$, and, with $\Delta = \delta(\{+, -\})$,

$$(158) \quad \Gamma_3 = \Delta - \min_{x \neq y} \delta(\{x, y\}), \quad \Gamma_M = \Delta \quad (M \geq 4).$$

The case $M = 2$ is exactly (157): it compares the seven two-block raw partitions and the three two-block erased partitions (one-block partitions are dominated). Equivalently, if D_M^+ and D_M^0 are the minimum sums of merge costs in the two alphabets, then

$$(159) \quad \Gamma_M = \Delta + D_M^0 - D_M^+.$$

These are common total Bayes risks, not two separately normalized memory regrets.

Proof. Conditional on a block message, the optimal forecast is z_B/p_B . Expanding squared loss gives block contribution $z_B - z_B^2/p_B$; summing proves the risk of every deterministic partition. For arbitrary random encoders and decoders, first replace a randomized forecast at each message by its conditional mean. Holding these at most M forecasts fixed, each raw symbol can be assigned deterministically to a forecast minimizing its conditional squared loss. This cannot increase risk. Reoptimizing each resulting block forecast also cannot increase risk. Thus a deterministic partition is optimal, and the finite maximum is attained. A fresh public seed independent of X, Y merely averages such risks and does not improve the optimum.

The identity (156) is obtained by expanding the weighted variance. The lossless raw risk is $\mu h - \sum_x z_x^2/p_x$; the lossless erased risk exceeds it by Δ . Adding the optimal merging distortion in each alphabet gives (159). Any erased code is a raw code after the deterministic map e , proving $\Gamma_M \geq 0$ directly. With one state both forecasts equal μh . With at least four states both alphabets are losslessly encoded. With three states the erased alphabet is lossless, and an optimal raw partition can be refined to exactly three blocks without increasing risk. Such a partition merges exactly one pair. This proves (158). The partition counts for $M = 2$ are the numbers of nontrivial subsets modulo complementation, namely $(2^4 - 2)/2 = 7$ and $(2^3 - 2)/2 = 3$. $\square$

Write $\mu_j = \mu(R^j)$, let $\mu_H(dR) = R\mu(dR)/\mu_1$, and put

$$v = \text{Var}_{\mu_H}(R^2), \quad \bar{c} = 1/2 + (\epsilon/\pi)\mu_H(R^2).$$

The exact lossless gain retained from [9] is

$$(160) \quad \Delta = \frac{2T\mu_1}{\mathfrak{a}} \frac{(\epsilon v/\mathfrak{a})^2}{\bar{c}(1 - \bar{c})} > 0.$$

Indeed conditional on a first hit the sign probability is $c(R) = 1/2 + \epsilon R^2/\pi$, and $\text{Cov}_{\mu_H}(h, c) = \epsilon v/\mathfrak{a}$. The two conditional-mean shifts are respectively this covariance divided by $\bar{c}$ and its negative divided by $1 - \bar{c}$. Their weighted squared sum is the covariance squared over $\bar{c}(1 - \bar{c})$.

Multiplication by the hit probability proves (160), which is also precisely the pair cost $\delta(\{+, -\})$. Full support makes $v > 0$. This is an explicit conditional-variance calculation, not a new general comparison-of-experiments principle [1].

Corollary G.2 (A genuine finite-resolution threshold). *Fix $T > 0$ and a full-support prior on I_R . There exists $\epsilon_* > 0$ such that, for $0 < \epsilon < \epsilon_*$,*

$$(161) \quad \Gamma_M = 0 \quad (M = 1, 2, 3), \quad \Gamma_M = \Delta > 0 \quad (M \geq 4).$$

Thus a four-state, two-bit controller realizes the strict gain, whereas all smaller state budgets have exactly zero gain in this amplitude range. This does not identify exact algebraic dimension with useful accuracy.

Proof. At $\epsilon = 0$, $b_+ = b_- = b_H$. The three distinct tag means satisfy

$$b_0 < \mu h < b_H < b_\partial.$$

The first strict inequality follows because k_0 is strictly decreasing and h strictly increasing, so $\text{Cov}_\mu(k_0, h) < 0$. For example this sign follows by integrating $(k_0(R) - k_0(R'))(h(R) - h(R'))/2$ against two independent prior draws. The second inequality uses the positive covariance of R and h . The last follows in the hit-tilted prior from the same positive covariance of R and h , since the placement posterior is its additional R -tilt. Full support makes all these covariances strict.

For three states the unique zero-distortion merging pattern at amplitude zero is to keep the three distinct tag means separate and merge the two signs. Every other pair has positive merge cost. For two states, no optimal partition at amplitude zero splits the equal-mean signs between its two blocks. To prove this, use the optimal block means as centers. Equal-mean points assigned to different centers must be equidistant from them; otherwise reassignment to the closer center strictly improves risk. If the centers are distinct, b_H is their midpoint. Each block must then contain a different extreme tag to place its mean strictly on the corresponding side of b_H . Move one sign into the other block, holding centers fixed; the cost is unchanged. The vacated-sign block still contains its extreme tag, so reoptimizing its center strictly reduces the cost, a contradiction. If the two centers agree, the achieved risk is the one-state risk, strictly worse than separating one extreme tag. Thus an optimal two-state partition never splits the signs.

There are finitely many partitions. At amplitude zero every partition that splits the signs and uses at most two states has a strictly positive risk gap to the optimum, as does every non-sign-merging three-state partition. All their risks depend continuously on ϵ because the cell masses stay positive. Therefore, for a sufficiently small common positive ϵ_* , none of these partitions is optimal for $M = 2, 3$. Every remaining partition keeps the signs together and has exactly the risk of the corresponding erased partition, because both $p_+ + p_-$ and $z_+ + z_-$ are independent of ϵ . This proves equality of risks for $M = 2, 3$; $M = 1$ and $M \geq 4$ follow from Theorem G.1 and (160). $\square$

The threshold is checkable from finitely many moment expressions: enumerate the partitions in (157), find their positive gaps at zero, and bound their continuous rational derivatives while cell masses stay bounded away from zero. No prior-uniform or horizon-uniform threshold is claimed. In particular (160) can be extremely small for a concentrated prior. The threshold theorem gives an exact resource comparison, not a promise of a practically large gain.

APPENDIX H. COMPILATION INTO FINITE NUMERICAL TABLES

A covering theorem supplies a finite set, but does not in general supply an algorithm for choosing that set. Moreover, a table with finitely many entries may still contain noncomputable

real numbers. We separate these issues. The construction below uses finite grids of feasible histories, ordinary rational comparisons, and approximations to a finite moment table. Neither rank detection nor a decision whether two exponents coincide is among its operations.

H.1. A compilation theorem for compact state systems. Let $T \geq 1$ be fixed, let $U = [\eta, 1 - \eta]^J$, with rational $0 < \eta < 1/2$, and let $\mathcal{X}$ be finite. For $0 \leq n \leq T$ let $S_n \subset [-B, B]^{d_n}$ be compact, with S_0 a singleton. There are transition maps

$$T_{n,x} : S_n \times U \longrightarrow S_{n+1}, \quad S_{n+1} = \bigcup_{x \in \mathcal{X}} T_{n,x}(S_n, U).$$

Norms in this section are maximum norms. Suppose

$$(162) \quad \|T_{n,x}(s, u) - T_{n,x}(s', u')\|_\infty \leq L_n \|s - s'\|_\infty + G_n \|u - u'\|_\infty.$$

Each stage has finitely many queries, with a prediction map $Q_n : S_n \rightarrow [0, 1]^{b_n}$ of Lipschitz constant K_n . The bounds and all dimensions are fixed, independent of the state budget. Write $H_n(w, u_1, \dots, u_n)$ for the exact state of the report word $w \in \mathcal{X}^n$. Iteration of (162) shows that its command dependence is A_n -Lipschitz, where

$$A_0 = 0, \quad A_{n+1} = L_n A_n + G_n.$$

For $M \geq 1$, define the unrestricted-centre covering radius

$$e_n(M) = \inf_{z_1, \dots, z_M \in \mathbb{R}^{d_n}} \sup_{s \in S_n} \min_i \|s - z_i\|_\infty.$$

No spectral representation or dimension formula for $e_n(M)$ is assumed.

Definition H.1 (Finite evaluation data). Given a rational grid history and a positive rational tolerance τ , an evaluation procedure returns a dyadic vector within τ of its exact state. It also returns dyadic query predictions within τ of the exact predictions at any chosen representative history. This is an offline procedure. In its effective version it is a specified algorithm with access to the numerical input oracles of the experiment. In a noneffective version the finitely many approximations are supplied data. Neither version gives an oracle to the running transducer.

Let $U_h \subset U$ be a finite rational h -net in the maximum norm, with $|U_h| \leq C_U(1 + h^{-1})^J$. At runtime an input includes an index of any $\bar{u} \in U_h$ satisfying $\|u - \bar{u}\|_\infty \leq h$. The guarantee below holds for every such choice. In particular it does not require deciding equality of an arbitrary real input to a rounding boundary. An input approximation, followed by rounding that rational approximation and clipping to U , supplies this interface with a fixed constant adjustment of h . For comparison with a memory lower bound, input coding is a fixed memoryless Borel function of the current command, or uses fresh independent rounding coins. It is not an additional history-dependent channel. The upper estimate itself holds for every name meeting the error bound.

Lemma H.2 (Greedy covering from approximate samples). *Let S be a compact subset of a normed space. Suppose a finite indexed list of points $s_j \in S$ is an a -net of S , and $\|\tilde{s}_j - s_j\| \leq \tau$. Farthest-first selection of at most M points from the rational list $(\tilde{s}_j)$, with fixed tie breaking, selects indices I for which*

$$\sup_{s \in S} \min_{i \in I} \|s - s_i\| \leq 2e_S(M) + a + 4\tau.$$

If there are at most M indices, select all of them. Distinct indices may have identical approximate vectors.

Proof. Let r be the final covering radius of the approximate list. When the list has more than M indices, the selected M points and a farthest point have mutual distances at least r . Every cover by M balls of radius b must put two of these $M + 1$ points in the same ball, and therefore $r \leq 2b$. This also handles $r = 0$. A cover of S with radius $e_S(M) + \epsilon$ covers the approximate list with radius $e_S(M) + \epsilon + \tau$, allowing its centres to lie outside S . Thus $r \leq 2e_S(M) + 2\tau$ after $\epsilon \downarrow 0$. For a given $s \in S$, choose s_j within a , and then a selected approximate point within r of $\tilde{s}_j$. The triangle inequality gives $a + 2\tau + r$. The case of a list of size at most M is immediate. This is the classical farthest-first argument [22]; the approximation allowance is recorded explicitly. $\square$

Theorem H.3 (Effective compilation without loss of covering order). *Under (162) and Definition H.1, there is a finite procedure producing stage-dependent integer transition tables and dyadic output tables, with at most M state indices at each stage, such that, on every input-report history,*

$$(163) \quad \|\hat{Q}_n - Q_n(s_n)\|_\infty \leq C_T \left(\max_{1 \leq j \leq n} e_j(M) + h + \tau \right).$$

The running procedure reads only its previous index, the current finite input code and report, and, at a checkpoint, the new query label. It stores no numerical state vector, representative history, or oracle. The constant depends only on the displayed bounds and the fixed horizon. The synthesis procedure does not need $e_n(M)$ or a covering certificate as input.

Proof. Enumerate all pairs of a report word in $\mathcal{X}^n$ and a command word in U_h^n . They are feasible histories by the hypotheses on the state system. Their exact images form an $A_n h$ -net of S_n . Evaluate every image with error at most τ and apply Lemma H.2. Associate each selected index with the *exact* state $c_{n,i}$ of its selected history, for analysis only. The stored approximations $\tilde{c}_{n,i}$ have error at most τ . The exact representatives cover S_n with radius

$$(164) \quad R_n \leq 2e_n(M) + A_n h + 4\tau.$$

For each representative history at stage n , each $\bar{u} \in U_h$ and each x , append $(\bar{u}, x)$ to that history and evaluate the resulting exact state $y = T_{n,x}(c_{n,i}, \bar{u})$ with error at most τ . Choose the next index by minimizing the maximum-norm distance between this approximation and the stored approximations at stage $n + 1$. All comparisons are between rational numbers. If j is selected, the triangle inequality, applied to a nearest exact centre, gives

$$\|y - c_{n+1,j}\|_\infty \leq R_{n+1} + 4\tau.$$

This specifies one integer for every entry of a finite transition table. The offline representative history is not a runtime variable. In particular no update is applied to an approximate vector outside the reachable set: y is defined and evaluated through an actual history.

Let $E_n = \|s_n - c_{n,i_n}\|_\infty$, where s_n denotes the actual uncompressed state, again only in the analysis. For any admitted finite input code,

$$(165) \quad E_0 = 0, \quad E_{n+1} \leq L_n E_n + G_n h + 2e_{n+1}(M) + A_{n+1} h + 8\tau.$$

Induction over the fixed horizon bounds E_n by a constant times the right-hand quantity in (163). Store a dyadic approximation to $Q_n(c_{n,i})$ with error at most τ and clip it to $[0, 1]$. Clipping cannot increase its error to a probability. The query error is at most $K_n E_n + \tau$, proving the theorem. There are finitely many evaluations and comparisons, so the procedure terminates whenever the evaluation procedure does. Nonuniqueness of representatives or ties does not require an equality test on exact data. $\square$

Proposition H.4 (Size of the finite program). *Put $V = |U_h|$ and $X = |\mathcal{X}|$. The transition table has at most $TMVX$ entries, each taking $\lceil \log_2(M+1) \rceil$ bits. If outputs are written to b binary places, the output tables take at most $M \sum_n b_n(b+1)$ bits, in addition to fixed metadata. Candidate enumeration uses at most $\sum_{n=0}^T (XV)^n$ histories. With cached candidate approximations, greedy selection uses $O(M \sum_{n=0}^T (XV)^n)$ fixed-dimensional distance operations, and transition selection uses $O(TM^2VX)$ such operations.*

Proof. The transition arguments are the stage, current representative, finite command code and report. Counting their combinations proves the first assertion. A query table has one b -place probability per representative and query. At stage n there are $(XV)^n$ grid histories. Maintain for each candidate its smallest distance to a chosen centre; one new centre updates this array with one pass, repeated at most M times. For each transition, comparison against at most M next centres gives the last count. Exact integer comparison, maximum and subtraction suffice. These counts exclude the running time of an arbitrary numerical oracle. $\square$

With a supplied clock the only persistent field has at most M values. An autonomous machine can retain the pair (n, i_n) , using at most $1 + (T+1)M$ values. The current input code, table address and returned output are transient data. They require $O(\log M + \log V + b + \log(T+1))$ bits for fixed dimensions. The finite read-only program is a separate, explicitly counted resource; it is not claimed to fit into $\log M$ bits.

The fixed fine-mesh realization below is retained in full. Its sufficient precision and program bounds are strengthened, without changing its construction or converse, in Section I.

H.2. Finite data for positive monomial experiments. Return to the experiment of Section 5, with $I = [0, 1]$, and put $T = N - 1$. Use the full formal raw moment vector $v_n = (\mathbb{E}[t^{\alpha \cdot a} \mid h])_{|\alpha|=N-n}$, including its constant entry. Coincident labels remain present. Its dimension is bounded by a constant depending only on r, N .

Choose a fixed bound

$$B_f \geq 1 + \sup_{g,x} \sum_{i=0}^{r-1} |f_i(g, x)|, \quad f_{g,x}(t) = \sum_i f_i(g, x) t^{a_i}.$$

This uses the coefficient realization, not exponent separation. For two reachable raw vectors, the update numerator and denominator have differences at most $B_f \|v - v'\|_\infty$. Their denominators are at least κ , and a numerator divided by its own denominator belongs to $[0, 1]$. Subtracting the two quotients gives a state Lipschitz bound $2B_f/\kappa$. For fixed posterior, the difference of report factors at two commands is at most $J \|g - g'\|_\infty$ pointwise, so the command bound is $2J/\kappa$. Consequently (162) holds uniformly on the calibration set. Physical query predictions are fixed linear combinations of raw moments, with uniformly bounded coefficients. For effective evaluation we fix a sufficiently small positive rational δ in the canonical probe basis $1/2, 1/2 + \delta t^{a_i}$. Its formal query coefficients are then rational. Alternatively certified approximations to a chosen fixed query coefficient table are included among the numerical inputs. This verifies every metric hypothesis of Theorem H.3.

Lemma H.5 (Certified finite evaluation from prior moments). *For a fixed horizon, dyadic approximations to the coefficients of the cell matrix and to*

$$(166) \quad m_\gamma(a) = \int_0^1 t^{\gamma \cdot a} d\mu(t), \quad \gamma \in \mathbb{N}_0^r, \quad |\gamma| = N,$$

suffice to evaluate every rational-command history state and every fixed finite product-query prediction with a prescribed error $\tau > 0$. Required input accuracy is $c_N \min\{\tau, 1\}$ for a positive constant determined by supplied coefficient and evidence bounds. The number of input moment

entries is independent of the history grid and of M . No equality decision on the exponents is required.

Proof. Expand an n -report product in formal homogeneous variables as $P(X) = \sum_{|\beta|=n} p_\beta X^\beta$. Its coefficient sum satisfies $\sum_\beta |p_\beta| \leq B_f^n$. A coordinate of the remaining raw state is

$$(167) \quad v_\alpha = \frac{\sum_{|\beta|=n} p_\beta m_{\alpha+\beta}}{\sum_{|\beta|=n} p_\beta m_{\beta+(N-n)e_0}}, \quad |\alpha| = N - n.$$

The denominator is the true prefix evidence and is at least κ^n . The formula remains exact when distinct formal labels have equal exponents. There is no cancellation-based inference of rank.

A perturbation of size δ in the input coefficients changes each factor coefficient vector by at most a fixed multiple of δ . The telescoping identity for products bounds the coefficient-sum error in P by $n(B_f + 1)^{n-1}C\delta$, for δ sufficiently small. Each prior moment lies in $[0, 1]$, and there are finitely many formal labels. Thus numerator and denominator errors are at most $D_N\delta$, where D_N is obtained from the finite bounds just displayed. Take $D_N\delta \leq \kappa^N/2$. The approximate denominator is then positive. If a true numerator is A , its denominator is Z , and $0 \leq A \leq Z$, direct subtraction gives

$$\left| \frac{\tilde{A}}{\tilde{Z}} - \frac{A}{Z} \right| \leq \frac{|\tilde{A} - A| + |\tilde{Z} - Z|}{\tilde{Z}} \leq 4D_N\kappa^{-N}\delta.$$

Choose δ to make this at most $\tau/2$ and round the result to a dyadic within $\tau/2$. Fixed product-query predictions are linear combinations of these raw entries, or the same product ratios; increase the accuracy constant by their fixed coefficient bounds. This gives the claimed finite procedure. $\square$

Theorem H.6 (Finite-precision realization of the intrinsic law). *Fix the experiment, the full-support prior, the horizon and compact calibration set of Section 5. Suppose the command cube is rationally specified, and positive rational lower bounds for the report likelihoods and positive exponents, together with upper coefficient bounds, are supplied. For each known calibration a and integer $M \geq 1$, finite approximations to the data in Lemma H.5 produce a deterministic finite-table transducer with at most M persistent labels per stage and*

$$(168) \quad \sup_{n, \mathbf{h}} \mathcal{R}(\hat{Q}_n, Q_n(\mathbf{h})) \leq C\{\Xi_N(M, a) + h^2 + \tau^2\}.$$

Here h is the command mesh and τ the certified error of the offline state and output evaluations. Constants are uniform in a, M . For meshes and evaluation errors at most $c/(M+1)$, this transducer has regret at most $C'\Xi_N(M, a)$, while every such transducer with the memoryless input interface has maximum unconditional checkpoint regret at least $c'\Xi_N(M, a)$ under the same exploration law as in the intrinsic theorem.

All numerical input and output precisions can be chosen $\log_2(M+1) + O(1)$ binary places. The transition and output program has $O(M^{J+1} \log(M+1))$ bits for the fixed experiment. Its synthesis uses polynomially many fixed-dimensional rational operations in M , with exponent allowed to depend on J, N . If the coefficients and moments have computable approximation procedures, the synthesis is a single uniform terminating algorithm using those procedures offline. No computability assumption is needed for the preceding existential intrinsic law for arbitrary full-support priors.

Proof. Theorem 5.3, its reachable-centre construction, and the fixed physical/raw metric comparison imply

$$e_n(M) \leq C_n \Psi_{n, N-n}(M, a)^{1/2}.$$

The raw update calculation above and Lemma H.5 verify all assumptions of Theorem H.3. Squaring (163), using $(u + v + w)^2 \leq 3(u^2 + v^2 + w^2)$, and averaging over the finite query menu prove (168). For $N \geq 2$, the $\ell = 1$ term at every nontrivial checkpoint gives

$$\Xi_N(M, a) \geq M^{-2}.$$

Hence command and evaluation errors of order $(M + 1)^{-1}$ are absorbed without determining any individual pivot or collision scale.

A dyadic grid with this mesh has $O(M^J)$ points. Coefficient and prior moment errors of size $c_N/(M + 1)$ suffice by Lemma H.5; the required binary precision is the stated logarithmic quantity. Proposition H.4 gives the storage and arithmetic counts. Dyadic input coefficients may be converted to rational product ratios of $O_N(\log(M + 1))$ bits, and all stored approximations have that precision. Integer comparisons are finite, including exact ties.

Finally the finite-table filter, even when its command input is a quantized approximation, is an admissible filter with at most M labels in the original experiment when input coding is memoryless. The exact-input filter can first apply that fixed coding rule. It has less, not more, information than one reading the exact command. The lower bound in Theorem 5.4 therefore applies unchanged. This is a lower bound for one filter over the common exploration law, not a combination of independently redesigned checkpoint filters. $\square$

Corollary H.7 (Calibration accuracy independent of collision gaps). *Suppose $a_i \geq a_* > 0$ for $i > 0$, the calibration entries are computable reals, and the prior has a procedure that approximates $\int t^b d\mu$ for positive rational b to requested accuracy. The preceding compiler needs only $\log_2(M + 1) + O(1)$ binary places of the calibration and of the resulting moments. The bound does not involve the smallest nonzero additive exponent gap and is valid at exact collisions.*

Proof. For $b, b' \geq a_*/2$, the mean value theorem and $\sup_{0 < t \leq 1} t^{a_*/2} |\log t| = 2/(ea_*)$ give

$$\left| \int (t^b - t^{b'}) d\mu \right| \leq \frac{2}{ea_*} |b - b'|.$$

Every nonconstant exponent in (166) is at least a_* . Approximate its constituent calibration entries so that its error is $O((M + 1)^{-1})$ and below $a_*/2$. Evaluate the corresponding positive rational exponent to the same accuracy. The constant entry is exactly one. There are finitely many labels, so the constants can be chosen uniformly. This argument neither compares two sums nor asks whether their difference vanishes. $\square$

Corollary H.8 (Digital realization of high-contact phases). *For the two-parameter family in Corollary 5.7, the same three-term regret law is attained by finite numerical transition and output tables under the finite-data assumptions above. Along $u = \theta$, $v = \theta + \theta^k$, the required numerical precision is $\log_2(M + 1) + O(1)$, with no extra term proportional to $\log(1/|v - u|)$. In particular the compiler remains valid before that smallest separation can be numerically resolved.*

Proof. Apply Theorem H.6 and Corollary H.7. The three-term expression and its contact orders are those of Corollary 5.7. The compiler never identifies the three collision lines or computes k ; its accuracy is set by M , and its performance follows from the complete attainable cover. The physical experiment and its full-history loss baseline are unchanged. $\square$

The construction is not a new interpolation theorem or a new farthest-first approximation algorithm. It connects a collision-uniform attainability theorem to a finite numerical state machine with a counted program. The distinction matters: a purely spectral identity supplies neither the attainable cover nor the positive-likelihood update bound. The horizon is fixed throughout. The program can be much larger than the persistent state, and the running time of a general moment oracle has not been bounded. For a noncomputable prior, finite advice exists

at each precision but a uniform algorithm for producing that advice is not asserted. These distinctions leave the arbitrary-prior mathematical classification intact.

H.3. Complete numerical realization statement. The following collects the intrinsic and effective conclusions. Its construction and resource bounds are proved in the surrounding appendices; no stronger numerical converse is implicit.

Theorem H.9 (Sharp causal memory and certified numerical realization). *Under the preceding fixed-experiment hypotheses, with $r \geq 2$, the checkpoint and causal regret laws have orders $\Psi_{n,m}$ and Ξ_N , respectively, uniformly in $a \in K$ and integers $M \geq 1$. Suppose the command cube is rationally specified and certified finite approximations to the cell coefficients and prior moments through formal degree N are available at requested tolerances, together with coefficient and positivity bounds. An algorithm constructs integer transition tables and dyadic output tables with at most M persistent labels at each stage and maximum worst-history regret at most $C\Xi_N(M, a)$. For every such filter with a fixed memoryless command-coding rule, the maximum unconditional checkpoint regret under the common exploration law is at least $c\Xi_N(M, a)$.*

The algorithm requires no collision stratum, additive-gap lower bound, or value of Ξ_N as input. Its selected numerical precision satisfies

$$b(M, a) = \frac{1}{2} \log_2(1/\Xi_N(M, a)) + O(1),$$

and a sufficient read-only program size is $O(M\Xi_N(M, a)^{-J/2} \log(M+1))$ bits. In particular, setting $d_0 = (r-1)\lfloor N/2 \rfloor$, one may uniformly use

$$b(M) \leq d_0^{-1} \log_2(M+1) + O(1), \quad P(M) = O(M^{1+J/d_0} \log(M+1)).$$

These numerical bounds are sufficient; no precision or program-length converse is asserted. The running filter retains only its index, not a posterior vector, historical tape, or numerical oracle. For computable numerical inputs the synthesis is uniform and terminating. All constants may depend on the fixed prior and horizon, but not on a or M .

APPENDIX I. CERTIFIED PRECISION AND PROFILE-ADAPTIVE COMPILATION

The geometric profile has a second use besides specifying the error of an M -state filter. It determines a sufficient numerical scale for constructing that filter. We first obtain a scale uniform over the calibration chamber. We then recover the actual covering scale from finite approximate histories, without evaluating a determinant or identifying a collision stratum. All precision and program-length estimates in this section are upper resource bounds. The matching converse remains the converse for persistent labels.

The separated-chain observation and its uniform resource consequence were recorded in the preceding technical note [25].

I.1. A separated family of future tests. Set $k = \lfloor N/2 \rfloor$ and

$$d_0 = (r-1)k, \quad \delta_K = \min_{a \in K} \min_{1 \leq i < r} (a_i - a_{i-1}), \quad \bar{\delta} = \delta_K/H.$$

We assume $r \geq 2$, so $d_0 \geq 1$. Compactness in the strictly ordered chamber gives $\delta_K > 0$. This is a one-step separation, not a lower bound on nonzero additive gaps.

Proposition I.1 (Separated-chain floor). *For every $a \in K$ and every integer $M \geq 1$,*

$$(169) \quad \Xi_N(M, a) \geq \bar{\delta}^{d_0-1} M^{-2/d_0}.$$

Proof. Put $m = N - k$. The future sumset mA contains

$$C_m = \{jD : 0 \leq j \leq m\} \cup \{jD + a_i : 0 \leq j < m, 1 \leq i \leq r - 2\}.$$

Each displayed exponent is a sum of at most m members of A ; padding by zeros produces a formal degree- m label. In the interval $[jD, (j+1)D]$ the chain is

$$jD < jD + a_1 < \cdots < jD + a_{r-2} < (j+1)D.$$

Thus C_m has $m(r-1) + 1$ distinct elements, with consecutive gaps at least δ_K . For $r = 2$ there are no interior terms, and the same count and separation hold. There are at least d_0 positive elements, since $m \geq k$. In particular $q_m \geq d_0$ and $p_{k,m} = d_0$. Choose d_0 positive elements and one formal label for each. Every pair of their normalized nodes is separated by at least $\bar{\delta}$. Their Vandermonde product is at least $\bar{\delta}^{d_0(d_0-1)/2}$. Keeping this one term in $\Psi_{k,m}$ proves (169). Other determinant terms, including all collision-sensitive ones, have not been removed. $\square$

Corollary I.2 (Uniform sufficient numerical resources). *Under the finite-data hypotheses of Theorem H.6, the complete regret order $\Xi_N(M, a)$ is realized with at most M persistent labels per stage using*

$$(170) \quad b(M) = \left\lceil \frac{\log_2(M+1)}{d_0} \right\rceil + O(1), \quad P(M) = O(M^{1+J/d_0} \log(M+1))$$

binary places and read-only program bits, respectively. With $T = N - 1$, a sufficient synthesis count is

$$O(M^{1+JT/d_0} + M^{2+J/d_0})$$

fixed-dimensional rational operations, excluding the running time of numerical input procedures. The constants have the same fixed-experiment and calibration-uniform scope as the intrinsic theorem.

Proof. Take h, τ bounded by a fixed multiple of $(M+1)^{-1/d_0}$. Proposition I.1 gives

$$h^2 + \tau^2 \leq CM^{-2/d_0} \leq C\bar{\delta}^{1-d_0} \Xi_N(M, a).$$

Theorem H.6 therefore gives the full upper law, not merely its separated-chain term. Its converse for memoryless input coding is unchanged. Lemma H.5 transfers this tolerance to the finitely many coefficient and moment inputs with only a fixed additive change of precision. The grid has $O(M^{J/d_0})$ points. Counting transition entries, output entries, candidates, and nearest-centre comparisons as in Proposition H.4 proves the remaining statements. The variable part of b is the least integer j for which $2^{jd_0} \geq M+1$, and can be found by integer comparisons. Neither δ_K nor an additive-gap inverse is required to choose it. $\square$

I.2. An observable certificate for the covering scale. Return to the general compact systems of Section H. Write $e(M) = \max_{1 \leq n \leq T} e_n(M)$ and choose a supplied rational upper bound $A \geq \max_n A_n$. Use coordinate product grids with a fixed indexing convention, so their mesh exponent is finite metadata rather than a list of real commands. The constants L_n, G_n, K_n may likewise be replaced by supplied rational upper bounds. For a command h -net, let $\tilde{S}_n$ be the indexed list of approximate grid-history states, each within τ of the corresponding true state. Let I_n be the indices chosen by farthest-first selection with budget M , and put

$$(171) \quad r_n = \max_{z \in \tilde{S}_n} \min_{i \in I_n} \|z - \tilde{s}_{n,i}\|_\infty, \quad r = \max_{1 \leq n \leq T} r_n.$$

These numbers are rational and obtained from the finite data. The true states of selected histories are denoted by $c_{n,i}$, only in the analysis and in offline verification.

Lemma I.3 (Two-sided finite certificate). *For each noninitial stage, and without any positivity assumption on $e_n(M)$,*

$$(172) \quad \max\{0, r_n/2 - \tau\} \leq e_n(M) \leq r_n + A_n h + 2\tau.$$

The selected true states cover S_n with radius at most the right-hand side. In particular, if $\tau = h$ and $c = A + 2$, then

$$(173) \quad \max\{0, r/2 - h\} \leq e(M) \leq r + ch.$$

Proof. If the approximate list has more than M entries, the selected M points together with a farthest point have mutual distances at least r_n . A cover of the true set by M balls of radius $e_n(M) + \epsilon$ covers this list by balls of radius $e_n(M) + \epsilon + \tau$. Two of the $M + 1$ points belong to one ball, hence $r_n \leq 2e_n(M) + 2\epsilon + 2\tau$. Let ϵ decrease to zero. If there are at most M entries, $r_n = 0$ and the lower inequality is immediate. Duplicate vectors cause no change in either argument. For the upper inequality, approximate an arbitrary history by a grid history. Its state changes by at most $A_n h$. Moving to the approximate sample, then to a selected approximate centre, and then to its true state costs at most $\tau + r_n + \tau$. This proves the covering claim and hence the upper inequality. Taking maxima proves (173). $\square$

Theorem I.4 (A posteriori precision selection). *Assume the compact-system and finite-evaluation hypotheses of Theorem H.3, and suppose $e(M) > 0$. For $b = 0, 1, \dots$, take $h_b = \tau_b = 2^{-b}$, construct the lists above, and stop at the first b satisfying*

$$(174) \quad r_b \geq 4ch_b, \quad c = A + 2.$$

Every comparison is rational. The procedure terminates without knowing $e(M)$. At termination it produces an M -label integer/dyadic program with all-history query error at most $C_{Te}(M)$. Moreover

$$(175) \quad h_b \asymp \min\{1, e(M)\}, \quad b = \log_2(2 + e(M)^{-1}) + O(1),$$

where the constants depend only on the fixed system bounds. The final finite certificate satisfies

$$(176) \quad \frac{3r_b}{8} \leq e(M) \leq \frac{5r_b}{4}.$$

A sufficient read-only program size is

$$(177) \quad O\left(M(1 + e(M)^{-1})^J \log(M + 1) + M \log(2 + e(M)^{-1})\right).$$

Proof. By (173), $r_b \geq e(M) - ch_b$. Consequently the test succeeds whenever $h_b \leq e(M)/(5c)$; such a b exists. At the first successful stage, $h_b \leq r_b/(4c) \leq r_b/8$, since $c \geq 2$. Substituting in the lower and upper bounds of (173) proves (176). It also gives $h_b \leq e(M)/(2c - 1)$, using $r_b \leq 2e(M) + 2h_b$.

If $b > 0$, failure at $b - 1$ and $h_{b-1} = 2h_b$ imply

$$e(M) \leq r_{b-1} + ch_{b-1} < 5ch_{b-1} = 10ch_b.$$

Thus $e(M)/(10c) < h_b \leq e(M)/(2c - 1)$ in this case. If $b = 0$, then $h_b = 1$ and the stopping condition and lower certificate give $e(M) \geq 2c - 1$. Since $e(M)$ is also bounded by a constant depending on B , the asserted comparison with $\min\{1, e(M)\}$ follows in all cases. Taking logarithms proves the precision statement.

Use the final selected histories in the transition construction of Theorem H.3. Equation (163) and $h_b = \tau_b \leq Ce(M)$ give the all-history error bound. The number of state indices is not increased during this construction: refinement stages are offline calculations, not persistent states. The final command grid has $O((1 + e(M)^{-1})^J)$ points. The transition and output counts

in Proposition H.4, with $b + O(1)$ places for outputs, give (177). No comparison of exact state coordinates, singular values, or exponents occurs in the stopping rule. $\square$

For $J \geq 1$, the total count over all dyadic refinements is bounded, up to fixed constants, by the last refinement count. Indeed the candidate and transition bounds are geometric sums in 2^{bJ} . A sufficient count is

$$O\left(M(1 + e(M)^{-1})^{JT} + M^2(1 + e(M)^{-1})^J\right).$$

This counts rational operations, not oracle running time or unit-cost operations on unrestricted real numbers. The number of refinements is $O(\log(2 + e(M)^{-1}))$. For fixed M the procedure does not decide whether an arbitrary compact set has zero covering radius.

Corollary I.5 (An absolute-tolerance version). *For any rational $\zeta \in (0, 1]$, stop instead when either (174) holds or $h_b \leq \zeta$. No assumption that $e(M) > 0$ is needed. The resulting all-history query error is at most $C_T(e(M) + \zeta)$, and $h_b \asymp \min\{1, e(M) + \zeta\}$.*

Proof. Termination follows from $h_b \downarrow 0$. If the radius condition holds, the previous theorem's stopping-stage estimates apply; if it holds at $b > 0$, failure of both tests at $b - 1$ also gives $h_b > \zeta/2$. If the radius condition fails at termination, the certificate gives $e(M) < 5ch_b$ and the tolerance condition gives $h_b \leq \zeta$; for $b > 0$ one again has $h_b > \zeta/2$. The initial-stage case is bounded by the fixed state diameter. In all cases $h_b \asymp \min\{1, e(M) + \zeta\}$. Apply (163) with $\tau_b = h_b$. $\square$

I.3. Verification of an individual finite program. A small covering radius does not check that a stored decoder is correct. The following residual certificate checks transitions and outputs separately. It is useful also when a compilation uses sharper, local numerical bounds than the universal covering estimate.

Proposition I.6 (Transition and decoder residual certificate). *Consider any integer transition table $j(n, i, u, x)$ and any stored probability vectors $q_{n,i}$. Suppose certified evaluations give $\tilde{c}_{n,i}$, $\tilde{y}_{n,i,u,x}$ and $\tilde{Q}_{n,i}$ within τ of, respectively, $c_{n,i}$, $T_{n,x}(c_{n,i}, \bar{u})$ and $Q_n(c_{n,i})$. Define the rational residual bounds*

$$d_n = \max_{i,u,x} \|\tilde{y}_{n,i,u,x} - \tilde{c}_{n+1,j(n,i,u,x)}\|_\infty + 2\tau,$$

$$o_n = \max_i \|q_{n,i} - \tilde{Q}_{n,i}\|_\infty + \tau.$$

Put $D_0 = 0$ and $D_{n+1} = L_n D_n + G_n h + d_n$. For every actual history and every admitted current input name,

$$(178) \quad \|\hat{Q}_n - Q_n(s_n)\|_\infty \leq K_n D_n + o_n.$$

The formula applies to a perturbed or corrupted table as well as to a correctly compiled one; the measured residuals, and therefore the certificate, then change.

Proof. Two state evaluation errors give $\|T_{n,x}(c_{n,i}, \bar{u}) - c_{n+1,j}\|_\infty \leq d_n$. One output evaluation error gives $\|q_{n,i} - Q_n(c_{n,i})\|_\infty \leq o_n$. The transition Lipschitz bound and $\|u - \bar{u}\|_\infty \leq h$ therefore imply inductively $\|s_n - c_{n,i_n}\|_\infty \leq D_n$. The query Lipschitz bound and the output inequality prove (178). All maxima are over finite tables; the justification for their numerical error allowances is the supplied evaluation certificate, not an assumption that approximate vectors are reachable states. $\square$

I.4. Finite input errors without consistency tests. We record explicit constants for the finite moment interface. They are not chosen to optimize numerical conditioning. Their purpose is to certify the data actually used by the offline algorithm.

Lemma I.7 (A finite coefficient–moment certificate). *Suppose each true report factor has coefficient ℓ^1 norm at most $B \geq 1$, and the coefficient matrix of J cells and r monomials is supplied with entrywise error at most ϵ . Every prior moment through formal degree N is supplied with error at most ϵ . Retain all formal labels, even when their exponents coincide. Put $C_f = Jr$ and assume $C_f\epsilon \leq 1$. For a prefix of length n , set*

$$D_n^{\text{in}} = nC_f(B+1)^{n-1} + (B+1)^n,$$

where the first term is zero when $n = 0$. The computed prefix evidence $\tilde{Z}$ and every computed raw-moment numerator $\tilde{A}$ have absolute errors at most ϵD_n^{in} . If $\epsilon D_n^{\text{in}} \leq \kappa^n/2$, then

$$(179) \quad \tilde{Z} \geq \kappa^n/2 > 0, \quad \left| \frac{\tilde{A}}{\tilde{Z}} - \frac{A}{Z} \right| \leq \frac{2\epsilon D_n^{\text{in}}}{\tilde{Z}} \leq 4\epsilon D_n^{\text{in}} \kappa^{-n}.$$

For a true probability query with exact formal coefficient ℓ^1 norm at most B_Q , replace the numerator error by $B_Q\epsilon D_n^{\text{in}}$ and the first numerator 2 on the right by $B_Q + 1$.

Proof. For an accepted cell or a rejection mixture, command entries have absolute value at most one. The error in a report factor has ℓ^1 norm at most $Jr\epsilon$. Thus every approximate factor has norm at most $B + 1$. Telescoping the product of n factors in the coefficient algebra, whose ℓ^1 norm is submultiplicative, bounds the product error by $nC_f\epsilon(B+1)^{n-1}$. Integrating the product error against true moments costs no further factor, since all true moments lie in $[0, 1]$. Integrating the approximate product against the moment errors costs at most $\epsilon(B+1)^n$. Shifting its formal indices for a raw moment changes neither coefficient norm. This proves both error estimates. Since $Z \geq \kappa^n$, the asserted positivity follows. Finally, with $q = A/Z \in [0, 1]$,

$$\frac{\tilde{A}}{\tilde{Z}} - q = \frac{(\tilde{A} - A) - q(\tilde{Z} - Z)}{\tilde{Z}}.$$

The two numerator errors prove (179). Multiplication by an exact query polynomial costs its coefficient norm, and the same identity proves the query version. Clipping an approximate probability to $[0, 1]$ does not increase its error. $\square$

The approximation table is not required to satisfy positivity identities or equality between coincident formal moments. Those properties hold for the true experiment; the error certificate is what is required of its numerical data. Formal products of degree less than N can be padded by the zero exponent to degree N . Conversely, keeping all degrees through N gives the same finite interface. For a prescribed evaluation tolerance τ , choosing $\epsilon \leq c_N\tau$ and allowing a fixed number of additional dyadic places proves the required certificate. Thus none of the refinement steps depends on deciding exact equality of exponent sums.

I.5. Recovery of the entire collision profile.

Lemma I.8 (Covering and prediction scales). *For the raw attainable state sets of the monomial experiment,*

$$(180) \quad e_n(M)^2 \asymp \Psi_{n, N-n}(M, a), \quad e(M)^2 \asymp \Xi_N(M, a),$$

uniformly over $a \in K$ and $M \geq 1$.

Proof. The global reachable-centre construction in Theorem 5.3 gives the upper inequality. For the reverse inequality take any M raw centres with covering radius $e_n(M) + \epsilon$. The fixed formal query matrix is a bounded linear map from raw coordinates to physical prediction coordinates. Send the index of a nearest raw centre and apply this matrix at the decoder; clip each resulting probability to $[0, 1]$. The worst-history mean squared error is at most $C(e_n(M) + \epsilon)^2$. This is an admissible one-checkpoint code, irrespective of whether its raw centres are reachable. The worst-history lower bound in Theorem 5.3 gives $c\Psi_{n,N-n} \leq C(e_n(M) + \epsilon)^2$. Let ϵ decrease to zero. There are finitely many checkpoints, so taking maxima proves the second assertion with uniform constants. This argument uses neither the new stopping rule nor its causal upper bound, and hence does not assume the conclusion to be proved. $\square$

Theorem I.9 (Profile-adaptive finite realization). *Under the finite-data hypotheses of Theorem H.6, with numerical data available at successively requested tolerances, one algorithm produces an M -label finite numerical filter without receiving Ξ_N , a collision stratum, or any additive-gap lower bound. Its maximum worst-history regret is at most $C\Xi_N(M, a)$, and the original common-exploration lower bound is $c\Xi_N(M, a)$ for every such filter with a memoryless input interface. The algorithm selects a precision satisfying*

$$(181) \quad b(M, a) = \frac{1}{2} \log_2(1/\Xi_N(M, a)) + O(1),$$

and has sufficient read-only program size

$$(182) \quad P(M, a) = O\left(M \Xi_N(M, a)^{-J/2} \log(M + 1)\right).$$

Equation (181) describes the precision selected by this algorithm, up to a uniform additive constant. It is not a lower bound on the precision of every possible realization.

Proof. By Lemma I.8 and Proposition I.1, $e(M) > 0$. All constants and finite evaluation hypotheses for the raw system were verified in Section H; Lemma I.7 provides explicit error allowances at each refinement. Apply Theorem I.4 and square its all-history query bound. Equation (180) gives the desired upper law. The state budget remains exactly the same upper budget M , so memoryless input coding is simulated by an exact-command filter and the intrinsic streaming converse applies unchanged.

Raw states lie in a unit cube, hence $e(M) \leq 1/2$. Also $M^{-2} \leq \Xi_N(M, a) \leq 1$. Taking logarithms of (180) and using (175) proves (181). Equations (177) and (169) absorb the output term into (182), and give $b(M, a) \leq d_0^{-1} \log_2(M + 1) + O(1)$. The supplied dyadic inputs and rounded outputs have $b + O(1)$ -place accuracy. The exhibited exact rational evaluator uses $O(b + 1)$ -bit intermediates, with a multiplicative constant depending on the fixed horizon and input format; it need not have only an additive overhead. Lemma J.10 makes this distinction explicit. If moments are obtained from approximate calibration entries, the derivative estimate in Corollary H.7 transfers the same tolerance. The synthesis is uniform when those input procedures are computable; otherwise it is a finite-advice construction with the same error bound. $\square$

Corollary I.10 (Program-size phases at intersecting collision lines). *In the four-cell, five-trial family of Corollary 5.7, write $\rho = \max\{|u|, |v|\}$ and $\tau_c = \min\{|u|, |v - u|, |v - 2u|\}$. Put*

$$F = \max\{M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau_c)^{2/9} M^{-2/9}\}.$$

The adaptive algorithm realizes regret of order F using $b = \frac{1}{2} \log_2(1/F) + O(1)$ places. A sufficient program bound is

$$(183) \quad O\left(\min\{M^{5/3}, \rho^{-1} M^{3/2}, (\rho^2 \tau_c)^{-4/9} M^{13/9}\} \log(M + 1)\right),$$

where a reciprocal term with zero denominator is omitted. In particular $b \leq \frac{1}{6} \log_2(M+1) + O(1)$ throughout the parameter square.

Proof. Here $r = J = 4$, $N = 5$, and $d_0 = 6$. The intrinsic two-parameter theorem gives $\Xi_N \asymp F$. Substitute this comparison in (181) and (182). The identity $MF^{-2} = \min\{M^{5/3}, \rho^{-1}M^{3/2}, (\rho^2\tau_c)^{-4/9}M^{13/9}\}$ proves (183), with the stated zero convention. The separated-chain term gives the uniform precision bound. $\square$

Along $u = \theta$, $v = \theta + \theta^k$, $k \geq 2$, the three sufficient program scales are respectively $M^{5/3}$, $\theta^{-1}M^{3/2}$, and $\theta^{-(4k+8)/9}M^{13/9}$, times $\log(M+1)$. Their crossovers are the same $M \asymp \theta^{-6}$ and $M \asymp \theta^{-(8k-2)}$ as for the error profile. This is a consequence for numerical realizations, not a new classification of collision arrangements or a converse for program length. In particular, the algorithm need not determine k or resolve $|v - u|$ before selecting its numerical scale.

APPENDIX J. VERIFIED CONSTRUCTION AND STABILITY OF FINITE ADVICE

The intrinsic law concerns an experiment, whereas a numerical certificate concerns specified data or a specified program. We now relate these statements without identifying them. The resulting construction is stable under a common, imperfect numerical description of several experiments. In particular, an exact additive relation need not be resolved when its effect lies below the accuracy of the supplied moments.

J.1. A construction certificate independent of the program residual. Use the compact finite-horizon system of Section H. The numerical value attached to each grid history is fixed during one compilation. Let $\tilde{s}_n(w)$ approximate its true state to accuracy τ in the maximum norm, including numerical rounding. Denote the selected histories by $w_{n,1}, \dots, w_{n,k_n}$, $k_n \leq M$, and put

$$\tilde{c}_{n,i} = \tilde{s}_n(w_{n,i}), \quad r_n = \max_{w \in \mathcal{H}_n^h} \min_{i \leq k_n} \|\tilde{s}_n(w) - \tilde{c}_{n,i}\|_\infty.$$

Here $\mathcal{H}_n^h$ is the complete finite list of grid histories. A history extending a selected history by a grid command and a report belongs to the next such list. We order the lists once and use the first index whenever a minimum or maximum is attained more than once.

Definition J.1 (Finite construction contract). A table satisfies the construction contract if its representatives are the farthest-first selection from the fixed numerical lists, its state counts agree with those representatives, every transition satisfies

$$(184) \quad j(n, i, u, x) = \min_{k \leq k_{n+1}} \operatorname{argmin} \|\tilde{s}_{n+1}(w_{n,i}(u, x)) - \tilde{c}_{n+1,k}\|_\infty,$$

and every output is the prescribed clipped dyadic rounding of the numerical query evaluation at its representative history. All numerical lists and their radii r_n are fixed before a table is checked. The histories and the contract are offline objects, not persistent state.

Theorem J.2 (Construction-to-covering implication). *Suppose the fixed state and query evaluations have errors at most τ , the command grid has radius h , and a table satisfies Definition J.1. If E_n bounds the distance from a true state to the true representative selected by the running filter, then one may take*

$$(185) \quad E_0 = 0, \quad E_{n+1} = L_n E_n + G_n h + r_{n+1} + 2\tau.$$

The query error at stage n is at most $K_n E_n + \tau$. Moreover,

$$(186) \quad r_n \leq 2e_n(M) + 2\tau, \quad \max_{i,u,x} \|T_n(c_{n,i}, \bar{u}, x) - c_{n+1,j(n,i,u,x)}\|_\infty \leq r_{n+1} + 2\tau.$$

The right side of the second inequality is fixed by the numerical history lists, not increased in response to a supplied transition table. Consequently, at fixed horizon the table has worst-history query error $O(e(M) + h + \tau)$.

Proof. The true initial state is the representative of the empty history, so $E_0 = 0$ even when its numerical name is imperfect. For a stored representative and a grid command, the numerical target in (184) belongs to $\mathcal{H}_{n+1}^h$. Its distance to the selected numerical centre is therefore at most r_{n+1} . Passing from the two numerical values to their two true values costs at most 2τ . This proves the second inequality in (186). The first is the farthest-first separation argument of Lemma I.3: if there is an unselected point, the selected points together with a farthest point are r_n -separated; transferring them to an arbitrary M -ball cover of the true set gives $r_n \leq 2e_n(M) + 2\tau$. If every point is selected, $r_n = 0$ and the inequality is immediate.

For an actual command, first compare the true current state with its true representative, at cost $L_n E_n$, and then replace the command by its grid code, at cost $G_n h$. Apply the just proved bound to the appended representative history. This gives (185). Compare a true query with its value at that representative, at cost $K_n E_n$, and then use the certified stored output, at cost τ . Iterating the recurrence finitely many times and using the first inequality in (186) proves the last assertion. At no step is an approximate vector passed to a transition map whose domain is the feasible state set. $\square$

The residual bound in Proposition I.6 remains valid for an arbitrary table. It is a different assertion: its right side is measured from that table. The following comparison makes the logical distinction quantitative. The example and its role as a transition diagnostic were given in [26]; we include the argument because the distinction is part of the numerical theorem.

Proposition J.3 (Residual validity and covering-order realization). *There is a fixed compact one-step Lipschitz system for which a sequence of M -label tables has exact outputs at all representatives and valid residual bounds, yet its worst-history squared prediction error divided by the squared optimal M -centre radius is $4M^2$. Thus these two properties, even together with knowledge of the optimal covering radius, do not imply the construction-to-covering conclusion. For the same system a table satisfying the nearest-centre construction has the covering-order bound of Theorem J.2.*

Proof. Take $U = [1/4, 3/4]$, $s_0 = 1/2$, $T(s, u) = (s + u)/2$, and $Q(s) = s$. The reachable interval $[3/8, 5/8]$ has length $1/4$. Since M balls of radius e have total length at most $2Me$, its covering radius is at least $1/(8M)$; equally spaced midpoints attain that value. Select representatives by farthest-first on an endpoint-containing command grid whose first history gives $3/8$, and store their exact query values. Replace every transition target by this first index. The output for command $3/4$ is now $3/8$, instead of $5/8$. The maximum error is exactly $1/4$, so the asserted squared ratio is $(1/4)^2 / (1/(8M))^2 = 4M^2$. Its measured transition residual is also $1/4$, hence its residual bound is true. None of these computations changes the optimal radius of the experiment. For the correct table, (184) and Theorem J.2 give error $O(e(M) + h)$; choose $h = O(e(M))$. For the dyadic grids with $8M$ segments used in the finite diagnostic, M a power of two, farthest-first gives maximum grid error at most $2e(M)$: after the endpoints, each new point bisects a longest interval between consecutive centres. Induction over the dyadic levels bounds its largest half-gap by $1/(4M)$; for $M = 1$ the endpoint error is $1/4$. $\square$

This proposition is not a limitation on finite approximation methods. It states exactly what a residual estimate does not assert about an arbitrary program. The additional input that identifies the sharp scale in this paper is the attained geometry of Theorem 5.3, not the error

recurrence itself. We next give a consequence of that geometry which also concerns imperfectly known experiments, rather than only a count of table entries.

J.2. Stability of attainable sets and their profiles. All state coordinates in this subsection are the formal raw coordinates; coincident labels are not removed. Two experiments have the same command and report alphabets and are compared on the same histories. Their moment tables contain every formal degree through N .

Lemma J.4 (Paired-history stability). *Consider two positive monomial experiments with uniformly bounded cell coefficient norms and report factors bounded below by $\kappa > 0$. Suppose the maximum entrywise difference of their cell coefficients and formal moment tables is at most $\delta \leq 1$. There is a constant C_* , depending only on the fixed finite format and positivity bounds, such that, at every history of length $n < N$,*

$$(187) \quad \|s_n(w) - s'_n(w)\|_\infty \leq C_*\delta.$$

For a common probe menu with fixed formal coefficients the query probabilities satisfy the same assertion, with another fixed constant. In particular,

$$(188) \quad d_H(S_n, S'_n) \leq C_*\delta, \quad |e_n(M) - e'_n(M)| \leq C_*\delta \quad (M \geq 1).$$

No lower bound on a nonzero difference of exponent sums occurs here.

Proof. Let B bound the coefficient norm of a report factor and put $C_f = Jr$. For a fixed command, the difference of the two factor coefficient vectors has norm at most $C_f\delta$. Telescoping a product of n factors and then changing its moments bounds each integrated numerator and its evidence by

$$D_n\delta, \quad D_n = nC_f(B+1)^{n-1} + (B+1)^n.$$

The same calculation applies after multiplication by any formal raw monomial of the remaining degree. If A/Z and A'/Z' are the two raw moments, then $Z, Z' \geq \kappa^n$ and $0 \leq A'/Z' \leq 1$. Consequently

$$\left| \frac{A}{Z} - \frac{A'}{Z'} \right| = \frac{|A - A' + (A'/Z')(Z' - Z)|}{Z} \leq 2D_n\kappa^{-n}\delta.$$

Take the maximum over the finitely many stages. A fixed formal query is a bounded linear combination of raw moments, proving its assertion. These estimates compare two actual experiments; the formal moment entries need not satisfy the same coincidence identities in the two.

Every reachable point is obtained from some history. Use that identical history in the other experiment to obtain (187) in both directions. This proves the Hausdorff bound. An arbitrary M -centre cover of one set, with radius within ϵ of its infimum, covers the other with additional radius at most $C_*\delta$. Interchange the sets and let ϵ decrease to zero to finish. $\square$

Uniformity over priors must be specified, not inferred from full support alone. Fix a full-support Borel probability μ_0 on $[0, 1]$ and constants $0 < c_- \leq 1 \leq c_+ < \infty$. Write

$$\mathcal{D} = \{\mu \text{ a probability} : c_-\mu_0 \leq \mu \leq c_+\mu_0\}.$$

These are inequalities of measures. They do not require μ_0 , or any member of $\mathcal{D}$, to have a Lebesgue density.

Lemma J.5 (Uniformity on a dominated prior family). *The constants in the intrinsic checkpoint and causal laws, and in Lemma I.8, can be chosen uniformly for $(a, \mu) \in K \times \mathcal{D}$. The finite-horizon transition and finite-input estimates have the same uniformity.*

Proof. The family $\mathcal{D}$ is weakly compact. Indeed, probabilities on $[0, 1]$ are weakly compact, and each measure inequality is closed: pass to the limit against a nonnegative continuous function and then use regularity of finite Borel measures. Every member has full support. The ordinary and complete confluent moment entries used in Lemma 4.3 depend continuously on (a, μ) on this compact family. For a logarithmic entry, its positive exponent is bounded away from zero; hence $t^c(\log t)^j$, extended by zero at the origin, varies continuously in the uniform norm on the relevant compact parameter set. The zero-exponent constant block has multiplicity one.

There are finitely many formal orderings and confluent block types. For each, the mixed-moment pairing is strictly positive by the confluent positivity lemma for every $\mu \in \mathcal{D}$. Equivalently, in its determinant integration formula a product of k copies of μ dominates $c_-^k \mu_0^{\otimes k}$; the ordered separated intervals used to prove strict positivity therefore retain positive mass. Compactness and continuity give the uniform lower singular-value bounds in the normalized attainable flag argument. The derivatives used in its local inverse and volume minorization are uniformly bounded, since all factor coefficients are bounded and the evidence is at least κ^n . The acquisition-prefix probability has this same lower bound independently of the prior.

For the global cover, changing the moments only changes real coefficients in the bounded-format formula. Neither its format nor its attainable dimension changes. The bounding rectangle, evidence bound, and finite query maps are uniformly bounded. Thus both the local lower estimate and the global upper estimate used in Theorem 5.3 are uniform. The causal argument uses only these estimates and finitely many uniform transition bounds. Finally, the proof of Lemma I.8 and the explicit moment error estimate use those same constants. This accounts for all prior-dependent inputs of the asserted estimates. $\square$

Corollary J.6 (Stability of the collision profile). *Fix the detector and let $(a, \mu), (a', \mu') \in K \times \mathcal{D}$. If their formal moment tables differ by at most δ , then, uniformly for all $M \geq 1$,*

$$(189) \quad \Xi_N(M, a) \leq C(\Xi_N(M, a') + \delta^2), \quad \Xi_N(M, a') \leq C(\Xi_N(M, a) + \delta^2).$$

For a fixed prior, δ may be replaced by $C\|a - a'\|_\infty$. In particular, equality of additive sums may change without producing an instability above the scale of the moment error.

Proof. Lemmas J.4 and J.5 give $e(M) \leq e'(M) + C_*\delta$ and the reverse inequality, uniformly. Square these inequalities and use Lemma I.8. For the calibration assertion, the constant formal label is independent of a . For every other label of degree at most N , its exponent is uniformly positive on K . The derivatives of $t^{\alpha \cdot a}$ with respect to positive-exponent coordinates are $\alpha_j t^{\alpha \cdot a} \log t$, uniformly bounded on $[0, 1]$. The line segment between two points of K retains a positive lower bound on the one-step exponents. Integrating the derivative along that segment proves the stated moment estimate for any probability prior. $\square$

J.3. One program from a common imperfect numerical name. Suppose a collection of compact systems shares its labelled history interface, its state-coordinate dimensions, and constants L_n, G_n, K_n . A *common numerical name of floor ρ* , $0 < \rho \leq 1$, supplies at scale $h = 2^{-b}$ one finite command grid and one numerical evaluation on its feasible histories. For every compatible system, the grid is an h -net and its state and query errors, including rounding, are at most

$$\tau_b = \max\{h, \rho\}.$$

The same data are used for all compatible systems. This quantifier is essential: a different exact prior hidden in a program is not a common numerical name. Write $e_{\mathcal{E}}(M)$ for the largest covering radius of a compatible system $\mathcal{E}$, and let $A = \max_n A_n$ be the history-net constant. Assume the state sets lie in fixed bounded cubes.

Theorem J.7 (Stable common-advice realization). *From such a common name one can construct one table satisfying Definition J.1, with at most M persistent labels at each stage, whose worst-history query error in every compatible system is at most*

$$(190) \quad C(e_{\mathcal{E}}(M) + \rho).$$

The construction does not select a compatible system. More precisely, put $c = A + 4$, and stop at the first dyadic scale for which

$$(191) \quad r_b \geq 4ch_b \quad \text{or} \quad h_b \leq \rho, \quad r_b = \max_{1 \leq n < N} r_{n,b}.$$

The stopping scale satisfies, simultaneously for every compatible system,

$$h_b \asymp \min\{1, e_{\mathcal{E}}(M) + \rho\}.$$

All constants depend only on the fixed horizon, bounds, and dimensions.

Proof. The second condition ensures finite termination. Until and including the first stopping scale, $\tau_b \leq 2h_b$: before crossing the floor it is h_b , and the first dyadic scale below ρ is greater than $\rho/2$ (with equality at the initial endpoint treated separately). For each compatible system, the finite-cover argument therefore gives

$$(192) \quad \max\{0, r_b/2 - 2h_b\} \leq e_{\mathcal{E}}(M) \leq r_b + (A + 4)h_b.$$

The numerical radius on both sides is common, although the true covering radius may depend on the system.

Suppose $b > 0$ is the first stopping index. At the previous scale both conditions failed. By (192) and $h_{b-1} = 2h_b$,

$$e_{\mathcal{E}}(M) < 10ch_b, \quad \rho < 2h_b.$$

Thus $h_b > (e_{\mathcal{E}}(M) + \rho)/(10c + 2)$. If the radius condition holds at b , use $r_b \leq 2e_{\mathcal{E}}(M) + 4h_b$ to obtain $h_b \leq e_{\mathcal{E}}(M)/(2c - 2)$. If it does not, the floor condition gives $h_b \leq \rho$. In both cases $h_b \leq e_{\mathcal{E}}(M) + \rho$. The initial stopping case follows from the fixed diameter bound and $\rho \leq 1$; if its radius condition holds, the same inequality bounds $e_{\mathcal{E}}$ below by a fixed positive number, and otherwise $\rho = 1$. This proves the asserted scale comparison.

The farthest-first selection and the independently checked nearest transitions use only the common numerical lists. In every compatible system Theorem J.2 bounds the query error by $C(e_{\mathcal{E}} + h_b + \tau_b)$. At the stopping scale the last two terms are $O(e_{\mathcal{E}} + \rho)$, proving (190). Every state remains an index of an actual representative history for that system. Neither its true coordinates nor its model identity are consulted at runtime. $\square$

Corollary J.8 (Finite moment uncertainty through collisions). *Fix a detector, K , N , and the prior envelope $\mathcal{D}$. Suppose a finite numerical moment table of degree at most N is within δ of every compatible $(a, \mu) \in K \times \mathcal{D}$. The cell coefficients may also be supplied to this numerical accuracy; the underlying detector is fixed. For all sufficiently small $\delta > 0$, one causal M -label filter constructed from these same data has maximum worst-history regret at most*

$$(193) \quad C(\Xi_N(M, a) + \delta^2)$$

in every compatible experiment. No collision decision or value of Ξ_N is supplied to the construction. Its selected scale is comparable to $\min\{1, \sqrt{\Xi_N(M, a) + \delta}\}$ for each such experiment.

Proof. Lemma I.7 bounds the error of evaluating any feasible history from the supplied finite data by $C_0\delta$, with C_0 uniform by Lemma J.5. Use $\rho = 2C_0\delta$, increasing C_0 if necessary to cover the fixed query coefficients. Exact rational evaluation of the supplied entries, followed by rounding at $b + 2$ binary places, has error at most $\rho/2 + h_b/8 \leq \max\{h_b, \rho\}$. Every compatible experiment therefore has the same valid numerical name. The rational command cube admits

the required grids. Apply Theorem J.7, square its bound, and use Lemma I.8, uniformly in the prior envelope. The physical probe map is the same fixed formal map for the fixed detector. The running program has the prescribed label budget throughout. The smallness restriction on δ merely ensures the printed positive-evidence and $\rho \leq 1$ bounds. $\square$

This is a robustness statement about one finite description, not an assumption that the approximate moment table is itself a positive moment sequence. It supplies a direct interpretation of the full collision law when even exact identities among supplied entries are unavailable. The additional δ^2 term is not only an artifact of propagating a norm bound.

Proposition J.9 (Sharp advice uncertainty in a positive experiment). *There is a fixed positive detector with $r = 2$ and $N = 2$, and a dominated full-support prior family, for which the minimax regret of one filter required to serve all priors compatible with a common δ -accurate moment table has order*

$$M^{-2} + \delta^2,$$

for all $M \geq 1$ and all sufficiently small $\delta > 0$. The lower bound holds under the same independent command exploration and a fixed finite spanning probe menu. It applies to programs receiving that common advice; it is not a precision converse for programs given different exact priors.

Proof. Take $a_0 = 0, a_1 = 1$ and the three detector cells

$$k_0(t) = \frac{1}{4}, \quad k_1(t) = \frac{3}{8} + \frac{1}{8}t, \quad k_2(t) = \frac{3}{8} - \frac{1}{8}t.$$

They are positive, sum to one, and span $\text{span}\{1, t\}$. Let $\eta = 1/4$, and take as reference prior uniform measure on $[0, 1]$, with envelope constants $1/2$ and $3/2$. For $\epsilon = 6\delta \leq 1/2$ consider the two densities $1 \pm \epsilon(2t - 1)$. Their moments are

$$m_k^\pm = \frac{1}{k+1} \pm \epsilon \frac{k}{(k+1)(k+2)}.$$

Since $k/((k+1)(k+2)) \leq 1/6$ for every nonnegative integer k , the uniform-prior moments form one common δ -accurate numerical name, even if more moments than required are supplied.

Upon reporting accepted cell 0, its likelihood $g_0/4$ is constant in t , so the posterior remains the prior. The joint density of this report and the observed command is identical in the two experiments; its probability is at least $\eta/4$. The probe obtained with command $(1/2, 3/8, 5/8)$ is the event of rejection, whose function is $q(t) = 1/2 + t/32$. Its two posterior means differ by $\delta/16$. For any possibly randomized prediction Y using the common advice, the distribution of Y conditional on the accepted-cell-0 history is the same in the two experiments. If the two means are q_-, q_+ ,

$$\frac{\mathbb{E}(Y - q_-)^2 + \mathbb{E}(Y - q_+)^2}{2} = \mathbb{E}\left(Y - \frac{q_- + q_+}{2}\right)^2 + \frac{(q_+ - q_-)^2}{4} \geq \frac{\delta^2}{1024}.$$

Integrate over those identical command histories and multiply by the fixed positive probability of selecting this probe from the spanning menu. The larger of the two unconditional regrets is at least $c\delta^2$, independently of M .

For either fixed prior, the intrinsic checkpoint lower bound is $c'M^{-2}$, uniformly in this envelope. Indeed, $n = m = 1$ and the attainable dimension is one, so $\Xi_2(M, a) = M^{-2}$. A common-advice filter is an admissible M -label filter for that fixed experiment; supplying less information does not invalidate its memory lower bound. Combining the two lower bounds gives a constant times $M^{-2} + \delta^2$. Corollary J.8 gives the matching upper bound simultaneously for all compatible priors in the envelope, proving the assertion. $\square$

J.4. Exact rational workspace.

Lemma J.10 (Input accuracy and exact intermediate lengths). *For the finite moment evaluator at fixed horizon and fixed format, $b + O(1)$ -place dyadic inputs suffice for $O(2^{-b})$ evaluation error. If evaluated with exact rational operations, every intermediate numerator and denominator has $O(b + 1)$ bits, with a constant depending on the fixed arithmetic circuit. This statement does not give an additive $O(1)$ overhead in exact intermediate length. In fact, for $x_b = (2^b - 1)/2^b$, the exact fractional length of x_b^N is Nb .*

Proof. The input-accuracy assertion is the positive-evidence error bound of Lemma I.7. For the exact length assertion, bound the lengths of the numerator and denominator of a reduced fraction by those of its unreduced representation. Multiplication or division adds the operand lengths, while addition forms two products and their sum, increasing the resulting length by at most one beyond those product bounds. The circuit for one fixed-degree moment quotient has a number of gates depending only on the fixed format. Induction over these gates gives $C(b + 1)$ bits for every intermediate. Its evidence is nonzero by the supplied certificate. Distances and comparisons of a fixed number of coordinates satisfy the same bound; selecting among more histories increases the number of operations and the index length, not the degree of a moment evaluation.

Finally, $2^b - 1$ is odd. The reduced denominator of x_b^N is thus exactly 2^{Nb} . This proves the stated exact length and explains why input/output accuracy and exact arithmetic storage cannot be identified. The rational-operation counts in the preceding sections exclude the cost of obtaining numerical advice; they are not bit-operation counts. $\square$

APPENDIX K. CONFORMANCE TO A FINITE SYNTHESIS REQUEST

The construction theorem assumes a specified horizon and a specified total numerical error. A table cannot certify these assumptions by choosing its own horizon or its own rounding scale. We record the finite interface that connects the mathematical implication to the returned program. This interface does not certify the accuracy of an arbitrary external moment oracle.

Definition K.1 (Bound synthesis specification). A synthesis specification consists of the requested horizon T , command and report alphabets with their fixed orders, state budget M , stage coordinate dimensions $d_0, \dots, d_T$, ordered query interfaces of sizes $b_0, \dots, b_T$, rounding precision v , raw evaluation error bound σ , and total error allowance τ , with

$$(194) \quad \sigma + 2^{-v-1} \leq \tau.$$

It also fixes the numerical state and query vectors on every grid history through T . All these objects are fixed from the caller's task and numerical input before synthesis is invoked. Their dimensions may be specified directly by the application, or read from that external input interface and checked on every history. They are never inferred from a returned transition table. Repeated evaluations use this same frozen name, with the same coordinate and query orders.

An independently enumerated check first compares a returned program's precision and all stage lengths with this specification. It then checks the complete numerical lists, the farthest-first representatives, every nearest-centre transition with the fixed tie rule, and every clipped rounded output as in Definition J.1. State and query dimensions are checked against the specification, not merely against one another. The structural check precedes every use of a program radius in a stopping certificate. Input fingerprints record identity and provenance; they are not proofs of numerical accuracy.

Proposition K.2 (Request-bound construction implication). *Suppose the numerical input fixed in Definition K.1 has state and query error at most σ on its specified histories, and the grid is an h -net. If the preceding check accepts, then the returned program implements all T requested stages, uses at most M labels per stage, and satisfies Theorem J.2 with the caller’s allowance τ . Its independently recomputed radii satisfy*

$$\max\{0, r_n/2 - \tau\} \leq e_n(M) \leq r_n + A_n h + 2\tau.$$

A returned object with a different precision or fewer stages is rejected before these inequalities are issued. Changing a precision field without changing its numerical tables does not avoid the entrywise checks.

Proof. The preliminary comparisons force the requested precision, stage lengths, and interface dimensions. Rounding a real number to the nearest v -place dyadic, with either prescribed rule at ties, changes it by at most 2^{-v-1} . The triangle inequality and (194) therefore give state and query errors at most τ ; clipping query outputs to $[0, 1]$ cannot increase error to a true probability. The remaining finite checks are precisely the hypotheses of Definition J.1 on those fixed numerical lists. Its theorem gives the recurrence and the label budget. Lemma I.3 gives the displayed bracket. A mismatch of precision or of any stage length fails a preliminary comparison. With correct metadata but different tables, acceptance still requires every output and transition to equal the independently rebuilt entry; if all entries agree, the program does implement the requested finite construction. None of these implications uses a residual bound increased after a faulty transition has been observed. $\square$

For the ordinary and common-name adaptive constructions take $v = b + 2$ and $\sigma = \tau_b/2$. Since $2^{-v-1} = h_b/8 \leq \tau_b/8$, (194) holds with the prescribed τ_b . The stopping bracket therefore uses an error allowance bound to the request before synthesis. Freezing these finite inputs and checking them is offline work. It adds no field to the running state, which remains one integer index. The earlier residual certificate for an arbitrary table remains valid, but does not assert conformance to a different synthesis request.

APPENDIX L. COMPARISON WITH FINITE APPROXIMATIONS AND INFORMATION STATES

L.1. Positive diagonal pairings and analytic matrix orders. The finite square necessity-and-sufficiency statement has an exact predecessor, not merely a common determinant identity. For $X = \{x_1, \dots, x_m\}$, let $A_{ik} = v_i(x_k)$ and $B_{jk} = \phi_j(x_k)$, with both matrices of row rank d . The pairing is $A \text{diag}(w) B^t$. Banaji–Pantea [33, Definition 2.24 and Lemma 2.36] characterize the corresponding common weak sign, with a nonzero product of maximal minors, by rank preservation under positive diagonal scalings. Their criterion applies because

$$\text{rank}(AD_1 B^t D_2 A) = d \iff \det(AD_1 B^t) \neq 0;$$

$D_2 A$ is onto. Positive weights and positive probability weights are equivalent by scalar normalization. The compact-space version uses full-support measures; posterior normalization and unconditional history mass are the additional experiment-specific conclusions. Andréief’s identity alone would not attribute this necessity result.

For rectangular evaluations $A \in \mathbb{R}^{d \times m}$ and $B \in \mathbb{R}^{k \times m}$, full column rank for every positive w is equivalent to

$$\ker A \cap \text{diag}(w) \text{im } B^t = \{0\} \quad \text{for every } w > 0.$$

The finite positive-diagonal/sign-vector alternative is the one in Müller et al. [34, Lemma 2.1]. Our product-cone formulation does not claim a new finite sign-vector theorem. Its compact-space rank alternatives specify full-support witnesses for entire kernels, and its quantitative

form returns to the actual history map. Proposition 9.6 shows why preserving only a fixed square subspace containing evidence would lose part of this experiment class.

Similarly, the Smith form and its singular-value orders are classical matrix facts; see Kaveh–Makhnatch [35, Theorem 1.1]. They are not themselves a memory theorem. Here the affine acquisition law supplies constants uniform through rank loss, while Theorem 12.1 realizes arbitrary small matrix germs inside one fixed positive physical experiment. Together these give all joint memory exponents in Corollary 12.4, including priors uniformly equivalent to a fixed full-support measure. A formal matrix factorization alone supplies neither physical realization nor a risk comparison uniform in the entire integer budget.

L.2. Acquisition, observation, and the resolution invariant. The two classifications distinguish the coordinates that can be acquired from the metric in which those coordinates are observed. In the monomial case let $z_a(h) = (\mathbb{E}[g_j^a | h])_j$ be the normalized complete Newton flag, and put $D_a = \text{diag}(d_j)$. Equation (17) and the fixed spanning query menu give the chain

$$h \mapsto z_a(h) \mapsto D_a z_a(h) \mapsto \text{conditional query probabilities.}$$

The last map is uniformly conditioned on its active coordinates. Each initial flag of length at most $p_{n,m}$ is attained with a uniform positive-history minorization, including at confluent parameter values. The d_j belong to observation: a zero pivot makes a normalized coordinate physically invisible, without invalidating its confluent extension. It is essential here to use a complete flag, rather than individual divided differences selected after the collision.

In the circular case the corresponding chain is

$$h \mapsto (S_j(h))_j \mapsto (c_j(h))_j = (\tau^j S_j(h))_j \mapsto (Y_j(h))_j = (\tau^{2j} S_j(h))_j.$$

The normalized S -flags have the uniform acquisition minorization; c consists of the posterior Fourier coefficients; the query metric is uniformly conditioned in Y . Each complex coordinate contributes two real directions. Thus acquisition and observation contribute separate factors of τ^j . At zero contrast the normalized extension remains an auxiliary device and every physical Y_j is zero. These statements are proved in Lemmas 13.2 and 13.3, not inferred from Fourier closure.

For nonincreasing real side orders $s_1, \dots, s_p$, write

$$\mathcal{Q}_M(s; p) = \max_{1 \leq \ell \leq p} \left(\frac{\prod_{i=1}^{\ell} s_i}{M} \right)^{2/\ell}.$$

The monomial law is $\mathcal{Q}_M(d; p_{n,m})$, since the initial Leja products are comparable to $\mathcal{V}_{m,\ell}$. The circular law is the same expression on the paired list $\tau^2, \tau^2, \dots, \tau^{2k}, \tau^{2k}$, $k = \min(n, m)$; its even products give $\mathcal{H}_{n,m}$ and its odd products add no larger term. This is a comparison of the two proved laws, not an assertion that an arbitrary positive detector admits either normal form. The product-tangent criterion supplies a sufficient attainment test; the identification of its tangent and its observation scales is a separate obligation in each experiment.

This identification is the structural content of the classification. Exact dimension records the number of nonzero directions. The full profile records their finite-resolution effect jointly in the label budget and the degenerating parameter, without first fixing a collision stratum. Its truncation records the restriction imposed by a finite past. The global covering and causal statements show that these are operational scales, rather than only local differential invariants. The separate ambiguity theorem tests the same observation flag against uncertainty of the prior, and consequently has no past-dimension truncation. No claim about a sharp memory law follows from a dimension count, a posterior recurrence, or bounded duality alone.

L.3. Relation to Bayesian Fourier filtering. Exact Fourier representations of Bayesian phase posteriors and their coefficient updates are established methods. Van den Berg [29, Section 2.2, equations (12)–(15)] treats trigonometric posterior representations, their exact updates, and the complexity caused by coefficient growth. De Neeve, Lebedev, Negnevitsky and Home [30] use reduced-contrast likelihoods and Fourier-coefficient recursions; see Section II, equations (1)–(3), of the cited preprint. Their numerical representations in Section IV address coefficient storage and computation in adaptive phase estimation. Fourier closure, neighboring-coefficient updates, and the use of a finite Fourier state are therefore not contributions claimed here.

The comparison with Theorem 13.1 concerns a different experiment, loss, and resource. Our four-cell lookup interface uses commands acting on the four detector cells, with real acceptance probabilities varying over the continuous gate cube; every acquisition trial and failure is counted. The finite detector, finite report alphabet, finite future query menu, and finite persistent-label alphabet do not make the command alphabet finite. A finite menu of physical future queries is revealed only after the persistent label has been formed. The loss is squared error in their conditional probabilities, and the resource is the cardinality M of the persistent state, not the number of stored real coefficients, the evolution time, or the accuracy of a phase estimator. A vector of finitely many exact real coefficients still takes continuously many values and does not by itself meet a finite-label constraint.

The additional quantitative conclusion is a uniform lower measure on the normalized acquisition flags, their identification with paired physical scales τ^{2j} , and matching unconditional and worst-history finite-label laws. The weighted update then realizes the maximum checkpoint law with one stage-compatible transducer, without access to a discarded prefix. These assertions are proved for each fixed N , uniformly in M and $0 \leq \tau \leq \tau_*(N, \eta, \rho)$. The contrast and model are known to the read-only program; its length and transient arithmetic are not counted as persistent labels. Neither the interval nor the comparison constants are claimed uniform as $N \rightarrow \infty$. The cited representation and update formulas are not identified with this minimax conclusion, nor is our theorem claimed to subsume the cited phase-estimation results. The circular causal construction is intrinsic; the effective compiler proved in the appendices concerns the monomial model.

The operational reconstruction in Section 14 is a converse to these forward metric classifications. It identifies the initial products from the full checkpoint regret curve, up to uniform constants; it does not claim a new Fourier representation or coefficient recursion.

L.4. Relation to finite approximation and information states. Finite approximation of a controlled model and finite realization of a prediction state are related, but their conclusions and numerical inputs must be distinguished. We make the comparison at the level of particular statements, rather than regarding finite-state approximation as a new principle.

Saldi, Yüksel and Linder [24, Theorem 5.2] in the explicitly linked arXiv version 3 bound the discounted performance loss of policies obtained from finite approximating Markov decision processes by $O(n^{-1/d})$. The result uses compact Euclidean state space, a Lipschitz cost, a transition kernel Lipschitz in the Wasserstein distance, and $\beta K_2 < 1$. Their finite model uses state quantization and averaged transition kernels; their Section 6 also studies order optimality. Our theorem does not strengthen that discounted-control conclusion. Here the supplied data are certified evaluations of feasible histories; the output is an integer transition program with at most a prescribed M labels at each stage. The scale selected in Theorem I.4 is the attainable covering radius itself, not a prescribed ambient-dimensional power. The collision law additionally identifies that scale and supplies a matching prediction converse for the same experiment.

Kara and Yüksel [5, Assumption 4 and Theorem 12] obtain an exponentially decreasing discounted policy-loss bound for finite-window feedback after a warm-up period. Their hypotheses

include transition regularity, a domination condition, a Dobrushin-type filter stability coefficient $\alpha < 1$, and a restriction on the discount. The finite-window description retains recent observations and actions. Our finite-horizon construction does not require filter contraction, but makes no corresponding infinite-horizon assertion. It enumerates histories only offline and retains an index, not a recent observation window. Its effective input is a finite coefficient–moment table in the monomial case. Window length, persistent-label budget, and numerical program length are different resources; no comparison of their optima follows simply by identifying these names.

Subramanian, Sinha, Seraj and Mahajan [8, Definition 7 and Theorem 9] give an all-history value approximation and policy-loss bound from reward and transition errors of an approximate information state. Their recursive error allowance combines reward error, an integral-probability-metric transition error, and the next-stage allowance; the policy bound has twice that allowance. They also treat action compression in Theorem 17. Thus neither an all-history error recurrence nor action compression is new here. The additional conclusion proved here is a finite synthesis with a prescribed label budget: approximate feasible histories give a computable two-sided covering certificate, which selects its own numerical mesh. In the monomial experiment the certificate is tied to the full collision-sensitive attainable profile by Lemma I.8, with both the finite program and its numerical advice separately accounted for.

These distinctions concern the displayed theorems, not an assertion that one of the cited theories is contained in another. The classical farthest-first estimate supplies the finite certificate; the attained posterior geometry supplies its interpretation as the sharp prediction scale. Their combination yields the resource phases of Corollary I.10 without an input identifying which phase is present. The geometric and numerical assertions require separate proofs, and the specializations in the appendices retain their own hypotheses.

Proposition J.3 isolates a narrower logical comparison: a truthful residual bound for a supplied table is not a construction theorem at the optimal covering scale. Its positive counterpart is Theorem J.2. This does not assert that a new finite-approximation principle is needed. The non-routine geometric input is the collision-uniform attainment of the entire exterior profile by positive histories, for every fixed full-support prior, together with a global cover of the same order. One consequence beyond table-size exponents is Corollary J.8: a single imperfect moment name produces one program achieving $O(\Xi_N + \delta^2)$ simultaneously in all compatible experiments in a stated prior envelope. Proposition J.9 identifies a positive experiment where the additional uncertainty term is necessary. These are finite-horizon prediction statements; no improvement to the cited infinite-horizon control results is asserted.

The common-moment problem considered here has a different quantifier: one fixed numerical name must determine one program for all compatible experiments. Theorem 16.1 gives a two-sided risk classification for interior names in this problem. The squared-loss two-point identity is elementary, and its use is not a new general lower-bound method. What enters its matching memory term is the full collision-uniform attained geometry, rather than the rank alone. The saturation formula in Corollary 16.4 makes this dependence explicit. It neither asserts a calibration-only uncertainty converse nor identifies a finite-program error recurrence as an additional geometric principle.

REFERENCES

- [1] D. Blackwell, *Equivalent comparisons of experiments*, Ann. Math. Statist. **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] S. Eliahou and E. Mazumdar, *Iterated sumsets and Hilbert functions*, J. Algebra **593** (2022), 274–294. doi:10.1016/j.jalgebra.2021.11.019.
- [3] J. Elias, *Sumsets and projective curves*, Mediterr. J. Math. **19** (2022), article 177. doi:10.1007/s00009-022-02108-0.
- [4] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics 1730, Springer, 2000. doi:10.1007/BFb0103945.

- [5] A. Kara and S. Yüksel, *Near optimality of finite memory feedback policies in partially observed Markov decision processes*, J. Mach. Learn. Res. **23** (2022), no. 11, 1–46. <https://jmlr.org/papers/v23/20-1152.html>.
- [6] M. L. Littman, R. S. Sutton and S. Singh, *Predictive representations of state*, Advances in Neural Information Processing Systems **14** (NIPS 2001). Original proceedings paper.
- [7] A. Pinkus, *Totally Positive Matrices*, Cambridge Tracts in Mathematics 181, Cambridge University Press, 2010.
- [8] J. Subramanian, A. Sinha, R. Seraj and A. Mahajan, *Approximate information state for approximate planning and reinforcement learning in partially observed systems*, J. Mach. Learn. Res. **23** (2022), no. 12, 1–83. <https://jmlr.org/papers/v23/20-1165.html>.
- [9] Q. Qi, *Finite-horizon information states of calibrated experiments*, English revision 5, September 6, 2026, source directory `papers/A1-english-v5` in TrillioniumFoundation/theta-theory, commit 20a26285a26c. Preserved verbatim in `papers/A1-english-v6/legacy/` in the repository.
- [10] *A1 English v5: referee report and claim audit*, owner-commissioned AI-assisted repository review, September 6, 2026, commit 171a7e20470d, `reviews/a1-english-v5-2026-09-06`. This is not a journal-appointed review or a proof certificate.
- [11] Q. Qi, *A1: strong finite-cylinder response on a common reporting space*, repository derivation, `round33/chapters/A1.tex`, commit 7f1bc9a42ba2.
- [12] C. de Boor, *Divided differences*, Surveys in Approximation Theory **1** (2005), 46–69. <https://arxiv.org/abs/math/0502036>.
- [13] Q. Qi, *Sparse observation algebras and finite-horizon memory*, English revision 6, September 6, 2026, commit 750a65ef6242.
- [14] *An exact dimension jump at fixed finite resolution*, technical note accompanying the owner-commissioned AI-assisted A1 v6 repository review, September 6, 2026, commit 414f8c43a623. This source introduced the near-collision family and its base-model bound; it did not assert the matching uniform crossover proved here.
- [15] D. Batenkov, B. Diederichs, G. Goldman and Y. Yomdin, *The spectral properties of Vandermonde matrices with clustered nodes*, arXiv:1909.01927v2 (2020), Theorems 2.2–2.3 and Corollary 2.1. <https://arxiv.org/abs/1909.01927>.
- [16] Y. Yomdin and G. Comte, *Tame Geometry with Application in Smooth Analysis*, Lecture Notes in Mathematics 1834, Springer, Berlin, 2004.
- [17] G. Comte and I. Halupczok, *Motivic Vitushkin invariants*, Compos. Math. **161** (2025), no. 5, 959–992. doi:10.1112/S0010437X25007031. The cited preprint is arXiv:2206.15412v2 (2024). The classical real variations and entropy inequality used here are recalled in the introduction, equations (4)–(5). <https://arxiv.org/abs/2206.15412>.
- [18] Y. Zhang and J. Kileel, *Covering Number of Real Algebraic Varieties and Beyond: Improved Bounds and Applications*, arXiv:2311.05116v4 (June 6, 2025), Lemma 2.18, p. 11. <https://arxiv.org/abs/2311.05116v4>.
- [19] *What the v7 theorems imply, and where their hypothesis matters*, technical note accompanying the owner-requested AI-assisted A1 v7 repository review, September 6, 2026, commit 72eb41e358bd. This note gives the necessary-budget deduction and the seven-trial past-limited boundary, not the attainable-flag classification proved here.
- [20] F. Leja, *Sur certaines suites liées aux ensembles plans et leur application à la représentation conforme*, Ann. Polon. Math. **4** (1957), no. 1, 8–13. <https://eudml.org/doc/208291>.
- [21] L. Bos, S. De Marchi, A. Sommariva and M. Vianello, *Computing multivariate Fekete and Leja points by numerical linear algebra*, SIAM J. Numer. Anal. **48** (2010), 1984–1999. doi:10.1137/090779024.
- [22] T. F. Gonzalez, *Clustering to minimize the maximum intercluster distance*, Theoret. Comput. Sci. **38** (1985), 293–306. doi:10.1016/0304-3975(85)90224-5.
- [23] *The classical spectral content and phase structure of A1 v9*, technical note accompanying the owner-requested AI-assisted repository review, September 6, 2026, commit 7a499e3cb323. The finite spectral comparison and isolated Hermite statement are attributed to this note; its provenance is not a journal endorsement.
- [24] N. Saldi, S. Yüksel and T. Linder, *On the asymptotic optimality of finite approximations to Markov decision processes with Borel spaces*, Math. Oper. Res. **42** (2017), 945–978. Theorem 5.2 and Section 6; arXiv:1503.02244v3.
- [25] *A stronger sufficient precision law for the A1 v10 compiler*, technical note accompanying the owner-requested AI-assisted repository review, September 6, 2026, commit 1aa4599eedca. The separated-chain floor and its uniform precision consequence are attributed to this note; it is not a journal report.

- [26] Independent referee-style diagnostic note, *Transition sensitivity and arithmetic precision: exact referee diagnostics*, repository review, September 6, 2026, commit [7492bf0d7423](#). The interval transition diagnostic is attributed to this note; it is not a journal report.
- [27] *Mathematical audit accompanying the A1 v14 referee report*, owner-requested AI-assisted repository assessment, September 6, 2026, commit [3d58bb33ae12](#), section A1. The bounded-test-vector reduction is attributed to this audit; it is not a report commissioned by a journal.
- [28] L. Chen, R. Lin and H. Zhang, *The smallest singular value of nonuniform Fourier matrices*, preprint, August 22, 2026, [arXiv:2608.21960v1](#).
- [29] E. van den Berg, *Efficient Bayesian phase estimation using mixed priors*, *Quantum* **5** (2021), 469. doi:10.22331/q-2021-06-07-469. Section 2.2, equations (12)–(15); author version [arXiv:2007.11629v2](#).
- [30] B. de Neeve, A. V. Lebedev, V. Negnevitsky and J. P. Home, *Time-adaptive phase estimation*, *Phys. Rev. Research* **7** (2025), 023070. doi:10.1103/PhysRevResearch.7.023070. The equation-level comparison refers to Section II, equations (1)–(3), and Section IV of [arXiv:2405.08930v1](#).
- [31] J. Bochnak, M. Coste and M.-F. Roy, *Real Algebraic Geometry*, *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*, vol. 36, Springer, Berlin, 1998. doi:10.1007/978-3-662-03718-8.
- [32] P. J. Forrester, *Meet Andréief, Bordeaux 1886, and Andreev, Kharkov 1882–83*, preprint, 2018. [arXiv:1806.10411](#).
- [33] M. Banaji and C. Pantea, *Some results on injectivity and multistationarity in chemical reaction networks*, *SIAM J. Appl. Dyn. Syst.* **15** (2016), 807–869. doi:10.1137/15M1034441.
- [34] S. Müller, E. Feliu, G. Regensburger, C. Conradi, A. Shiu and A. Dickenstein, *Sign conditions for injectivity of generalized polynomial maps with applications to chemical reaction networks and real algebraic geometry*, *Found. Comput. Math.* **16** (2016), 69–97. doi:10.1007/s10208-014-9239-3.
- [35] K. Kaveh and P. Makhnatch, *Invariant factors as limit of singular values of a matrix*, preprint, [arXiv:1811.07706v2](#), 2022. [arXiv:1811.07706v2](#).
- [36] R. de Wolf, *A brief introduction to Fourier analysis on the Boolean cube*, *Theory of Computing Library, Graduate Surveys* **1** (2008), 1–20. doi:10.4086/toc.gs.2008.001. The author-hosted version, Section 2.2, gives the product-character expansion used here.